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(54) **RESOURCE RECLAIMING IN A TENANCY
HAVING MULTIPLE SERVICE
SUBSCRIPTIONS**

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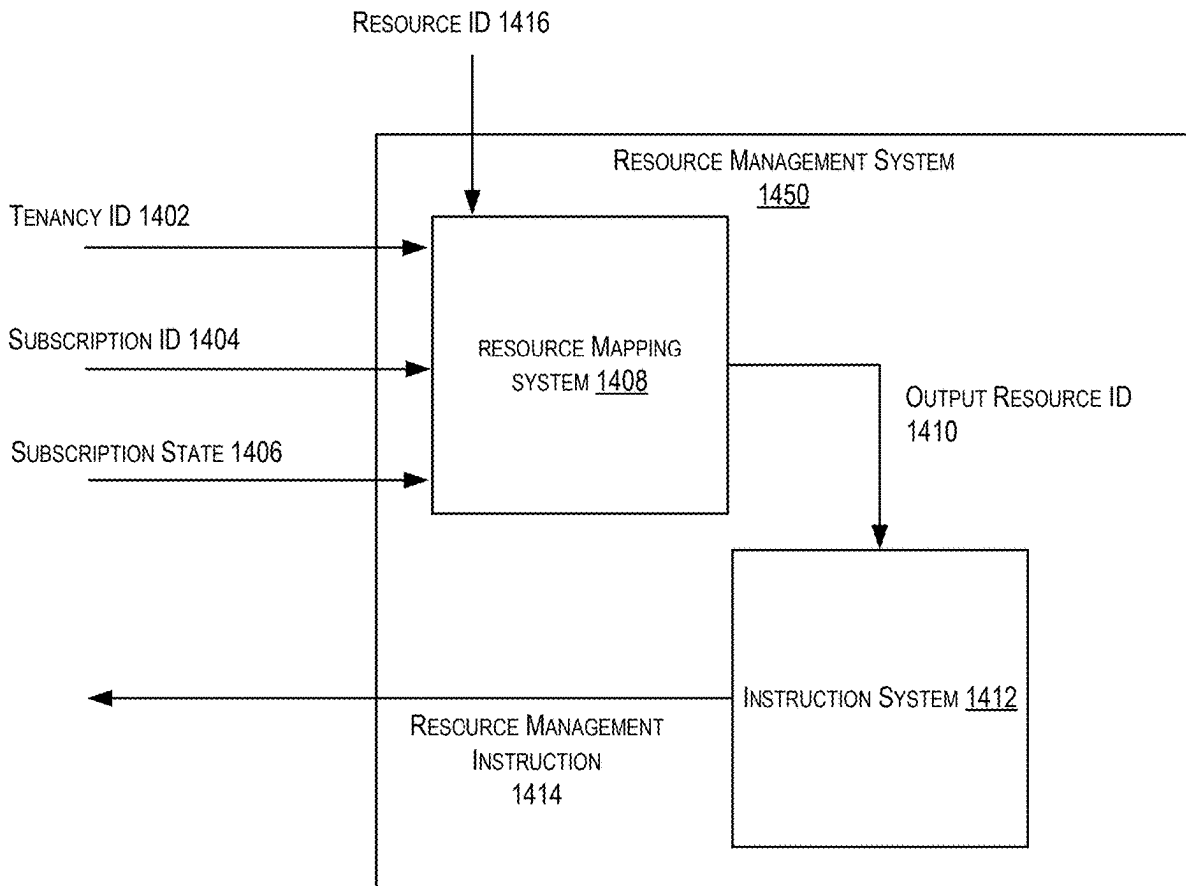
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(57) **ABSTRACT**

Techniques including associating a subscription of a user account with a cloud service provider (CSP) with a virtual resource of the CSP. The techniques further include enabling a device associated with the user account to access the virtual resource, the virtual resource available and accessible as part of a private cloud associated with the user account. The techniques further include determining a change to the subscription. The techniques further include disassociating, based on the change, the virtual resource from the subscription such that the virtual resource is unavailable via the private cloud in association with the subscription and such that the virtual resource is available and accessible for use in association with a different subscription.



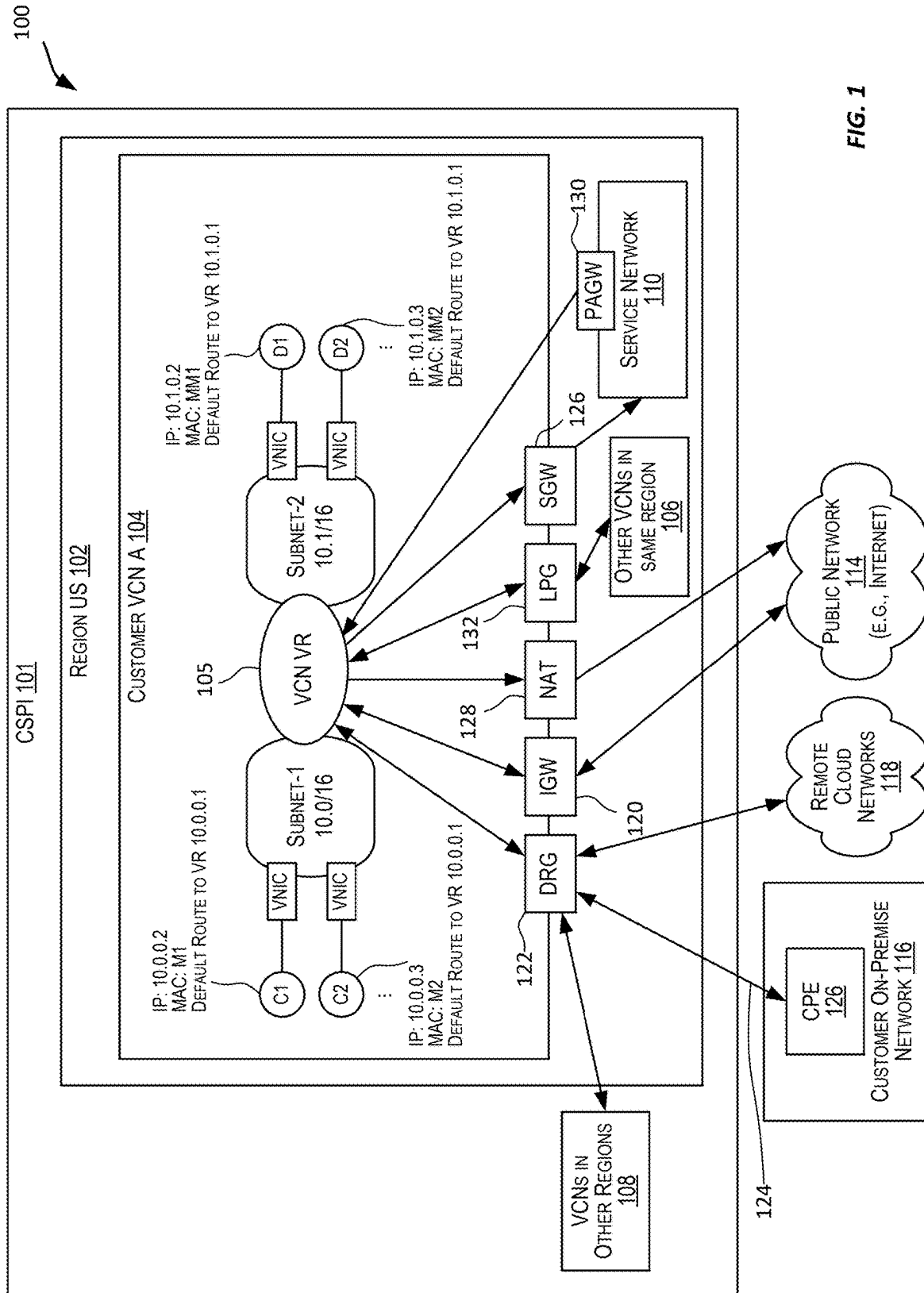
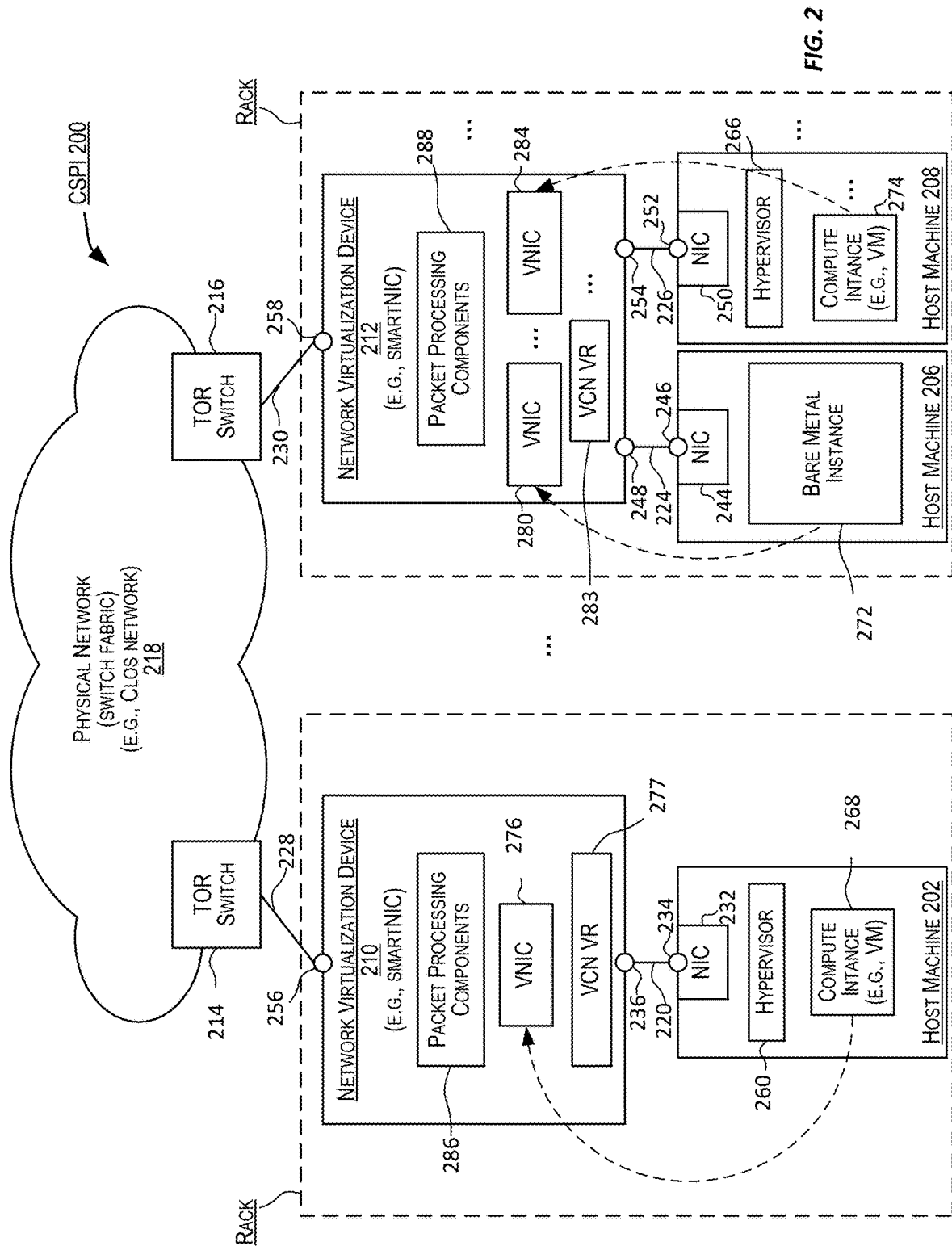


FIG. 1



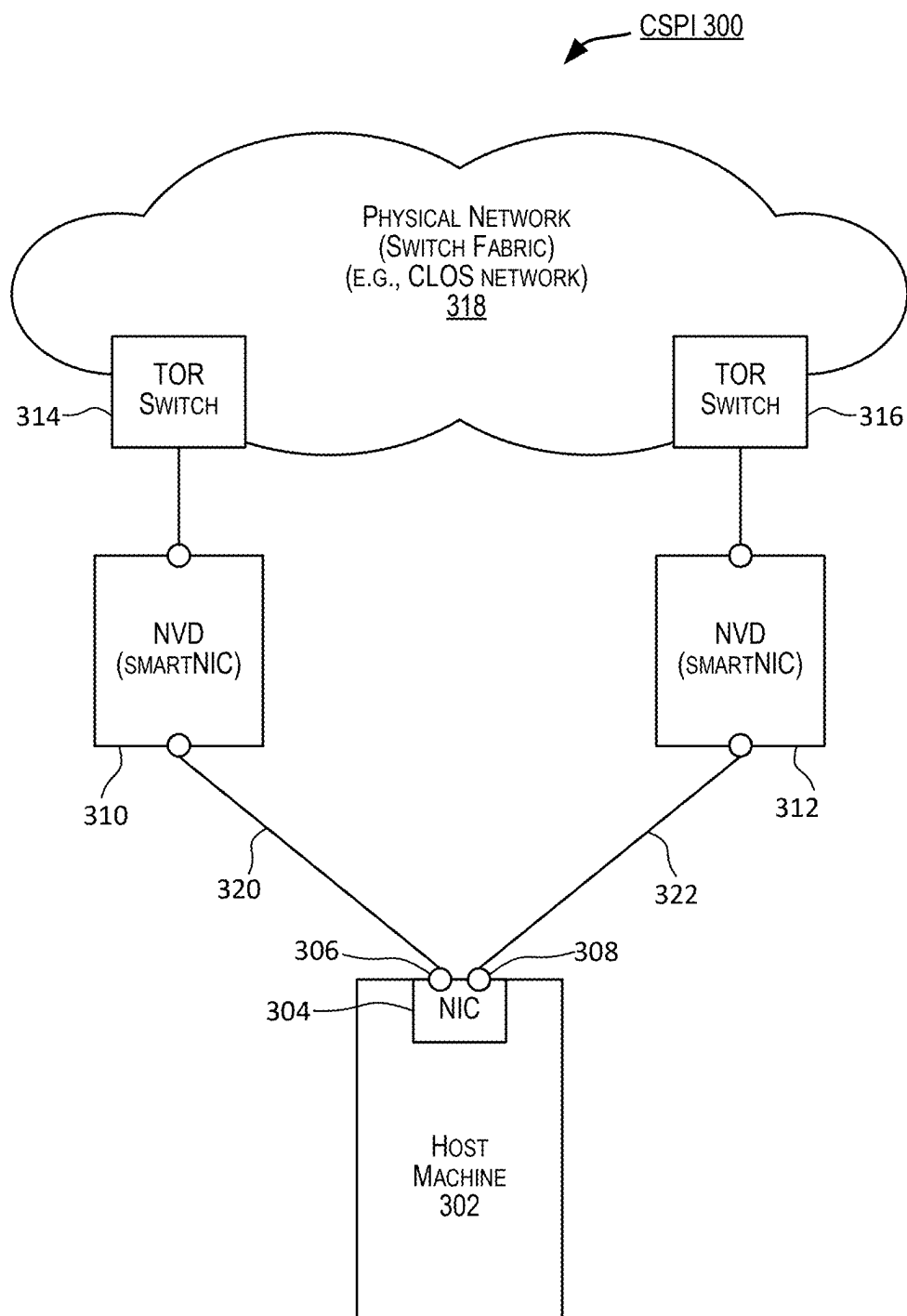


FIG. 3

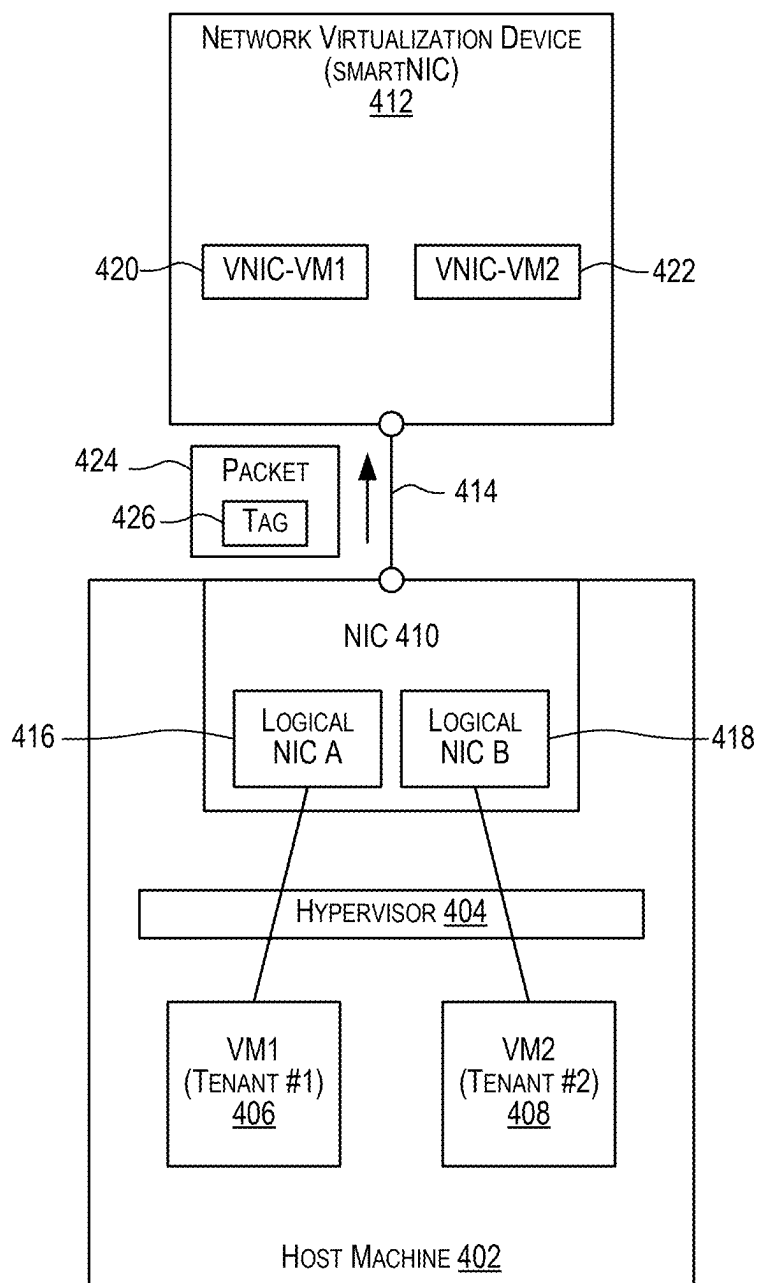


FIG. 4

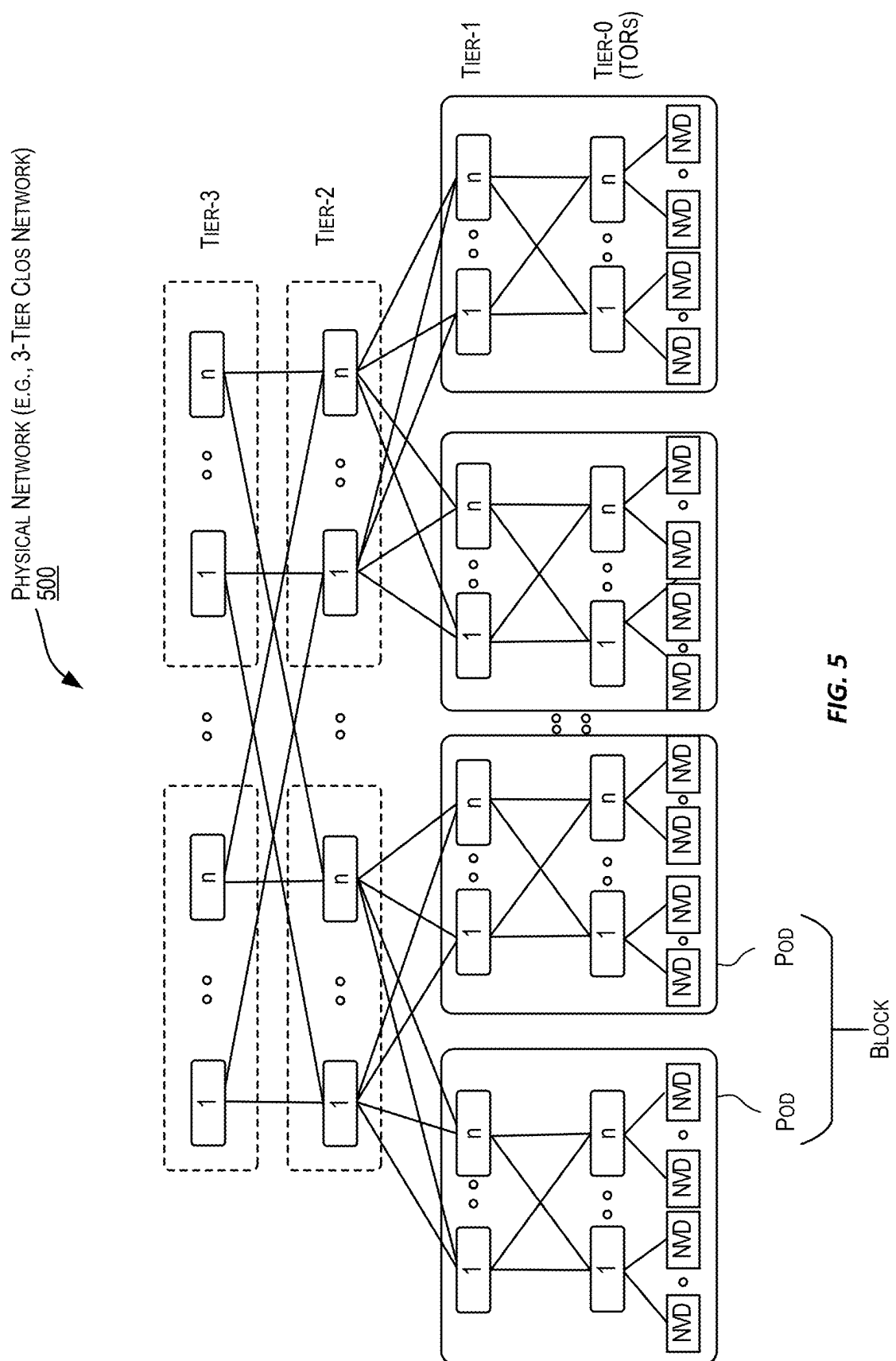
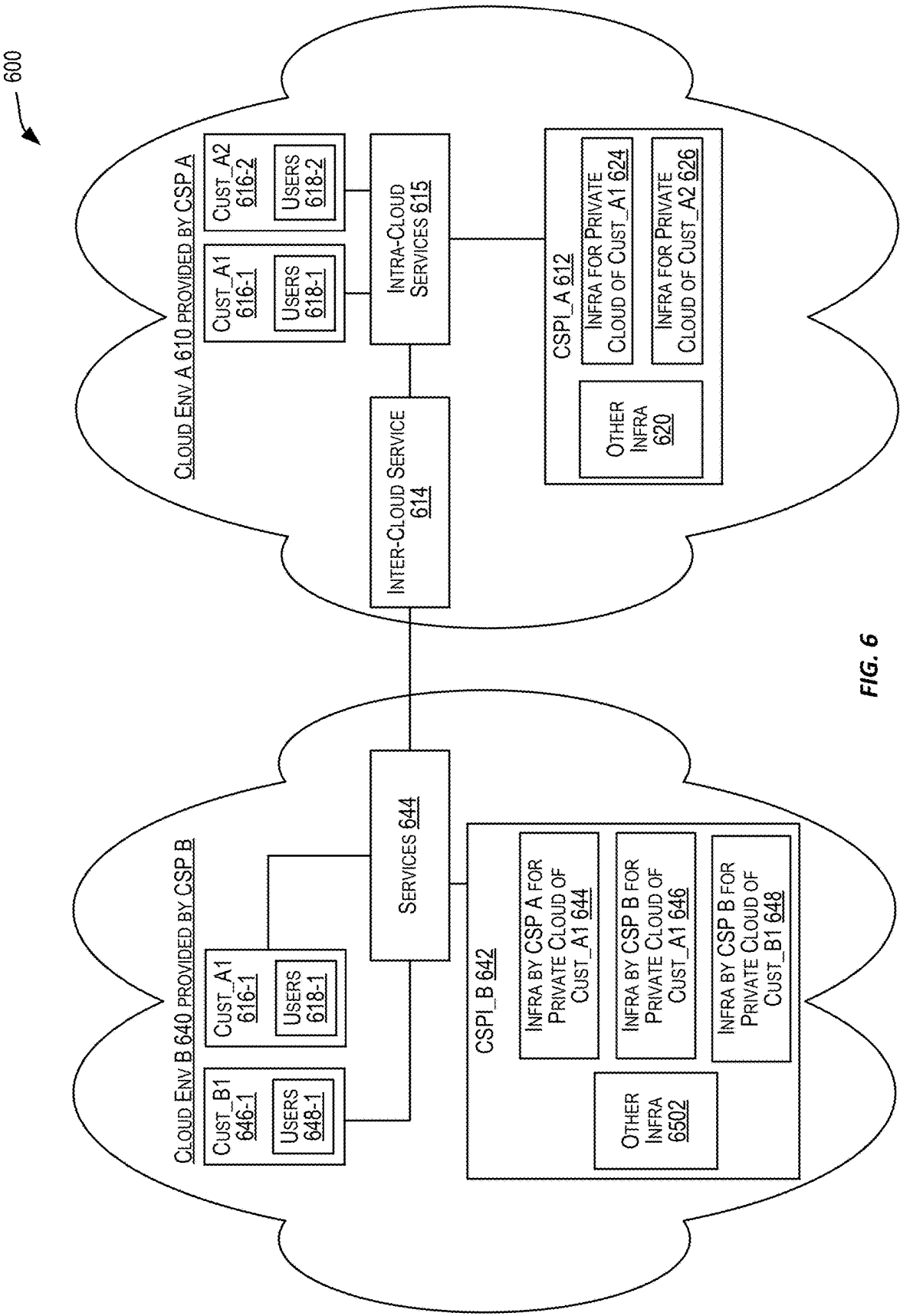


FIG. 5



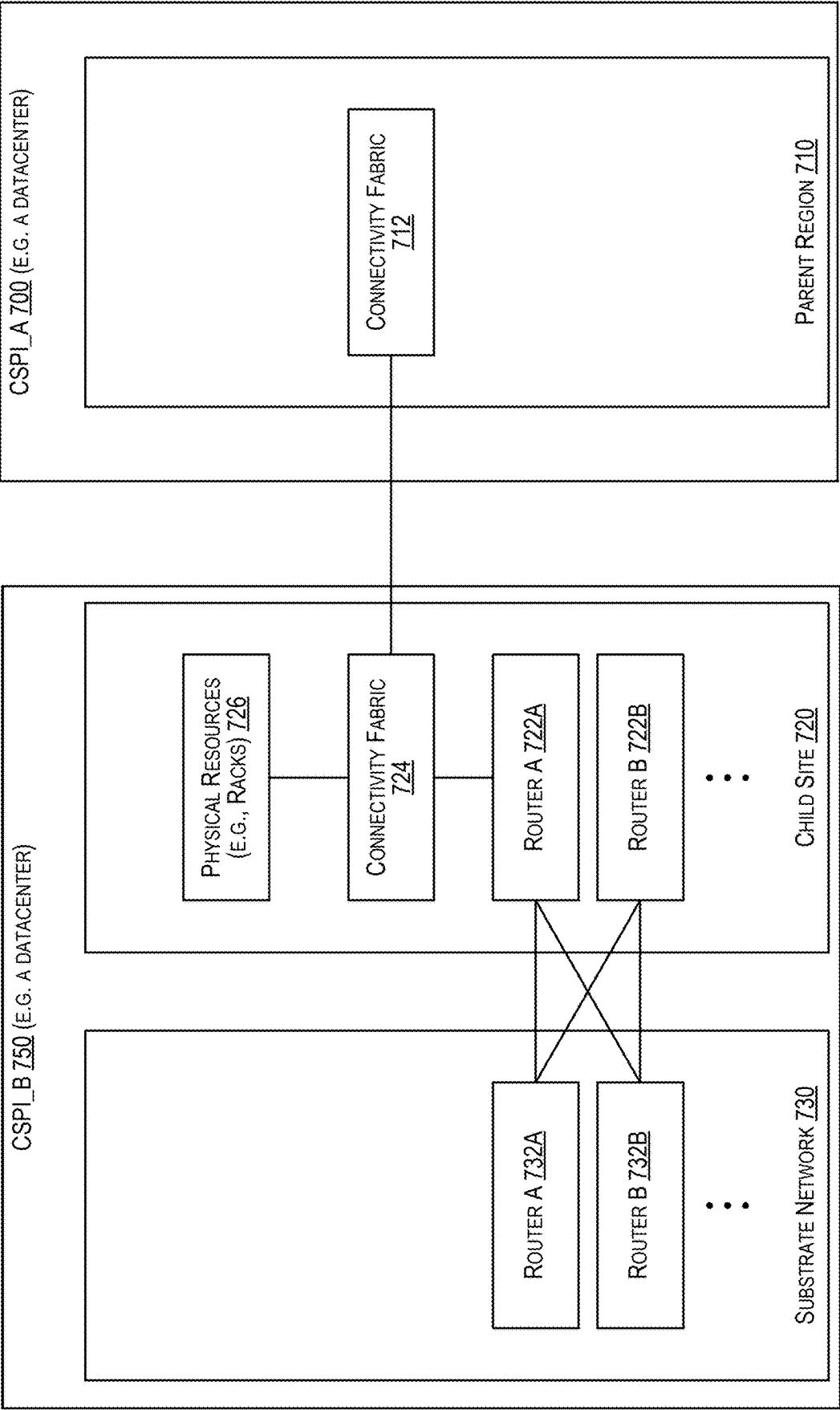


FIG. 7

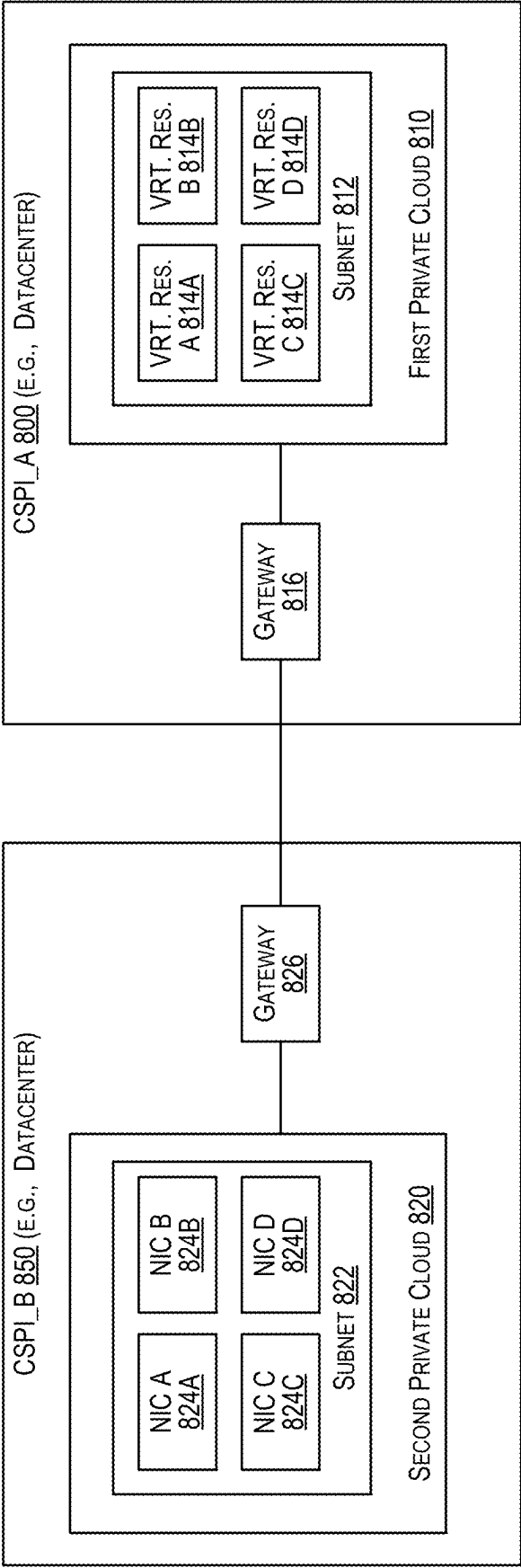


FIG. 8

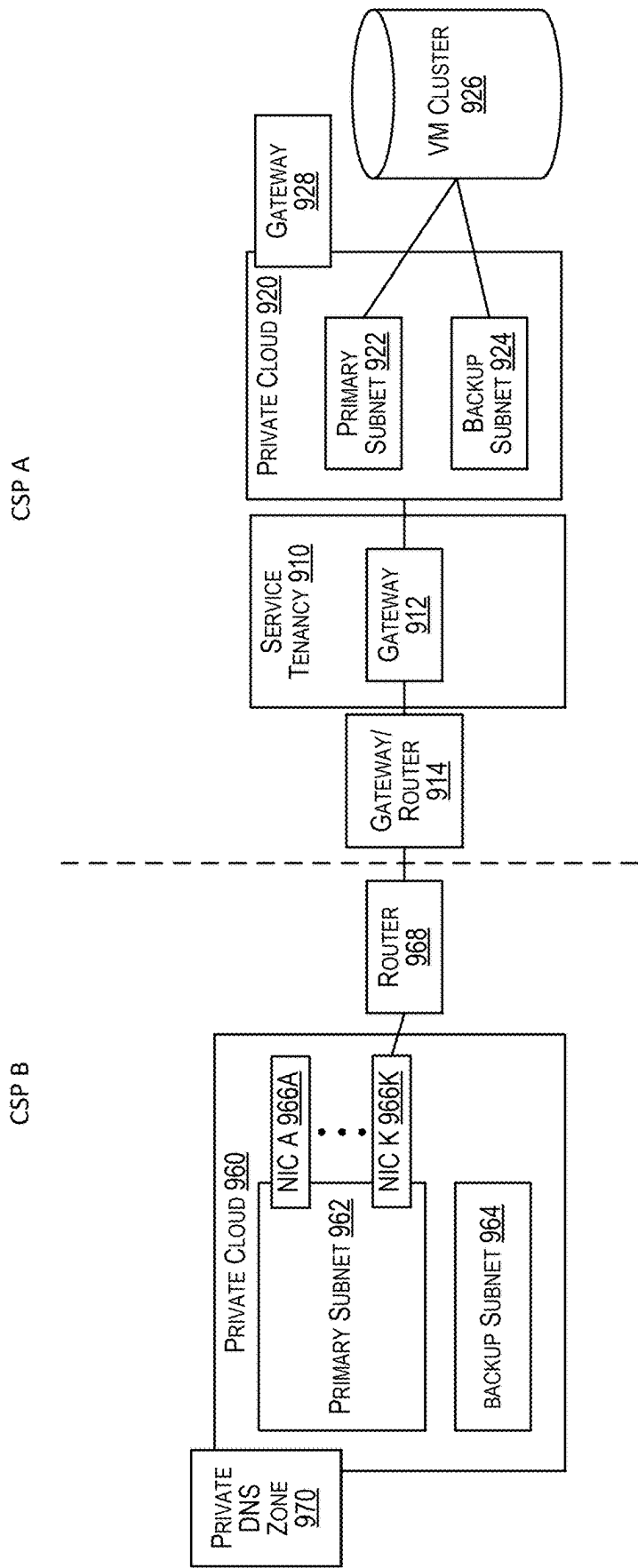


FIG. 9

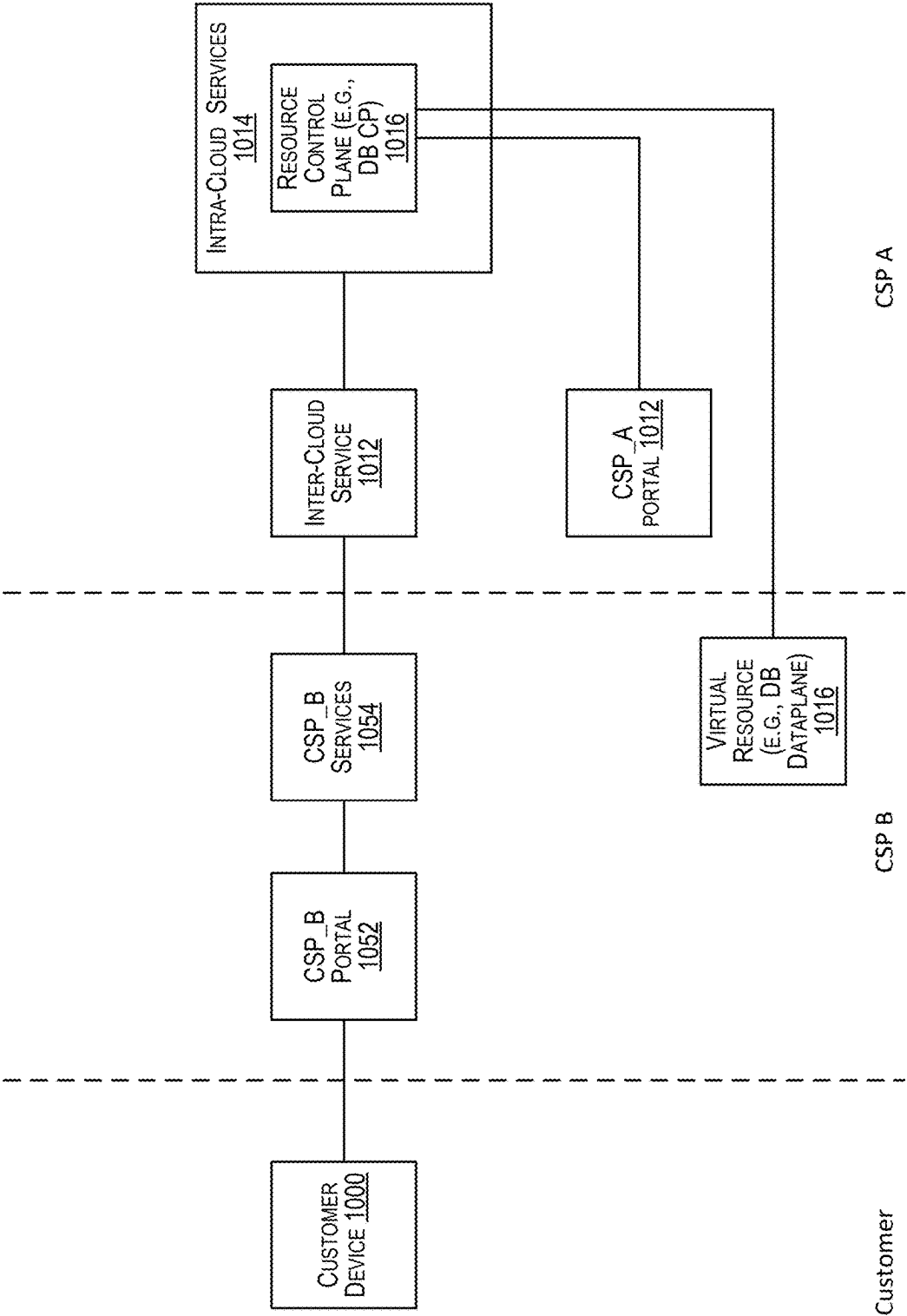


FIG. 10

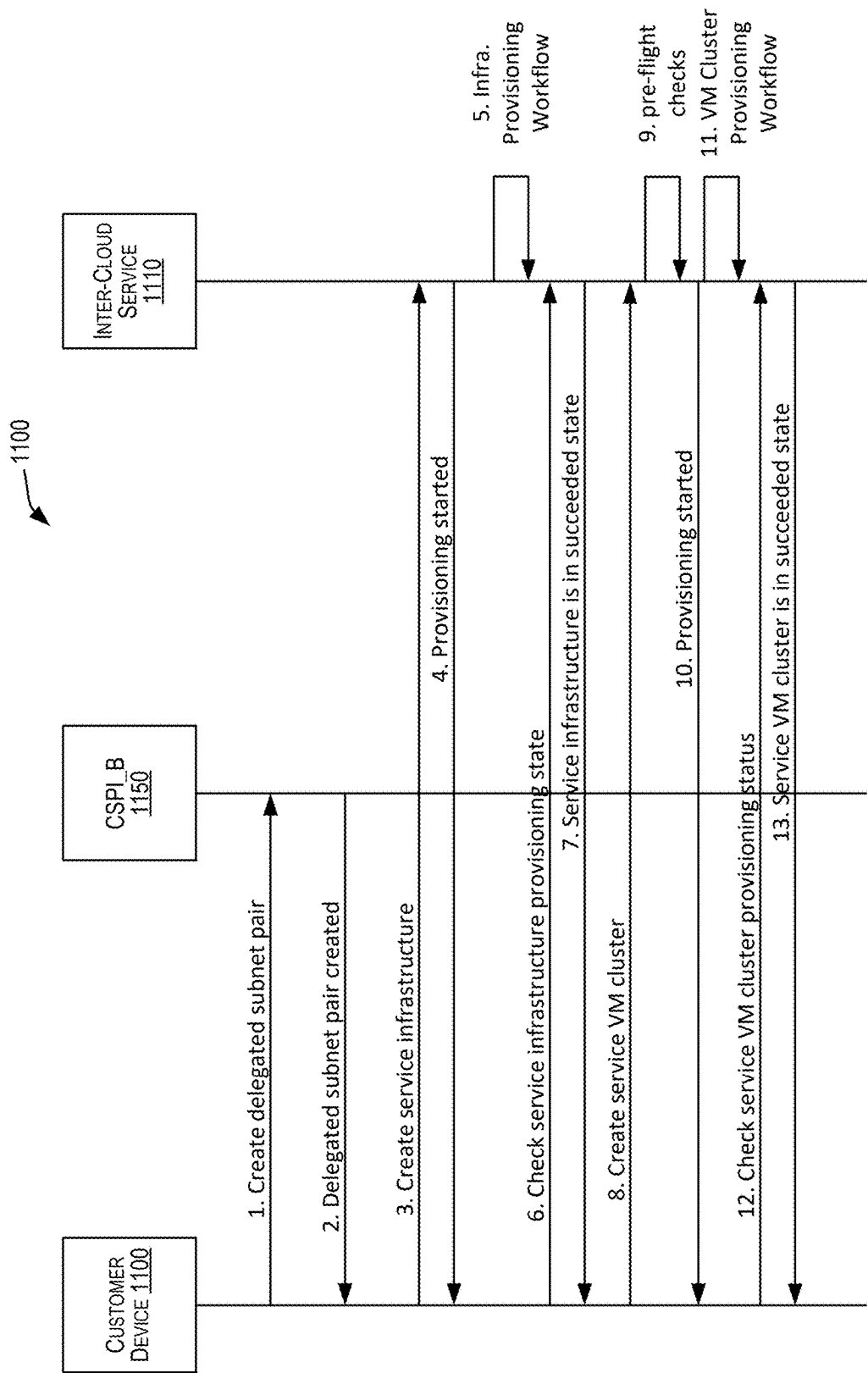


FIG. 11

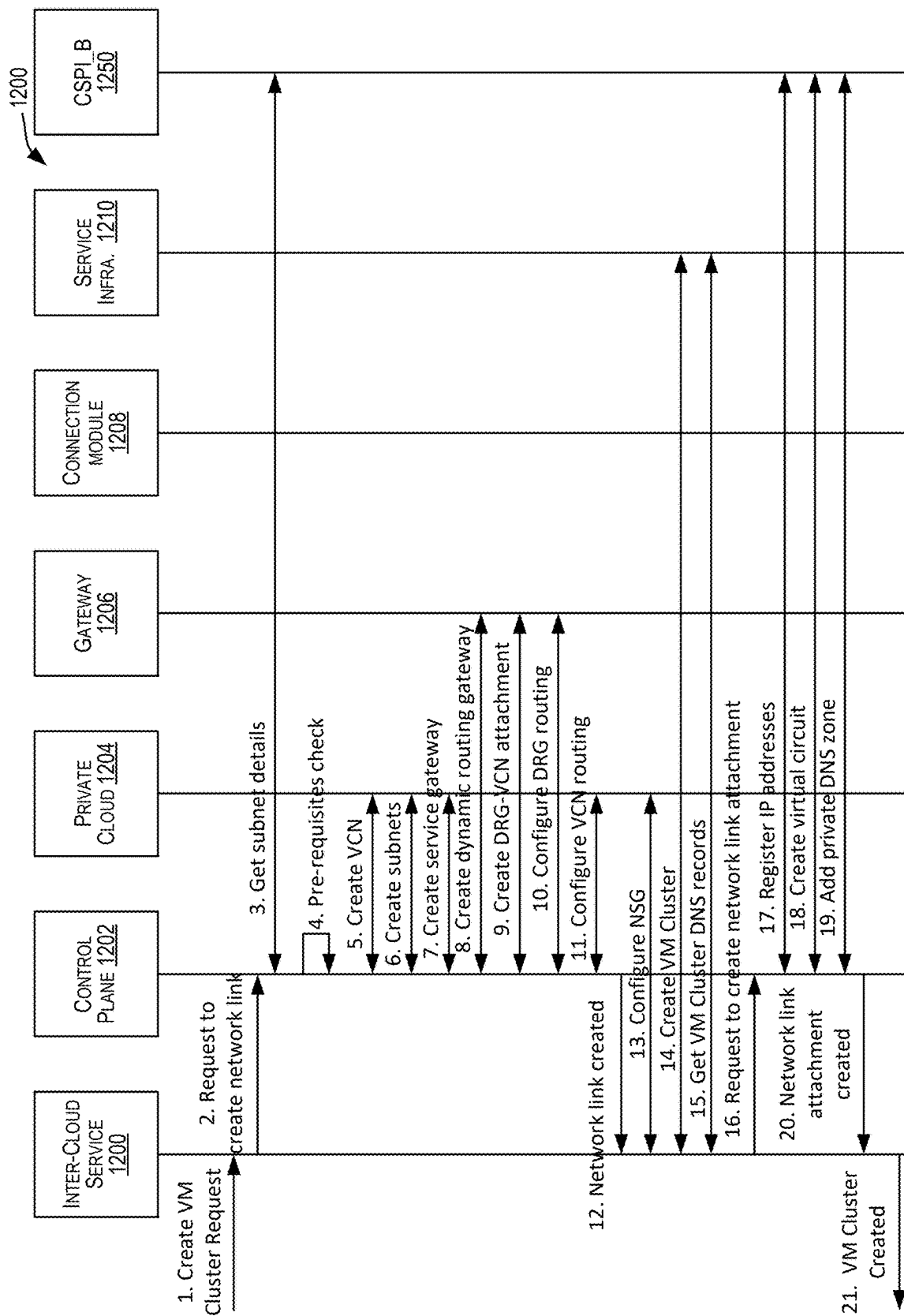


FIG. 12

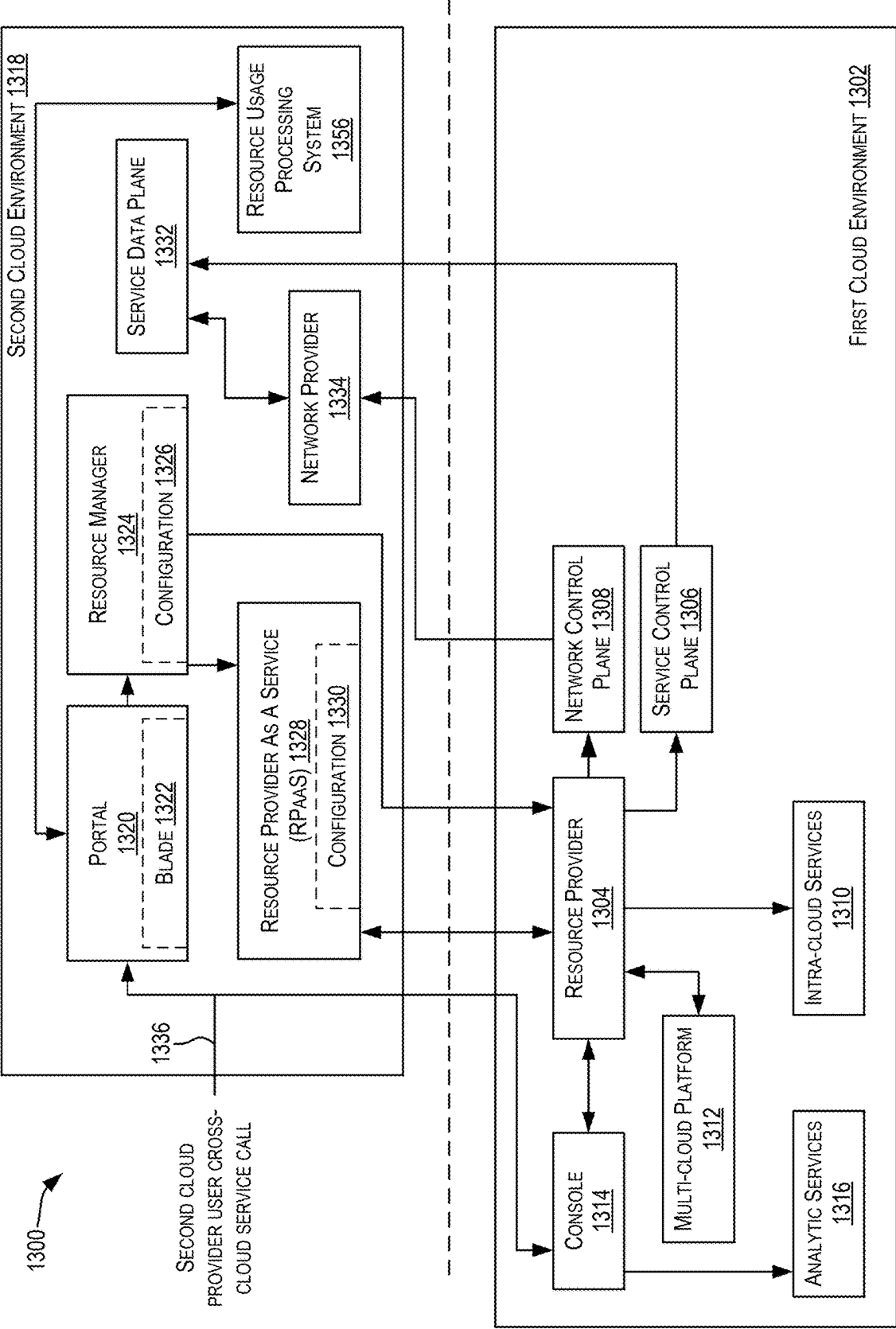


FIG. 13

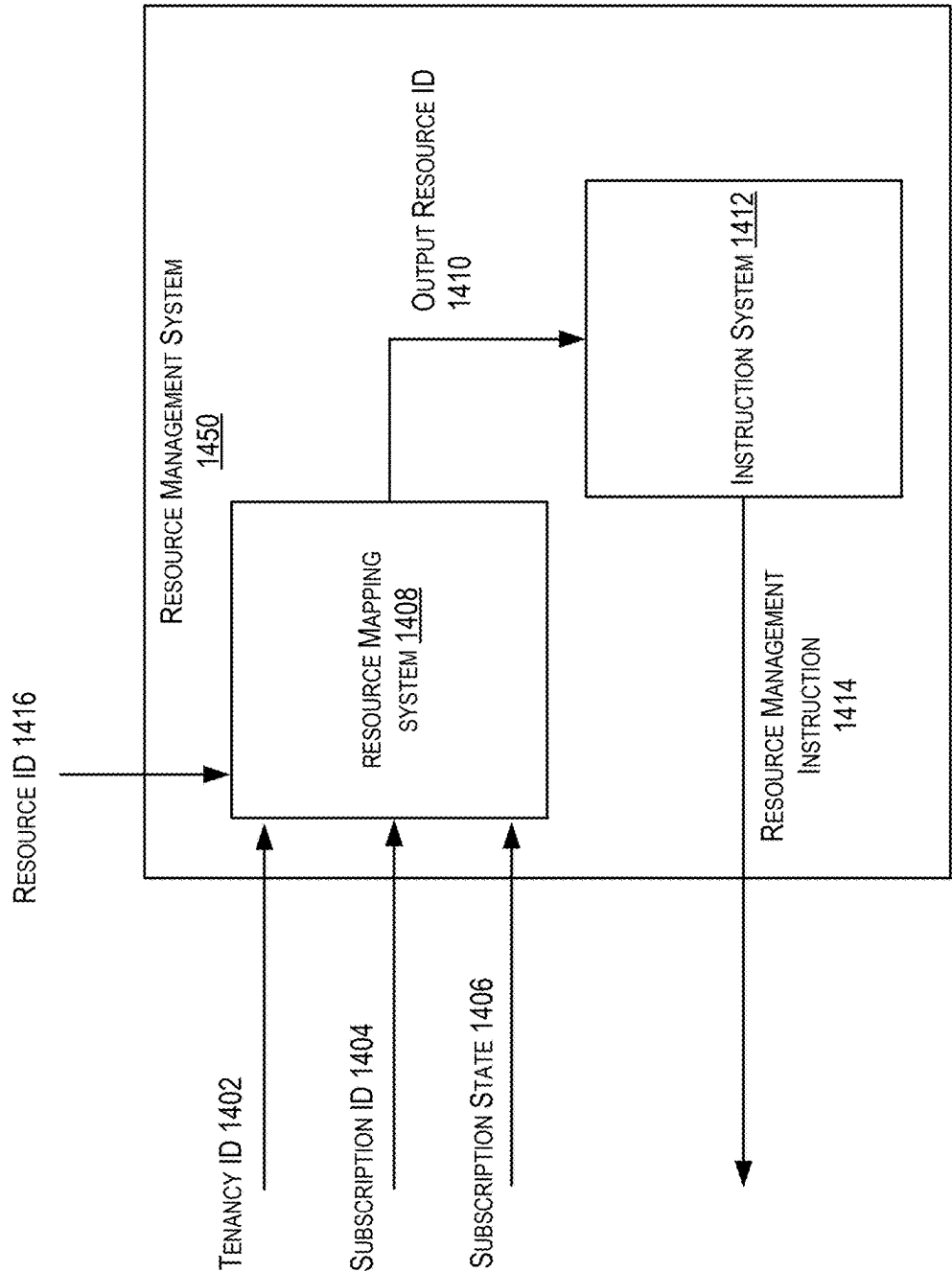


FIG. 14

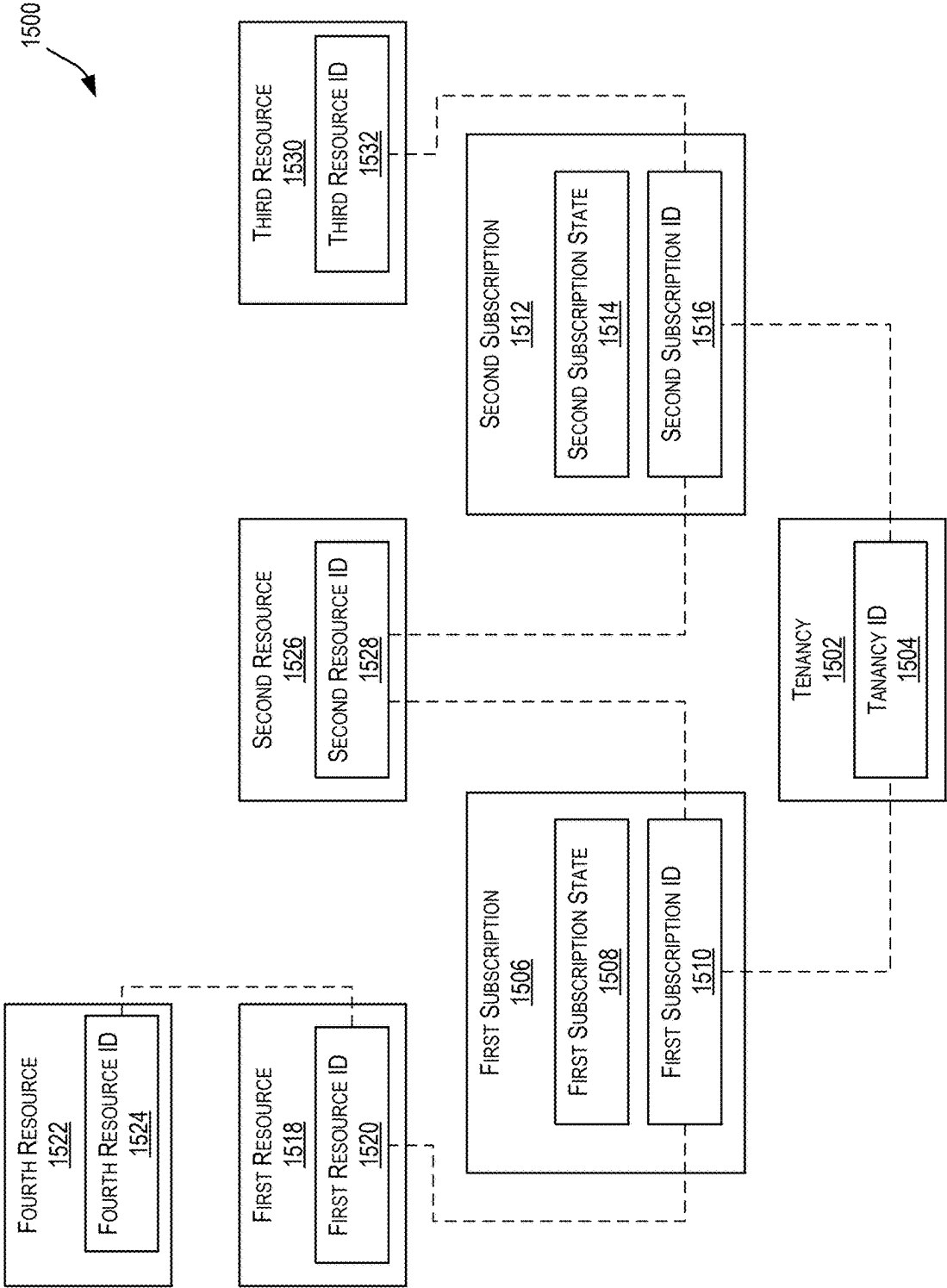


FIG. 15

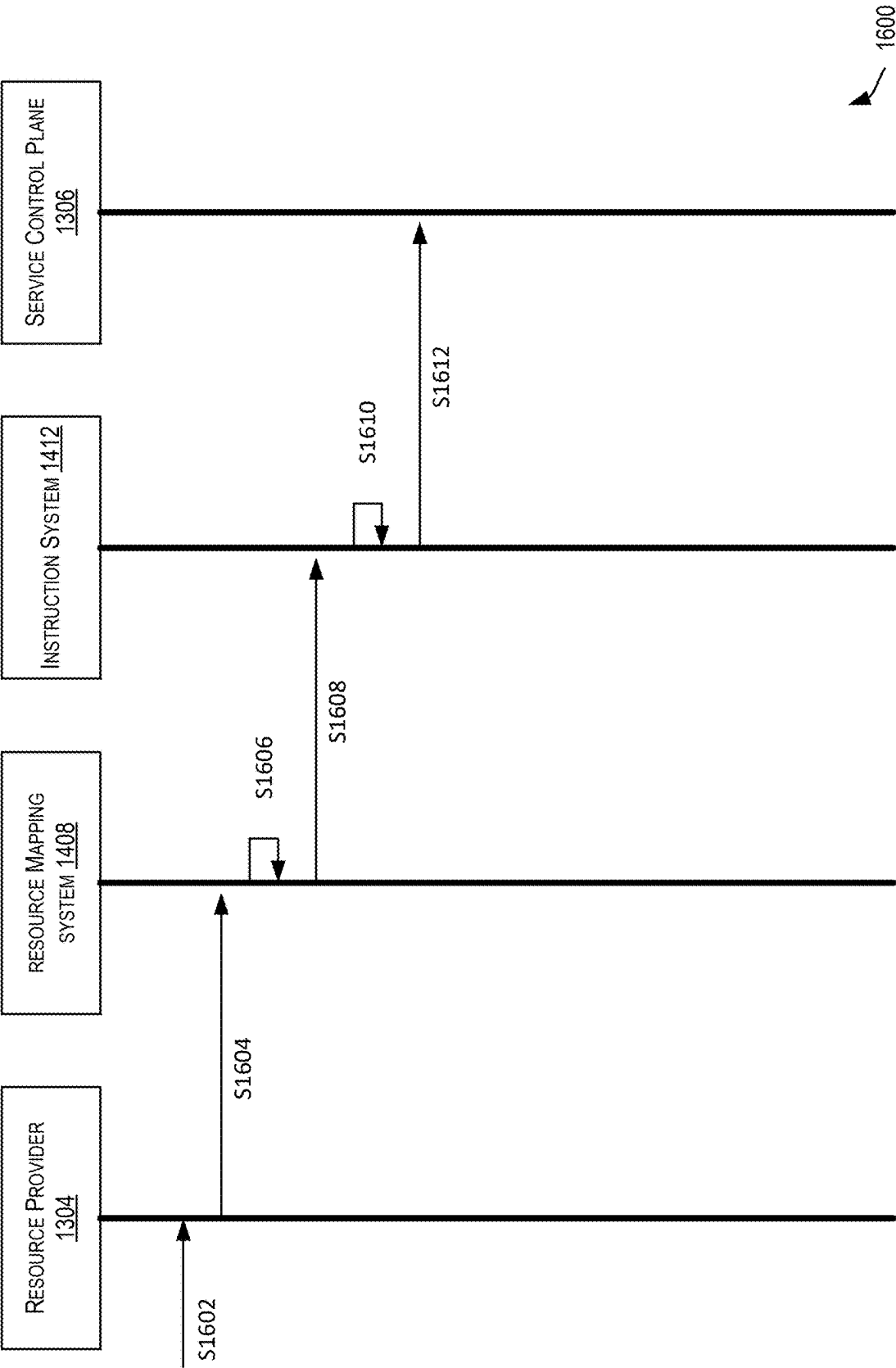


FIG. 16

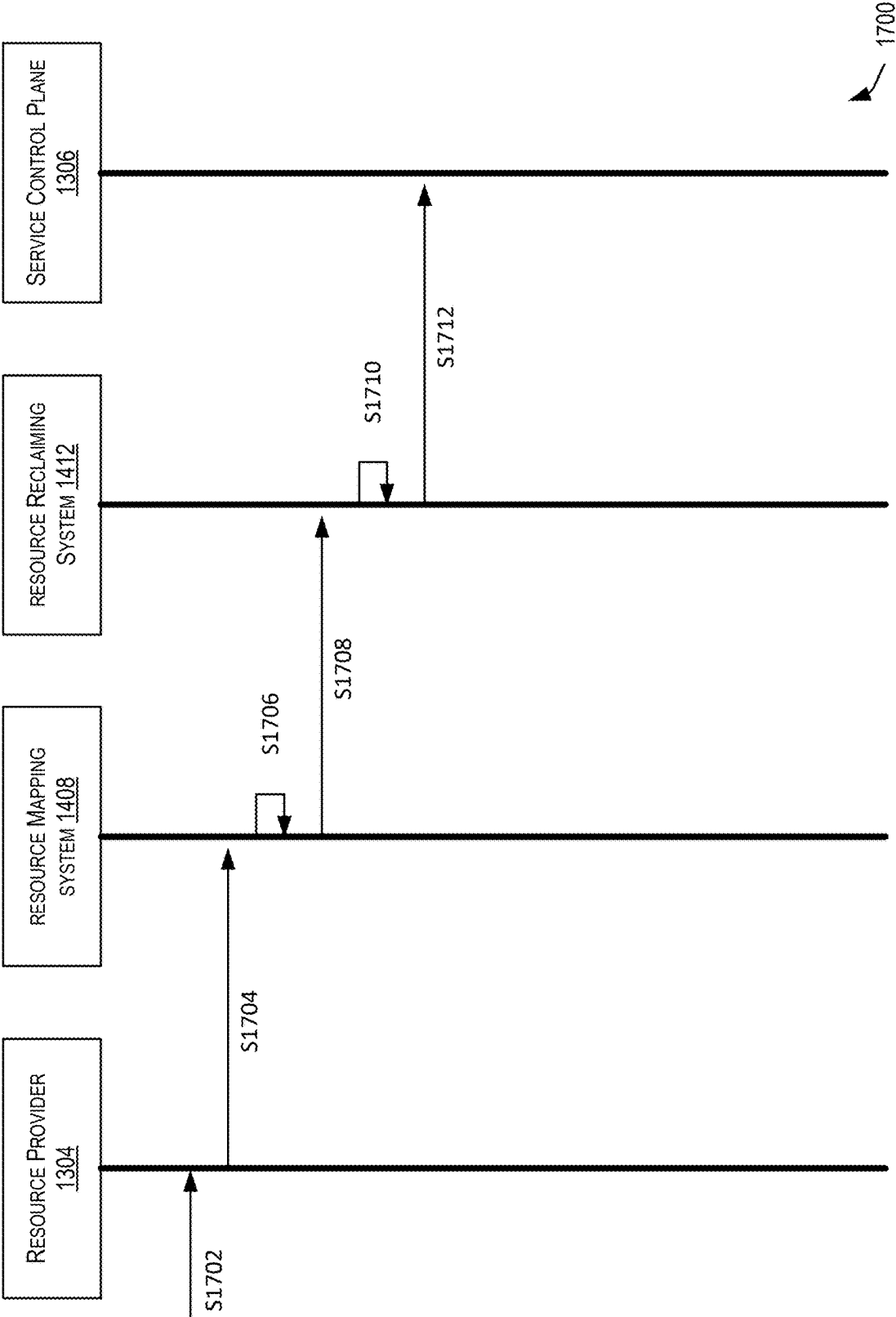


FIG. 17

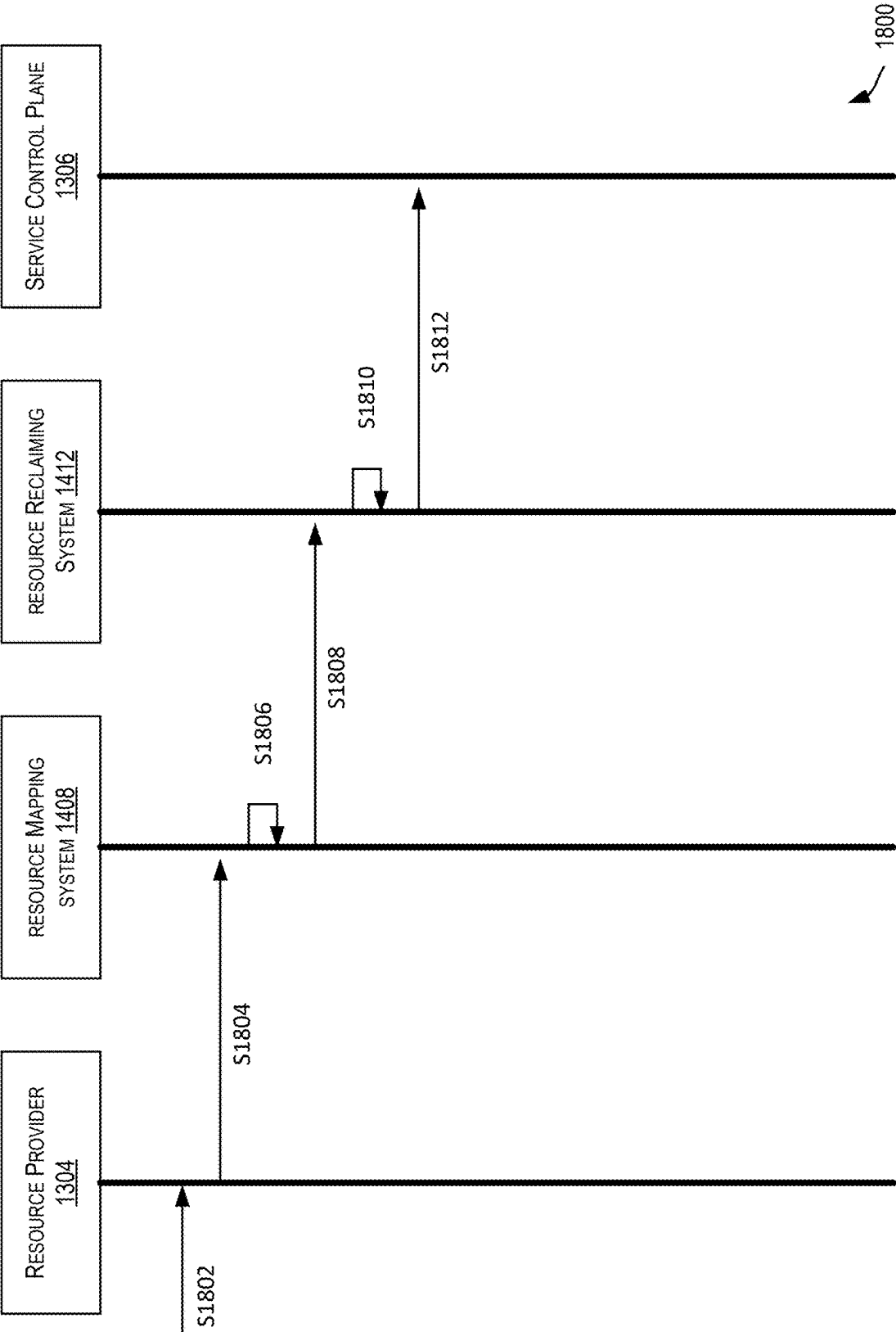


FIG. 18

1900

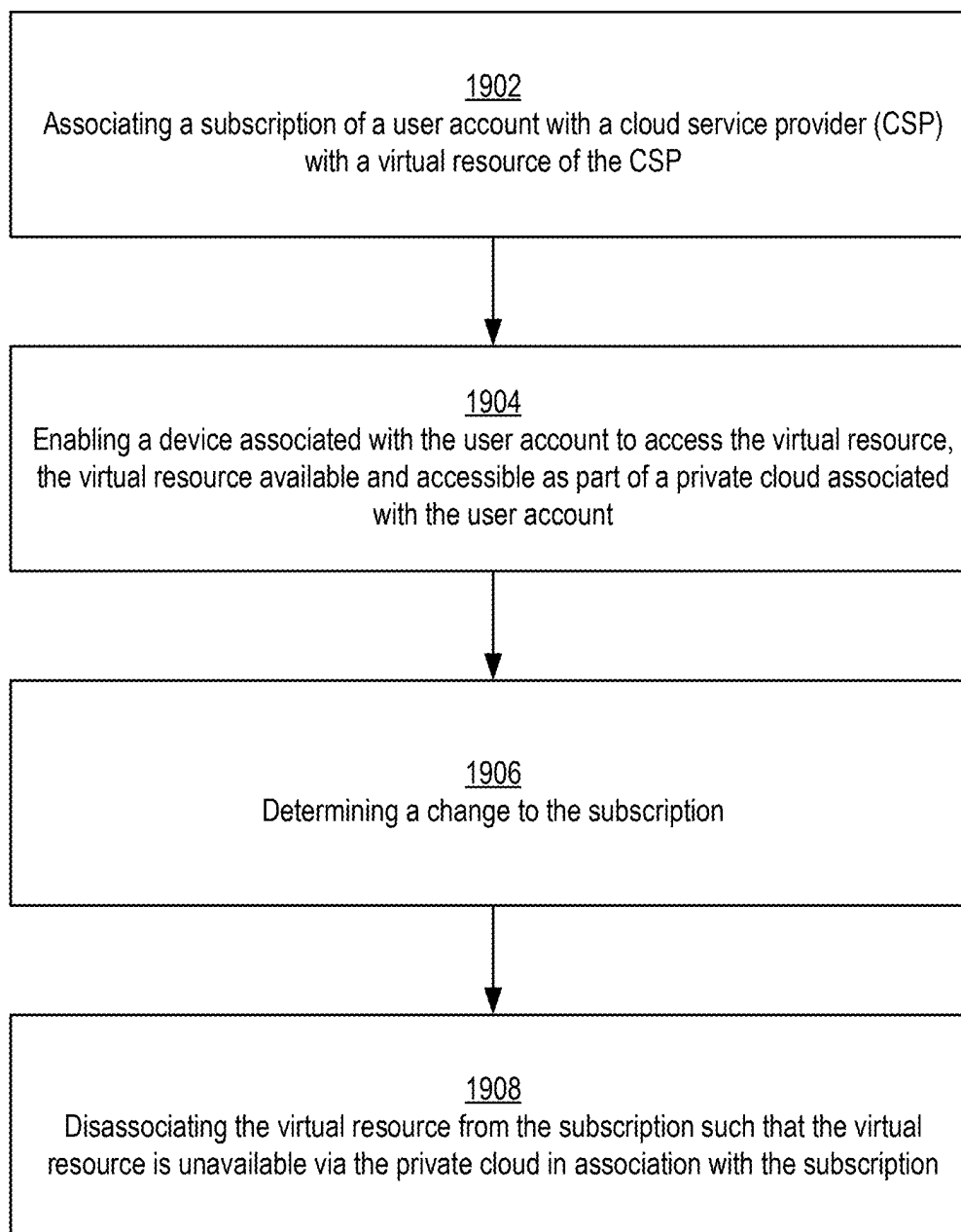


FIG. 19

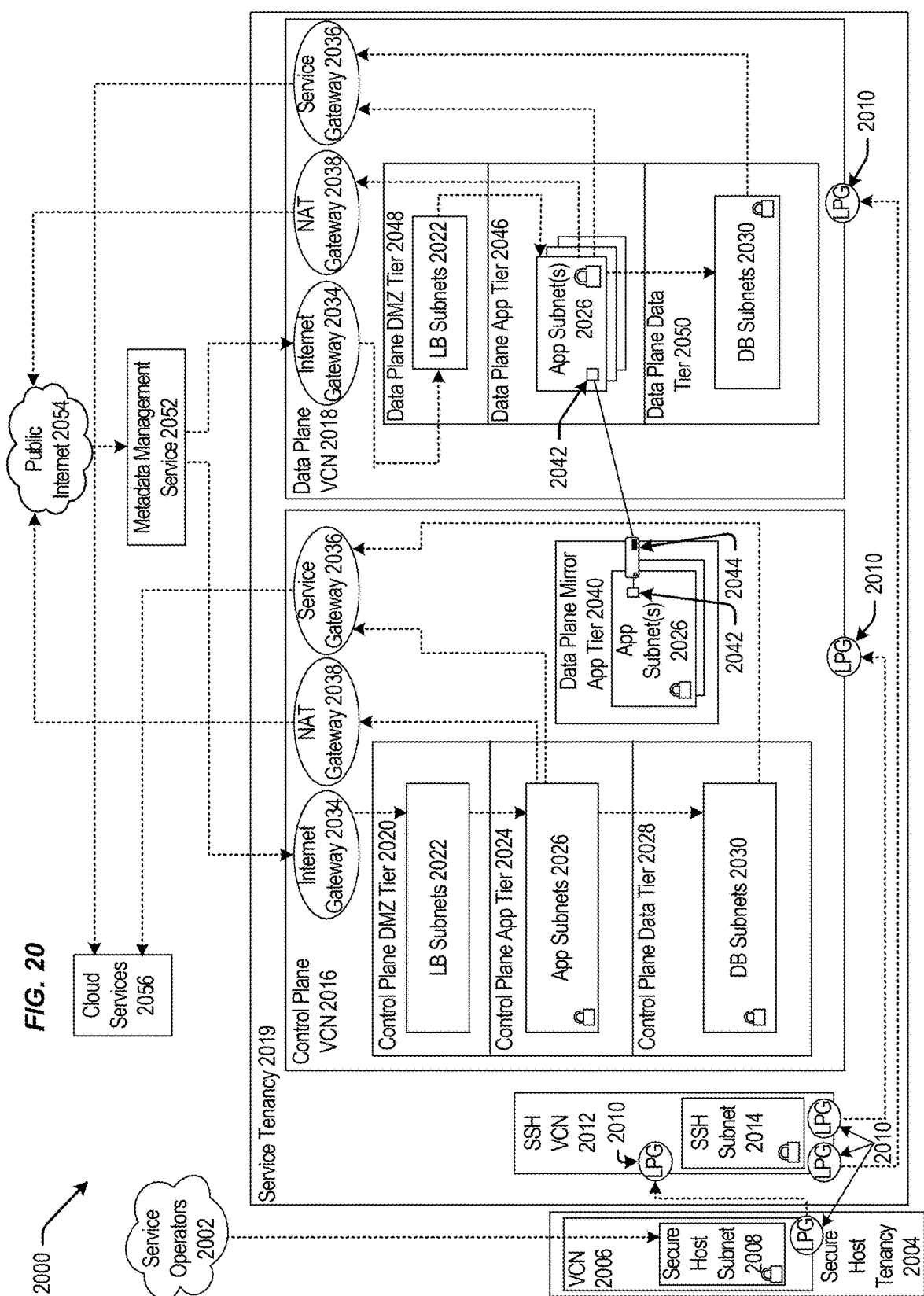
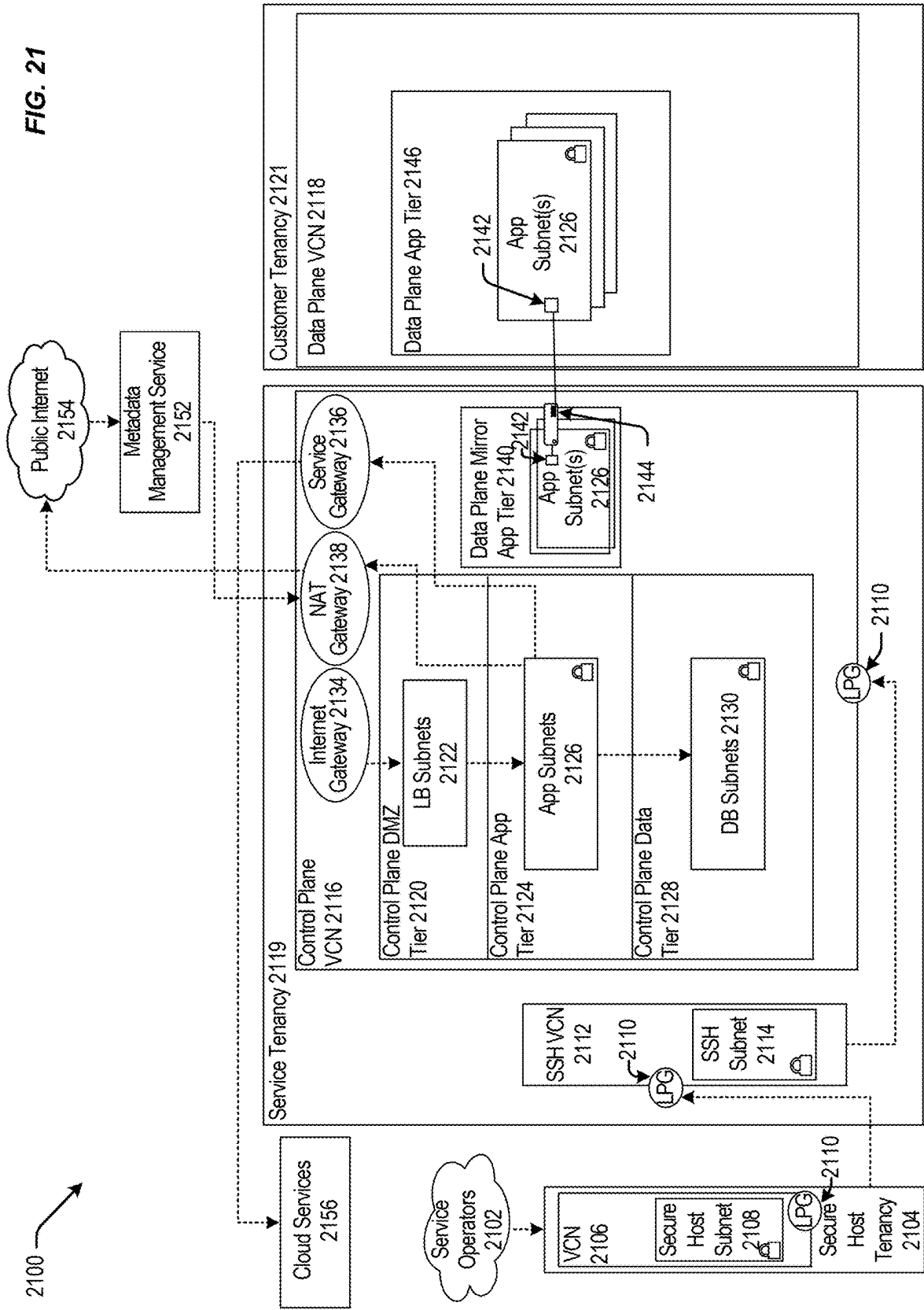
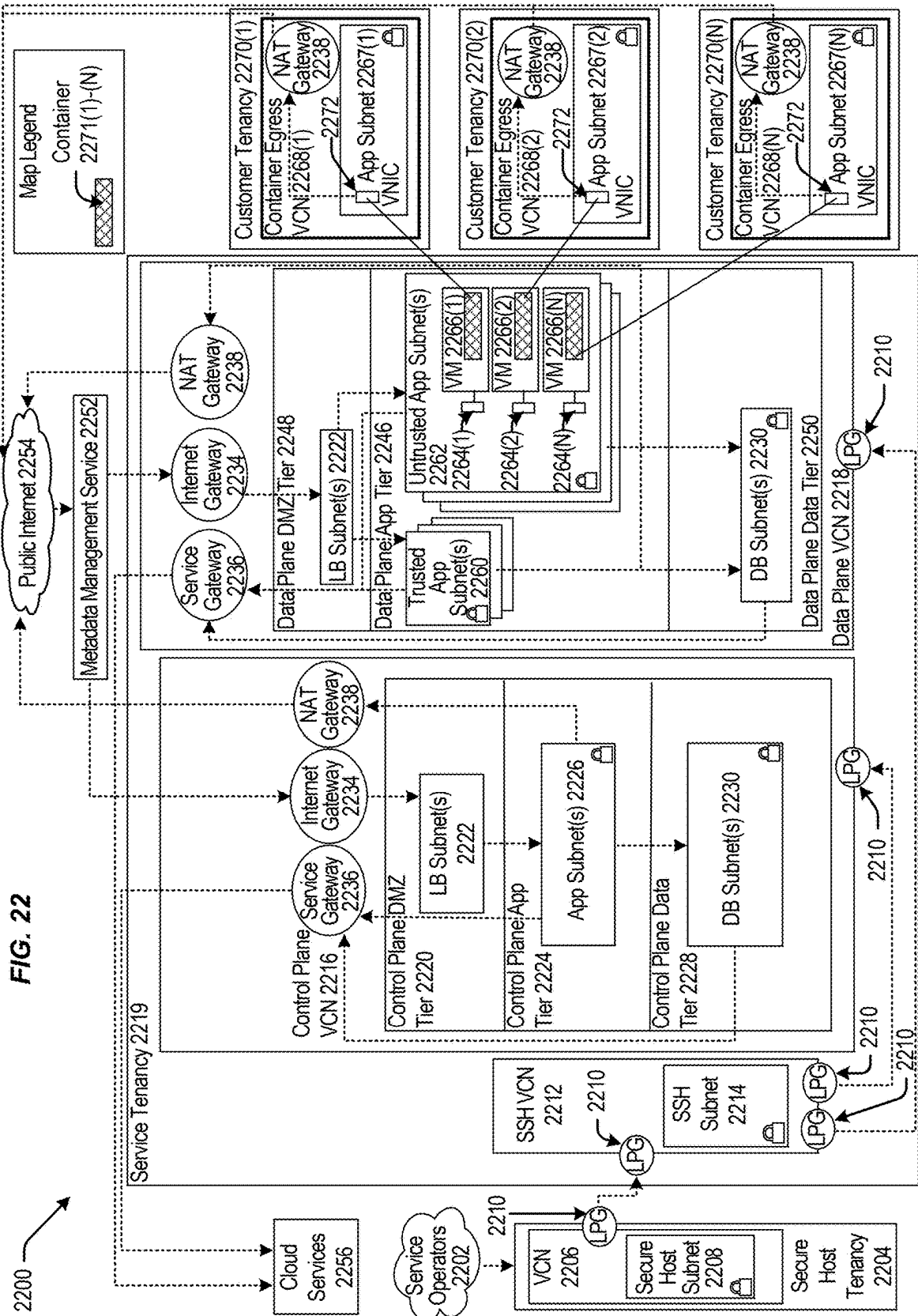
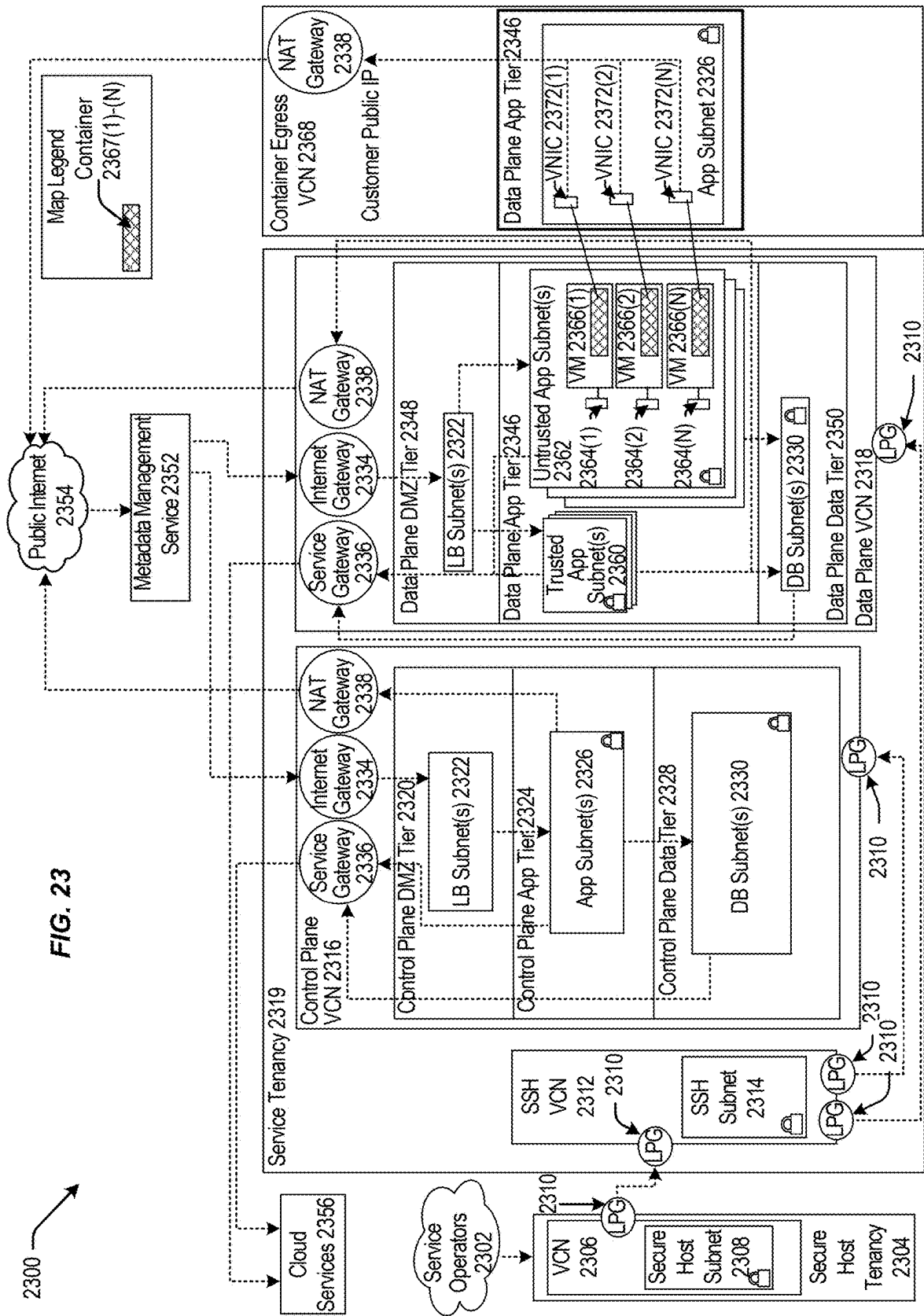


FIG. 21







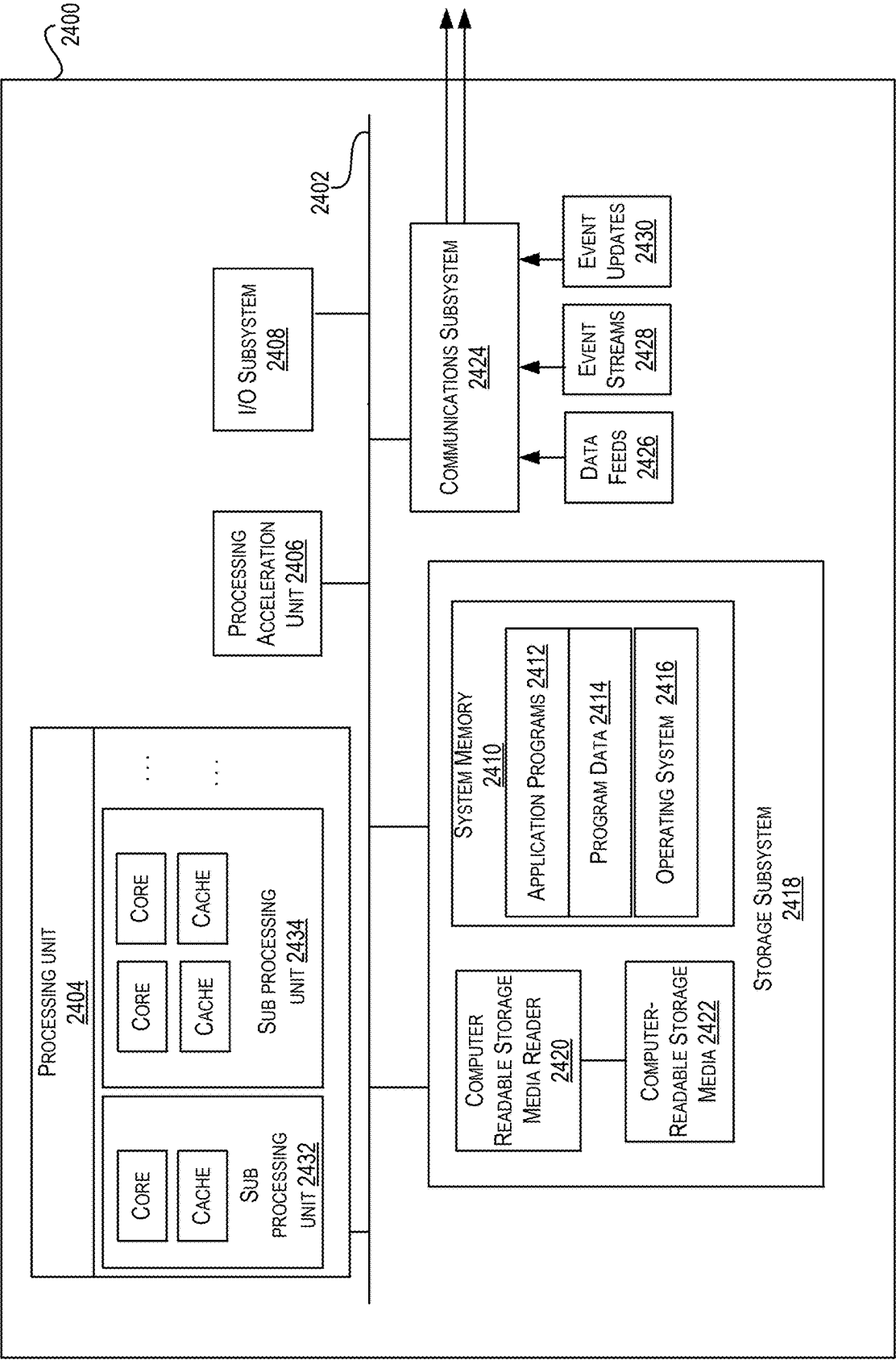


FIG. 24

RESOURCE RECLAIMING IN A TENANCY HAVING MULTIPLE SERVICE SUBSCRIPTIONS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a non-provisional application of and claims the benefit and priority under 35 U.S.C. 119 (e) of U.S. Provisional Application No. 63/561,610, filed Mar. 5, 2024 entitled “MANAGING METERING, LIMITS, AND RESOURCE RECLAIMING IN A TENANCY HAVING MULTIPLE SERVICE SUBSCRIPTIONS,” the entire contents of which is incorporated herein by reference for all purposes.

BACKGROUND

[0002] The last few years have seen a dramatic increase in the adoption of cloud services and this trend is only going to increase. Various different cloud environments are being provided by different cloud service providers (CSPs), each cloud environment providing a set of one or more cloud services. The set of cloud services offered by a cloud environment may include one or more different types of services including but not restricted to Software-as-a-Service (SaaS) services, Infrastructure-as-a-Service (IaaS) services, Platform-as-a-Service (PaaS) services, and others.

[0003] While various different cloud environments are currently available, each cloud environment provides a closed ecosystem for its subscribing customers. As a result, a customer of a cloud environment is restricted to using the services offered by that cloud environment. There is no easy way for a customer subscribing to a cloud environment provided by a CSP to, via that cloud environment, use a service offered provided by a different CSP. Embodiments discussed herein address these and other issues.

BRIEF SUMMARY

[0004] Disclosed herein are techniques for provisioning a cloud service of a first cloud service provider using a control plane of a second cloud service provider.

[0005] In some embodiments, a method comprises associating a subscription of a user account with a cloud service provider (CSP) with a virtual resource of the CSP; enabling a device associated with the user account to access the virtual resource, the virtual resource available and accessible as part of a private cloud associated with the user account; determining a change to the subscription; and disassociating, based on the change, the virtual resource from the subscription such that the virtual resource is unavailable via the private cloud in association with the subscription and such that the virtual resource is available and accessible for use in association with a different subscription.

[0006] In some embodiments, a system comprises one or more storage media storing instructions and one or more processors configured to execute the instructions to cause the system to: associate a subscription of a user account with a cloud service provider (CSP) with a virtual resource of the CSP; enable a device associated with the user account to access the virtual resource, the virtual resource available and accessible as part of a private cloud associated with the user account; determine a change to the subscription; and disassociate, based on the change, the virtual resource from the subscription such that the virtual resource is unavailable via

the private cloud in association with the subscription and such that the virtual resource is available and accessible for use in association with a different subscription.

[0007] In some embodiments, a one or more non-transitory computer-readable media storing instructions which, when executed by one or more processors, cause a system to perform operations comprising: associating a subscription of a user account with a cloud service provider (CSP) with a virtual resource of the CSP; enabling a device associated with the user account to access the virtual resource, the virtual resource available and accessible as part of a private cloud associated with the user account; determining a change to the subscription; and disassociating, based on the change, the virtual resource from the subscription such that the virtual resource is unavailable via the private cloud in association with the subscription and such that the virtual resource is available and accessible for use in association with a different subscription.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Features, embodiments, and advantages of the present disclosure are better understood when the following Detailed Description is read with reference to the accompanying drawings.

[0009] FIG. 1 is a high-level diagram of a distributed environment showing a virtual or overlay cloud network hosted by a cloud service provider infrastructure, according to certain embodiments.

[0010] FIG. 2 depicts a simplified architectural diagram of the physical components in the physical network within a cloud service provider infrastructure (CSPI), according to certain embodiments.

[0011] FIG. 3 shows an example arrangement within CSPI where a host machine is connected to multiple network virtualization devices (NVDs), according to certain embodiments.

[0012] FIG. 4 depicts connectivity between a host machine and an NVD for providing I/O virtualization for supporting multitenancy, according to certain embodiments.

[0013] FIG. 5 depicts a simplified block diagram of a physical network provided by a CSPI, according to certain embodiments.

[0014] FIG. 6 depicts a simplified high-level diagram of a distributed environment comprising multiple cloud environments provided by different cloud service providers (CSPs), according to certain embodiments.

[0015] FIG. 7 depicts an exemplary physical architecture for providing a cross-cloud service based on infrastructure distributed between multiple CSPs, according to some embodiments.

[0016] FIG. 8 depicts an exemplary virtual architecture for providing a cross-cloud service based on infrastructure distributed between multiple CSPs, according to some embodiments.

[0017] FIG. 9 depicts exemplary virtual resources provisioned by a first CSP for a customer of a second CSP, according to some embodiments.

[0018] FIG. 10 depicts an exemplary architecture for provisioning and managing a cross-cloud service based on an infrastructure distributed between multiple CSPs, according to some embodiments, according to some embodiments.

[0019] FIG. 11 depicts an exemplary user experience flow to provision resources, according to some embodiments.

[0020] FIG. 12 depicts an exemplary control plane provisioning flow, according to some embodiments.

[0021] FIG. 13 depicts an example of an architecture that includes a resource management system for cross-cloud services between multiple cloud environments, according to some embodiments.

[0022] FIG. 14 depicts an example of a resource management system, according to some embodiments.

[0023] FIG. 15 depicts an example of a relationship mapping between a tenancy, subscriptions, and resources, according to some embodiments.

[0024] FIG. 16 depicts an example flow for suspending a resource, according to some embodiments.

[0025] FIG. 17 depicts an example flow for unsuspending a resource, according to some embodiments.

[0026] FIG. 18 depicts an example flow for terminating a resource, according to some embodiments.

[0027] FIG. 19 depicts an example process for reclaiming a resource of a cloud environment, according to some embodiments.

[0028] FIG. 20 is a block diagram illustrating one pattern for implementing a cloud infrastructure as a service system, according to certain embodiments.

[0029] FIG. 21 is a block diagram illustrating another pattern for implementing a cloud infrastructure as a service system, according to certain embodiments.

[0030] FIG. 22 is a block diagram illustrating another pattern for implementing a cloud infrastructure as a service system, according to certain embodiments.

[0031] FIG. 23 is a block diagram illustrating another pattern for implementing a cloud infrastructure as a service system, according to certain embodiments.

[0032] FIG. 24 is a block diagram illustrating an example computer system, according to certain embodiments.

DETAILED DESCRIPTION

[0033] In the following description, for the purposes of explanation, specific details are set forth in order to provide a thorough understanding of certain embodiments. However, it will be apparent that various embodiments may be practiced without these specific details. The figures and description are not intended to be restrictive. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs.

[0034] The present disclosure relates generally to improved cloud architectures, and more particularly to techniques for providing services based on infrastructure distributed between multiple cloud service providers. In an example, a first cloud service provider (CSP) provides first services (e.g., a database service, such as EXADATA available from ORACLE), some of which can be available from a first cloud service provider infrastructure (CSPI) of the first CSP (e.g., Oracle Cloud Infrastructure-OCI). A second cloud service provider (CSP) provides second services (e.g., virtual instances, such as ones available from AZURE), some of which can be available from a second cloud service provider infrastructure (CSPI) of the second CSP (e.g., AZURE). At least a first service of the first services can be made available to customers of the second CSP via the second CSPI. In an example, hardware and/or software configured by the first CSPI to provide the first service are deployed at the second CSPI.

[0035] For example, at the physical level, the second CSPI can include a first set of computing resources of the first CSPI (e.g., a rack of servers optimized for a service of the first CSP). The first CSPI can include a second set of computing resources of the first CSPI (e.g., servers within a region). The second CSPI can also include a third set of computing resources of the second CSPI (e.g., servers within an availability zone). The first set of computing resources can be connected to the second set of computing resources and the third set of computing resources.

[0036] At the virtual level, the first set of resources and the second set of resource can host a first cloud for a customer with the first CSP. In comparison, the third set of resources can host a second cloud for the customer with the second CSP. The two clouds can be connected (e.g., via a peering connection that uses virtual routers). A service provided by the first CSP to the customer can have a first resource hosted in the first set of computing sources and a second resources hosted in the second set of computing resources. This service can be available to the customer via the second cloud. The use of the first resource enables the reduction of latency. The offering of the service of the first CSP via the second CSP can enable a better experience by extending the service to customers of the second CSP that may be more familiar with the configuring and managing of clouds with the second CSP.

[0037] To illustrate, consider the example of Exadata service (an Oracle OCI database service) made available via Microsoft AZURE. In this example, at the physical level, an OCI child site belonging to an OCI region is co-located in an AZURE datacenter. The OCI child site is connected to the AZURE datacenter via a FastConnect router (and a Microsoft MeetMe ToR router at the AZURE side). The OCI child site is also connected to the parent OCI region via fiber optics connection.

[0038] At the virtual level, the AZURE datacenter hosts a customer cloud (e.g., VNET), whereas the OCI's parent region and child site collectively host a customer VCN. Latency critical resources for the OCI service (e.g., a data plane resource for the Exadata service) are hosted in the child site, whereas other supporting resources (e.g., a control plane resource, a customer facing console resource, etc.) are hosted in the parent region.

[0039] On the OCI side, the VNET and the VCN are connected via a virtual router (e.g., a dynamic routing gateway (DRG) that supports FastConnect. AZURE also provides a virtual router that supports MeetMe). In the VNET, a delegate subnet can be configured. In the VCN, an Exadata service subnet can be configured. Both subnets use the same internet protocol (IP) address range. The VNET can store mapping information that maps an IP address that is within the IP address range and that is used in the VNET to an IP address of an Exadata service at the VCN. Traffic to the IP address (e.g., from a compute instance of the VNET) is sent from the VNET to the VCN given the one-to-one IP address mapping.

[0040] As such, from a customer perspective, the customer perceives an Exadata service within its VNET having an IP address. However, in effect this Exadata service is in the VCN at that IP address. Further, by hosting at least a part of the Exadata service in the child site (e.g., the data plane), the latency associated with the Exadata service can be reduced.

[0041] Referring back to the above architecture, the first set of computing resources can be referred to as belonging

to a child site, whereas the second set of computing resources can be referred to as belonging to a parent site. The third set of computing resources can be referred to as datacenter resources. The child site of the first CSP is deployed in a datacenter (or the second CSPI) of the second CSP. In comparison, the parent site of the first CSP is deployed in the first CSPI of the first CSP.

[0042] A child site may enable low latency services of the first CSP to be offered to customers via the second CSP. However, a deployment mechanism may be needed to control deployment of resources of the first CSP in the child site and in the parent site.

[0043] Embodiments of the present disclosure relate to such a deployment mechanism. In an example, the deployment mechanism includes a control plane of the first CSP, where the control plane is not hosted in a child site. Customer input at the second CSPI can be received by the control plane and can mapped to a customer ID with the first CSPI. Based on the customer ID, the control plane can provision resources at the first CSPI, such as by creating a cloud network for the customer and related connectivity resources (e.g., a dynamic routing gateway-DRG) for the cloud network. The cloud network can be also connected to a virtual machine (VM) cluster that would host the service. The VM cluster can be hosted in a child site, whereas other resources of the cloud network can be hosted in a parent site. Thereafter, the control plane can inter-connect the cloud network to the customer's cloud network with the second CSP, such that the two cloud networks of the customer are inter-connected. As part of this inter-connection, the control plane can provide internet protocol (IP) addresses of the VM cluster and, possibly, corresponding DNS records to the second CSPI such that these IP addresses, and possibly DNS records, are available to the customer at their cloud network with the second CSP.

[0044] To illustrate, consider again the example of OCI and AZURE. Azure customer input (e.g., selecting their VNET and a subnet) is received from an Azure resource manager (ARM). Oracle resource provide (ORP) can transform this input into an OCI ID for the customer and pass this information to the control plane. The control plane performs two main processes. The first process is to provision the relevant OCI resources. The second process is to connect these resources to the customer's VNET.

[0045] Under the first process, the control plane creates the customer's VCN in a customer tenancy (e.g., at a parent site) and creates a subnet within the VCN (and possibly a backup subnet). The Azure and OCI subnets (one in the customer's Azure VNET and one in the customer's OCI VCN) use the same CIDR. The control plane also creates a DRG in the customer tenancy and attaches the DRG to the VCN. It also sets up the routing information for the DRG and the VCN (such that these two resources are inter-connected). ORP can also provision a VM cluster (e.g., in a child site). IP address(es) of this cluster from the Classless Inter-Domain Routing (CIDR) and are mapped to corresponding DNS records.

[0046] Under the second process, the control plane registers these IP addresses with Azure (on the MeetMeRouter). It also creates a virtual circuit between the DRG and the MeetMeRouter and sends the DNS records such that a private DNS zone can be set up in Azure.

[0047] The first figure below is a sequence diagram illustrating these two processes, with the steps involved in each

(the first process corresponding to the "NetworkLink Provisioning" and the second process to "NetworkLinkAttachment Provisioning"). The figure after illustrates the different resources.

[0048] In the interest of clarity of explanation, embodiments of the present disclosure are described in connection with particular CSPs (e.g., ORACLE and MICROSOFT) and services (e.g., EXADATA Service). However, the embodiments are not limited as such and, instead, similarly and equivalently apply to any CSPs, CSPIs, and services in a multi-cloud environment.

Examples of Cloud Networks

[0049] The term cloud service is generally used to refer to a service that is made available by a cloud services provider (CSP) to users or customers on demand (e.g., via a subscription model) using systems and infrastructure (cloud infrastructure) provided by the CSP. Typically, the servers and systems that make up the CSP's infrastructure are separate from the customer's own on-premises servers and systems. Customers can thus avail themselves of cloud services provided by the CSP without having to purchase separate hardware and software resources for the services. Cloud services are designed to provide a subscribing customer easy, scalable access to applications and set of computing resources without the customer having to invest in procuring the infrastructure that is used for providing the services.

[0050] There are several cloud service providers that offer various types of cloud services. There are various different types or models of cloud services including Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS), Infrastructure-as-a-Service (IaaS), and others.

[0051] A customer can subscribe to one or more cloud services provided by a CSP. The customer can be any entity such as an individual, an organization, an enterprise, and the like. When a customer subscribes to or registers for a service provided by a CSP, a tenancy or an account is created for that customer. The customer can then, via this account, access the subscribed-to one or more cloud resources associated with the account.

[0052] As noted above, infrastructure as a service (IaaS) is one particular type of cloud computing service. In an IaaS model, the CSP provides infrastructure (referred to as cloud services provider infrastructure or CSPI) that can be used by customers to build their own customizable networks and deploy customer resources. The customer's resources and networks are thus hosted in a distributed environment by infrastructure provided by a CSP. This is different from traditional computing, where the customer's resources and networks are hosted by infrastructure provided by the customer.

[0053] The CSPI may comprise interconnected high-performance compute resources including various host machines, memory resources, and network resources that form a physical network, which is also referred to as a substrate network or an underlay network. The resources in CSPI may be spread across one or more data centers that may be geographically spread across one or more geographical regions. Virtualization software may be executed by these physical resources to provide a virtualized distributed environment. The virtualization creates an overlay network (also known as a software-based network, a software-defined network, or a virtual network) over the physical

network. The CSPI physical network provides the underlying basis for creating one or more overlay or virtual networks on top of the physical network. The physical network (or substrate network or underlay network) comprises physical network devices such as physical switches, routers, computers and host machines, and the like. An overlay network is a logical (or virtual) network that runs on top of a physical substrate network. A given physical network can support one or multiple overlay networks. Overlay networks typically use encapsulation techniques to differentiate between traffic belonging to different overlay networks. A virtual or overlay network is also referred to as a virtual cloud network (VCN). The virtual networks are implemented using software virtualization technologies (e.g., hypervisors, virtualization functions implemented by network virtualization devices (NVDs) (e.g., smartNICs), top-of-rack (TOR) switches, smart TORs that implement one or more functions performed by an NVD, and other mechanisms) to create layers of network abstraction that can be run on top of the physical network. Virtual networks can take on many forms, including peer-to-peer networks, IP networks, and others. Virtual networks are typically either Layer-3 IP networks or Layer-2 VLANs. This method of virtual or overlay networking is often referred to as virtual or overlay Layer-3 networking. Examples of protocols developed for virtual networks include IP-in-IP (or Generic Routing Encapsulation (GRE)) Virtual Extensible LAN (VXLAN-IETF RFC 7348), Virtual Private Networks (VPNs) (e.g., MPLS Layer-3 Virtual Private Networks (RFC 4364)), VMware's NSX, GENEVE (Generic Network Virtualization Encapsulation), and others.

[0054] For IaaS, the infrastructure (CSPI) provided by a CSP can be configured to provide virtualized set of computing resources over a public network (e.g., the Internet). In an IaaS model, a cloud computing services provider can host the infrastructure components (e.g., servers, storage devices, network nodes (e.g., hardware), deployment software, platform virtualization (e.g., a hypervisor layer), or the like). In some cases, an IaaS provider may also supply a variety of services to accompany those infrastructure components (e.g., billing, monitoring, logging, security, load balancing and clustering, etc.). Thus, as these services may be policy-driven, IaaS users may be able to implement policies to drive load balancing to maintain application availability and performance. CSPI provides infrastructure and a set of complementary cloud services that enable customers to build and run a wide range of applications and services in a highly available hosted distributed environment. CSPI offers high-performance compute resources and capabilities and storage capacity in a flexible virtual network that is securely accessible from various networked locations such as from a customer's on-premises network. When a customer subscribes to or registers for an IaaS service provided by a CSP, the tenancy created for that customer is a secure and isolated partition within the CSPI where the customer can create, organize, and administer their cloud resources.

[0055] Customers can build their own virtual networks using compute, memory, and networking resources provided by CSPI. One or more customer resources or workloads, such as compute instances, can be deployed on these virtual networks. For example, a customer can use resources provided by CSPI to build one or multiple customizable and private virtual network(s) referred to as virtual cloud networks (VCNs). A customer can deploy one or more cus-

tomers resources, such as compute instances, on a customer VCN. Compute instances can take the form of virtual machines, bare metal instances, and the like. The CSPI thus provides infrastructure and a set of complementary cloud services that enable customers to build and run a wide range of applications and services in a highly available virtual hosted environment. The customer does not manage or control the underlying physical resources provided by CSPI but has control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., firewalls).

[0056] The CSP may provide a console that enables customers and network administrators to configure, access, and manage resources deployed in the cloud using CSPI resources. In certain embodiments, the console provides a web-based user interface that can be used to access and manage CSPI. In some implementations, the console is a web-based application provided by the CSP.

[0057] CSPI may support single-tenancy or multi-tenancy architectures. In a single tenancy architecture, a software (e.g., an application, a database) or a hardware component (e.g., a host machine or a server) serves a single customer or tenant. In a multi-tenancy architecture, a software or a hardware component serves multiple customers or tenants. Thus, in a multi-tenancy architecture, CSPI resources are shared between multiple customers or tenants. In a multi-tenancy situation, precautions are taken, and safeguards put in place within CSPI to ensure that each tenant's data is isolated and remains invisible to other tenants.

[0058] In a physical network, a network endpoint ("endpoint") refers to a computing device or system that is connected to a physical network and communicates back and forth with the network to which it is connected. A network endpoint in the physical network may be connected to a Local Area Network (LAN), a Wide Area Network (WAN), or other type of physical network. Examples of traditional endpoints in a physical network include modems, hubs, bridges, switches, routers, and other networking devices, physical computers (or host machines), and the like. Each physical device in the physical network has a fixed network address that can be used to communicate with the device. This fixed network address can be a Layer-2 address (e.g., a MAC address), a fixed Layer-3 address (e.g., an IP address), and the like. In a virtualized environment or in a virtual network, the endpoints can include various virtual endpoints such as virtual machines that are hosted by components of the physical network (e.g., hosted by physical host machines). These endpoints in the virtual network are addressed by overlay addresses such as overlay Layer-2 addresses (e.g., overlay MAC addresses) and overlay Layer-3 addresses (e.g., overlay IP addresses). Network overlays enable flexibility by allowing network managers to move around the overlay addresses associated with network endpoints using software management (e.g., via software implementing a control plane for the virtual network). Accordingly, unlike in a physical network, in a virtual network, an overlay address (e.g., an overlay IP address) can be moved from one endpoint to another using network management software. Since the virtual network is built on top of a physical network, communications between components in the virtual network involves both the virtual network and the underlying physical network. In order to facilitate such communications, the components of CSPI are configured to learn and store mappings that map overlay

addresses in the virtual network to actual physical addresses in the substrate network, and vice versa. These mappings are then used to facilitate the communications. Customer traffic is encapsulated to facilitate routing in the virtual network.

[0059] Accordingly, physical addresses (e.g., physical IP addresses) are associated with components in physical networks and overlay addresses (e.g., overlay IP addresses) are associated with entities in virtual or overlay networks. A physical IP address is an IP address associated with a physical device (e.g., a network device) in the substrate or physical network. For example, each NVD has an associated physical IP address. An overlay IP address is an overlay address associated with an entity in an overlay network, such as with a compute instance in a customer's virtual cloud network (VCN). Two different customers or tenants, each with their own private VCNs can potentially use the same overlay IP address in their VCNs without any knowledge of each other. Both the physical IP addresses and overlay IP addresses are types of real IP addresses. These are separate from virtual IP addresses. A virtual IP address is typically a single IP address that represents or maps to multiple real IP addresses. A virtual IP address provides a 1-to-many mapping between the virtual IP address and multiple real IP addresses. For example, a load balancer may use a VIP to map to or represent multiple servers, each server having its own real IP address.

[0060] The cloud infrastructure or CSPI is physically hosted in one or more data centers in one or more regions around the world. The CSPI may include components in the physical or substrate network and virtualized components (e.g., virtual networks, compute instances, virtual machines, etc.) that are in a virtual network built on top of the physical network components. In certain embodiments, the CSPI is organized and hosted in realms, regions, and availability domains. A region is typically a localized geographic area that contains one or more data centers. Regions are generally independent of each other and can be separated by vast distances, for example, across countries or even continents. For example, a first region may be in Australia, another one in Japan, yet another one in India, and the like. CSPI resources are divided among regions such that each region has its own independent subset of CSPI resources. Each region may provide a set of core infrastructure services and resources, such as, compute resources (e.g., bare metal servers, virtual machine, containers and related infrastructure, etc.); storage resources (e.g., block volume storage, file storage, object storage, archive storage); networking resources (e.g., virtual cloud networks (VCNs), load balancing resources, connections to on-premise networks), database resources; edge networking resources (e.g., DNS); and access management and monitoring resources, and others. Each region generally has multiple paths connecting it to other regions in the realm.

[0061] Generally, an application is deployed in a region (i.e., deployed on infrastructure associated with that region) where it is most heavily used, because using nearby resources is faster than using distant resources. Applications can also be deployed in different regions for various reasons, such as redundancy to mitigate the risk of region-wide events such as large weather systems or earthquakes, to meet varying requirements for legal jurisdictions, tax domains, and other business or social criteria, and the like.

[0062] The data centers within a region can be further organized and subdivided into availability domains (ADs).

An availability domain may correspond to one or more data centers located within a region. A region can be composed of one or more availability domains. In such a distributed environment, CSPI resources are either region-specific, such as a virtual cloud network (VCN), or availability domain-specific, such as a compute instance.

[0063] ADs within a region are isolated from each other, fault tolerant, and are configured such that they are very unlikely to fail simultaneously. This is achieved by the ADs not sharing critical infrastructure resources such as networking, physical cables, cable paths, cable entry points, etc., such that a failure at one AD within a region is unlikely to impact the availability of the other ADs within the same region. The ADs within the same region may be connected to each other by a low latency, high bandwidth network, which makes it possible to provide high-availability connectivity to other networks (e.g., the Internet, customers' on-premises networks, etc.) and to build replicated systems in multiple ADs for both high-availability and disaster recovery. Cloud services use multiple ADs to ensure high availability and to protect against resource failure. As the infrastructure provided by the IaaS provider grows, more regions and ADs may be added with additional capacity. Traffic between availability domains is usually encrypted.

[0064] In certain embodiments, regions are grouped into realms. A realm is a logical collection of regions. Realms are isolated from each other and do not share any data. Regions in the same realm may communicate with each other, but regions in different realms cannot. A customer's tenancy or account with the CSP exists in a single realm and can be spread across one or more regions that belong to that realm. Typically, when a customer subscribes to an IaaS service, a tenancy or account is created for that customer in the customer-specified region (referred to as the "home" region) within a realm. A customer can extend the customer's tenancy across one or more other regions within the realm. A customer cannot access regions that are not in the realm where the customer's tenancy exists.

[0065] An IaaS provider can provide multiple realms, each realm catered to a particular set of customers or users. For example, a commercial realm may be provided for commercial customers. As another example, a realm may be provided for a specific country for customers within that country. As yet another example, a government realm may be provided for a government, and the like. For example, the government realm may be catered for a specific government and may have a heightened level of security than a commercial realm. For example, Oracle Cloud Infrastructure (OCI) currently offers a realm for commercial regions and two realms (e.g., FedRAMP authorized and IL5 authorized) for government cloud regions.

[0066] In certain embodiments, an AD can be subdivided into one or more fault domains. A fault domain is a grouping of infrastructure resources within an AD to provide anti-affinity. Fault domains allow for the distribution of compute instances such that the instances are not on the same physical hardware within a single AD. This is known as anti-affinity. A fault domain refers to a set of hardware components (computers, switches, and more) that share a single point of failure. A compute pool is logically divided up into fault domains. Due to this, a hardware failure or compute hardware maintenance event that affects one fault domain does not affect instances in other fault domains. Depending on the embodiment, the number of fault domains for each AD may

vary. For instance, in certain embodiments each AD contains three fault domains. A fault domain acts as a logical data center within an AD.

[0067] When a customer subscribes to an IaaS service, resources from CSPI are provisioned for the customer and associated with the customer's tenancy. The customer can use these provisioned resources to build private networks and deploy resources on these networks. The customer networks that are hosted in the cloud by the CSPI are referred to as virtual cloud networks (VCNs). A customer can set up one or more virtual cloud networks (VCNs) using CSPI resources allocated for the customer. A VCN is a virtual or software defined private network. The customer resources that are deployed in the customer's VCN can include compute instances (e.g., virtual machines, bare-metal instances) and other resources. These compute instances may represent various customer workloads such as applications, load balancers, databases, and the like. A compute instance deployed on a VCN can communicate with publicly accessible endpoints ("public endpoints") over a public network such as the Internet, with other instances in the same VCN or other VCNs (e.g., the customer's other VCNs, or VCNs not belonging to the customer), with the customer's on-premise data centers or networks, and with service endpoints, and other types of endpoints.

[0068] The CSP may provide various services using the CSPI. In some instances, customers of CSPI may themselves act like service providers and provide services using CSPI resources. A service provider may expose a service endpoint, which is characterized by identification information (e.g., an IP Address, a DNS name and port). A customer's resource (e.g., a compute instance) can consume a particular service by accessing a service endpoint exposed by the service for that particular service. These service endpoints are generally endpoints that are publicly accessible by users using public IP addresses associated with the endpoints via a public communication network such as the Internet. Network endpoints that are publicly accessible are also sometimes referred to as public endpoints.

[0069] In certain embodiments, a service provider may expose a service via an endpoint (sometimes referred to as a service endpoint) for the service. Customers of the service can then use this service endpoint to access the service. In certain implementations, a service endpoint provided for a service can be accessed by multiple customers that intend to consume that service. In other implementations, a dedicated service endpoint may be provided for a customer such that only that customer can access the service using that dedicated service endpoint.

[0070] In certain embodiments, when a VCN is created, it is associated with a private overlay Classless Inter-Domain Routing (CIDR) address space, which is a range of private overlay IP addresses that are assigned to the VCN (e.g., 10.0/16). A VCN includes associated subnets, route tables, and gateways. A VCN resides within a single region but can span one or more or all of the region's availability domains. A gateway is a virtual interface that is configured for a VCN and enables communication of traffic to and from the VCN to one or more endpoints outside the VCN. One or more different types of gateways may be configured for a VCN to enable communication to and from different types of endpoints.

[0071] A VCN can be subdivided into one or more sub-networks such as one or more subnets. A subnet is thus a unit

of configuration or a subdivision that can be created within a VCN. A VCN can have one or multiple subnets. Each subnet within a VCN is associated with a contiguous range of overlay IP addresses (e.g., 10.0.0.0/24 and 10.0.1.0/24) that do not overlap with other subnets in that VCN, and which represent an address space subset within the address space of the VCN.

[0072] Each compute instance is associated with a virtual network interface card (VNIC), that enables the compute instance to participate in a subnet of a VCN. A VNIC is a logical representation of physical Network Interface Card (NIC). In general, a VNIC is an interface between an entity (e.g., a compute instance, a service) and a virtual network. A VNIC exists in a subnet, has one or more associated IP addresses, and associated security rules or policies. A VNIC is equivalent to a Layer-2 port on a switch. A VNIC is attached to a compute instance and to a subnet within a VCN. A VNIC associated with a compute instance enables the compute instance to be a part of a subnet of a VCN and enables the compute instance to communicate (e.g., send and receive packets) with endpoints that are on the same subnet as the compute instance, with endpoints in different subnets in the VCN, or with endpoints outside the VCN. The VNIC associated with a compute instance thus determines how the compute instance connects with endpoints inside and outside the VCN. A VNIC for a compute instance is created and associated with that compute instance when the compute instance is created and added to a subnet within a VCN. For a subnet comprising a set of compute instances, the subnet contains the VNICs corresponding to the set of compute instances, each VNIC attached to a compute instance within the set of computer instances.

[0073] Each compute instance is assigned a private overlay IP address via the VNIC associated with the compute instance. This private overlay IP address is assigned to the VNIC that is associated with the compute instance when the compute instance is created and used for routing traffic to and from the compute instance. All VNICs in a given subnet use the same route table, security lists, and DHCP options. As described above, each subnet within a VCN is associated with a contiguous range of overlay IP addresses (e.g., 10.0.0.0/24 and 10.0.1.0/24) that do not overlap with other subnets in that VCN, and which represent an address space subset within the address space of the VCN. For a VNIC on a particular subnet of a VCN, the private overlay IP address that is assigned to the VNIC is an address from the contiguous range of overlay IP addresses allocated for the subnet.

[0074] In certain embodiments, a compute instance may optionally be assigned additional overlay IP addresses in addition to the private overlay IP address, such as, for example, one or more public IP addresses if in a public subnet. These multiple addresses are assigned either on the same VNIC or over multiple VNICs that are associated with the compute instance. Each instance however has a primary VNIC that is created during instance launch and is associated with the overlay private IP address assigned to the instance—this primary VNIC cannot be removed. Additional VNICs, referred to as secondary VNICs, can be added to an existing instance in the same availability domain as the primary VNIC. All the VNICs are in the same availability domain as the instance. A secondary VNIC can be in a

subnet in the same VCN as the primary VNIC, or in a different subnet that is either in the same VCN or a different one.

[0075] A compute instance may optionally be assigned a public IP address if it is in a public subnet. A subnet can be designated as either a public subnet or a private subnet at the time the subnet is created. A private subnet means that the resources (e.g., compute instances) and associated VNICs in the subnet cannot have public overlay IP addresses. A public subnet means that the resources and associated VNICs in the subnet can have public IP addresses. A customer can designate a subnet to exist either in a single availability domain or across multiple availability domains in a region or realm.

[0076] As described above, a VCN may be subdivided into one or more subnets. In certain embodiments, a Virtual Router (VR) configured for the VCN (referred to as the VCN VR or just VR) enables communications between the subnets of the VCN. For a subnet within a VCN, the VR represents a logical gateway for that subnet that enables the subnet (i.e., the compute instances on that subnet) to communicate with endpoints on other subnets within the VCN, and with other endpoints outside the VCN. The VCN VR is a logical entity that is configured to route traffic between VNICs in the VCN and virtual gateways (“gateways”) associated with the VCN. A VCN VR is a Layer-3/IP Layer concept. In one embodiment, there is one VCN VR for a VCN where the VCN VR has potentially an unlimited number of ports addressed by IP addresses, with one port for each subnet of the VCN. In this manner, the VCN VR has a different IP address for each subnet in the VCN that the VCN VR is attached to. The VR is also connected to the various gateways configured for a VCN. In certain embodiments, a particular overlay IP address from the overlay IP address range for a subnet is reserved for a port of the VCN VR for that subnet. For example, consider a VCN having two subnets with associated address ranges 10.0/16 and 10.1/16, respectively. For the first subnet within the VCN with address range 10.0/16, an address from this range is reserved for a port of the VCN VR for that subnet. In some instances, the first IP address from the range may be reserved for the VCN VR. For example, for the subnet with overlay IP address range 10.0/16, IP address 10.0.0.1 may be reserved for a port of the VCN VR for that subnet. For the second subnet within the same VCN with address range 10.1/16, the VCN VR may have a port for that second subnet with IP address 10.1.0.1. The VCN VR has a different IP address for each of the subnets in the VCN.

[0077] In some other embodiments, each subnet within a VCN may have its own associated VR that is addressable by the subnet using a reserved or default IP address associated with the VR. The reserved or default IP address may, for example, be the first IP address from the range of IP addresses associated with that subnet. The VNICs in the subnet can communicate (e.g., send and receive packets) with the VR associated with the subnet using this default or reserved IP address. In such an embodiment, the VR is the ingress/egress point for that subnet. The VR associated with a subnet within the VCN can communicate with other VRs associated with other subnets within the VCN. The VRs can also communicate with gateways associated with the VCN. The VR function for a subnet is running on or executed by one or more NVDs executing VNICs functionality for VNICs in the subnet.

[0078] Route tables, security rules, and DHCP options may be configured for a VCN. Route tables are virtual route tables for the VCN and include rules to route traffic from subnets within the VCN to destinations outside the VCN by way of gateways or specially configured instances. A VCN’s route tables can be customized to control how packets are forwarded/routed to and from the VCN. DHCP options refers to configuration information that is automatically provided to the instances when they boot up.

[0079] Security rules configured for a VCN represent overlay firewall rules for the VCN. The security rules can include ingress and egress rules, and specify the types of traffic (e.g., based upon protocol and port) that is allowed in and out of the instances within the VCN. The customer can choose whether a given rule is stateful or stateless. For instance, the customer can allow incoming SSH traffic from anywhere to a set of instances by setting up a stateful ingress rule with source CIDR 0.0.0.0/0, and destination TCP port 22. Security rules can be implemented using network security groups or security lists. A network security group consists of a set of security rules that apply only to the resources in that group. A security list, on the other hand, includes rules that apply to all the resources in any subnet that uses the security list. A VCN may be provided with a default security list with default security rules. DHCP options configured for a VCN provide configuration information that is automatically provided to the instances in the VCN when the instances boot up.

[0080] In certain embodiments, the configuration information for a VCN is determined and stored by a VCN Control Plane. The configuration information for a VCN may include, for example, information about the address range associated with the VCN, subnets within the VCN and associated information, one or more VRs associated with the VCN, compute instances in the VCN and associated VNICs, NVDs executing the various virtualization network functions (e.g., VNICs, VRs, gateways) associated with the VCN, state information for the VCN, and other VCN-related information. In certain embodiments, a VCN Distribution Service publishes the configuration information stored by the VCN Control Plane, or portions thereof, to the NVDs. The distributed information may be used to update information (e.g., forwarding tables, routing tables, etc.) stored and used by the NVDs to forward packets to and from the compute instances in the VCN.

[0081] In certain embodiments, the creation of VCNs and subnets are handled by a VCN Control Plane (CP), and the launching of compute instances is handled by a Compute Control Plane. The Compute Control Plane is responsible for allocating the physical resources for the compute instance and then calls the VCN Control Plane to create and attach VNICs to the compute instance. The VCN CP also sends VCN data mappings to the VCN data plane that is configured to perform packet forwarding and routing functions. In certain embodiments, the VCN CP provides a distribution service that is responsible for providing updates to the VCN data plane. Examples of a VCN Control Plane are also depicted in FIGS. 6, 7, 8, and 9 (see references 616, 716, 816, and 916) and described below.

[0082] A customer may create one or more VCNs using resources hosted by CSPI. A compute instance deployed on a customer VCN may communicate with different endpoints. These endpoints can include endpoints that are hosted by CSPI and endpoints outside CSPI.

[0083] Various different architectures for implementing cloud-based service using CSPI are depicted in FIGS. 1-5 and 13-24, and are described below. FIG. 1 is a high-level diagram of a distributed environment 100 showing an overlay or customer VCN hosted by CSPI according to certain embodiments. The distributed environment depicted in FIG. 1 includes multiple components in the overlay network. Distributed environment 100 depicted in FIG. 1 is merely an example and is not intended to unduly limit the scope of claimed embodiments. Many variations, alternatives, and modifications are possible. For example, in some implementations, the distributed environment depicted in FIG. 1 may have more or fewer systems or components than those shown in FIG. 1, may combine two or more systems, or may have a different configuration or arrangement of systems.

[0084] As shown in the example depicted in FIG. 1, distributed environment 100 comprises CSPI 101 that provides services and resources that customers can subscribe to and use to build their virtual cloud networks (VCNs). In certain embodiments, CSPI 101 offers IaaS services to subscribing customers. The data centers within CSPI 101 may be organized into one or more regions. One example region “Region US” 102 is shown in FIG. 1. A customer has configured a customer VCN c/o Oracle International Corporation for region 102. The customer may deploy various compute instances on VCN 104, where the compute instances may include virtual machines or bare metal instances. Examples of instances include applications, database, load balancers, and the like.

[0085] In the embodiment depicted in FIG. 1, customer VCN 104 comprises two subnets, namely, “Subnet-1” and “Subnet-2”, each subnet with its own CIDR IP address range. In FIG. 1, the overlay IP address range for Subnet-1 is 10.0/16 and the address range for Subnet-2 is 10.1/16. A VCN Virtual Router 105 represents a logical gateway for the VCN that enables communications between subnets of the VCN 104, and with other endpoints outside the VCN. VCN VR 105 is configured to route traffic between VNICs in VCN 104 and gateways associated with VCN 104. VCN VR 105 provides a port for each subnet of VCN 104. For example, VR 105 may provide a port with IP address 10.0.0.1 for Subnet-1 and a port with IP address 10.1.0.1 for Subnet-2.

[0086] Multiple compute instances may be deployed on each subnet, where the compute instances can be virtual machine instances, and/or bare metal instances. The compute instances in a subnet may be hosted by one or more host machines within CSPI 101. A compute instance participates in a subnet via a VNIC associated with the compute instance. For example, as shown in FIG. 1, a compute instance C1 is part of Subnet-1 via a VNIC associated with the compute instance. Likewise, compute instance C2 is part of Subnet-1 via a VNIC associated with C2. In a similar manner, multiple compute instances, which may be virtual machine instances or bare metal instances, may be part of Subnet-1. Via its associated VNIC, each compute instance is assigned a private overlay IP address and a MAC address. For example, in FIG. 1, compute instance C1 has an overlay IP address of 10.0.0.2 and a MAC address of M1, while compute instance C2 has a private overlay IP address of 10.0.0.3 and a MAC address of M2. Each compute instance in Subnet-1, including compute instances C1 and C2, has a default route to VCN VR 105 using IP address 10.0.0.1, which is the IP address for a port of VCN VR 105 for Subnet-1.

[0087] Subnet-2 can have multiple compute instances deployed on it, including virtual machine instances and/or bare metal instances. For example, as shown in FIG. 1, compute instances D1 and D2 are part of Subnet-2 via VNICs associated with the respective compute instances. In the embodiment depicted in FIG. 1, compute instance D1 has an overlay IP address of 10.1.0.2 and a MAC address of MM1, while compute instance D2 has a private overlay IP address of 10.1.0.3 and a MAC address of MM2. Each compute instance in Subnet-2, including compute instances D1 and D2, has a default route to VCN VR 105 using IP address 10.1.0.1, which is the IP address for a port of VCN VR 105 for Subnet-2.

[0088] VCN A 104 may also include one or more load balancers. For example, a load balancer may be provided for a subnet and may be configured to load balance traffic across multiple compute instances on the subnet. A load balancer may also be provided to load balance traffic across subnets in the VCN.

[0089] A particular compute instance deployed on VCN 104 can communicate with various different endpoints. These endpoints may include endpoints that are hosted by CSPI 200 and endpoints outside CSPI 200. Endpoints that are hosted by CSPI 101 may include: an endpoint on the same subnet as the particular compute instance (e.g., communications between two compute instances in Subnet-1); an endpoint on a different subnet but within the same VCN (e.g., communication between a compute instance in Subnet-1 and a compute instance in Subnet-2); an endpoint in a different VCN in the same region (e.g., communications between a compute instance in Subnet-1 and an endpoint in a VCN in the same region 106 or 110, communications between a compute instance in Subnet-1 and an endpoint in service network 110 in the same region); or an endpoint in a VCN in a different region (e.g., communications between a compute instance in Subnet-1 and an endpoint in a VCN in a different region 108). A compute instance in a subnet hosted by CSPI 101 may also communicate with endpoints that are not hosted by CSPI 101 (i.e., are outside CSPI 101). These outside endpoints include endpoints in the customer’s on-premises network 116, endpoints within other remote cloud hosted networks 118, public endpoints 114 accessible via a public network such as the Internet, and other endpoints.

[0090] Communications between compute instances on the same subnet are facilitated using VNICs associated with the source compute instance and the destination compute instance. For example, compute instance C1 in Subnet-1 may want to send packets to compute instance C2 in Subnet-1. For a packet originating at a source compute instance and whose destination is another compute instance in the same subnet, the packet is first processed by the VNIC associated with the source compute instance. Processing performed by the VNIC associated with the source compute instance can include determining destination information for the packet from the packet headers, identifying any policies (e.g., security lists) configured for the VNIC associated with the source compute instance, determining a next hop for the packet, performing any packet encapsulation/decapsulation functions as needed, and then forwarding/routing the packet to the next hop with the goal of facilitating communication of the packet to its intended destination. When the destination compute instance is in the same subnet as the source compute instance, the VNIC associated with the source

compute instance is configured to identify the VNIC associated with the destination compute instance and forward the packet to that VNIC for processing. The VNIC associated with the destination compute instance is then executed and forwards the packet to the destination compute instance.

[0091] For a packet to be communicated from a compute instance in a subnet to an endpoint in a different subnet in the same VCN, the communication is facilitated by the VNICs associated with the source and destination compute instances and the VCN VR. For example, if compute instance C1 in Subnet-1 in FIG. 1 wants to send a packet to compute instance D1 in Subnet-2, the packet is first processed by the VNIC associated with compute instance C1. The VNIC associated with compute instance C1 is configured to route the packet to the VCN VR 105 using default route or port 10.0.0.1 of the VCN VR. VCN VR 105 is configured to route the packet to Subnet-2 using port 10.1.0.1. The packet is then received and processed by the VNIC associated with D1 and the VNIC forwards the packet to compute instance D1.

[0092] For a packet to be communicated from a compute instance in VCN 104 to an endpoint that is outside VCN 104, the communication is facilitated by the VNIC associated with the source compute instance, VCN VR 105, and gateways associated with VCN 104. One or more types of gateways may be associated with VCN 104. A gateway is an interface between a VCN and another endpoint, where another endpoint is outside the VCN. A gateway is a Layer-3/IP layer concept and enables a VCN to communicate with endpoints outside the VCN. A gateway thus facilitates traffic flow between a VCN and other VCNs or networks. Various different types of gateways may be configured for a VCN to facilitate different types of communications with different types of endpoints. Depending upon the gateway, the communications may be over public networks (e.g., the Internet) or over private networks. Various communication protocols may be used for these communications.

[0093] For example, compute instance C1 may want to communicate with an endpoint outside VCN 104. The packet may be first processed by the VNIC associated with source compute instance C1. The VNIC processing determines that the destination for the packet is outside the Subnet-1 of C1. The VNIC associated with C1 may forward the packet to VCN VR 105 for VCN 104. VCN VR 105 then processes the packet and as part of the processing, based upon the destination for the packet, determines a particular gateway associated with VCN 104 as the next hop for the packet. VCN VR 105 may then forward the packet to the particular identified gateway. For example, if the destination is an endpoint within the customer's on-premise network, then the packet may be forwarded by VCN VR 105 to Dynamic Routing Gateway (DRG) gateway 122 configured for VCN 104. The packet may then be forwarded from the gateway to a next hop to facilitate communication of the packet to its final intended destination.

[0094] Various different types of gateways may be configured for a VCN. Examples of gateways that may be configured for a VCN are depicted in FIG. 1 and described below. Examples of gateways associated with a VCN are also depicted in FIGS. 24, 20, 21, and 22 (for example, gateways referenced by reference numbers 2434, 2436, 2438, 2034, 2036, 2038, 2134, 2136, 2138, 2234, 2236, and 2238) and described below. As shown in the embodiment

depicted in FIG. 1, a Dynamic Routing Gateway (DRG) 122 may be added to or be associated with customer VCN 104 and provides a path for private network traffic communication between customer VCN 104 and another endpoint, where the another endpoint can be the customer's on-premise network 116, a VCN 108 in a different region of CSPI 101, or other remote cloud networks 118 not hosted by CSPI 101. Customer on-premise network 116 may be a customer network or a customer data center built using the customer's resources. Access to customer on-premise network 116 is generally very restricted. For a customer that has both a customer on-premise network 116 and one or more VCNs 104 deployed or hosted in the cloud by CSPI 101, the customer may want their on-premise network 116 and their cloud based VCN 104 to be able to communicate with each other. This enables a customer to build an extended hybrid environment encompassing the customer's VCN 104 hosted by CSPI 101 and their on-premises network 116. DRG 122 enables this communication. To enable such communications, a communication channel 124 is set up where one endpoint of the channel is in customer on-premise network 116 and the other endpoint is in CSPI 101 and connected to customer VCN 104. Communication channel 124 can be over public communication networks such as the Internet or private communication networks. Various different communication protocols may be used such as IPsec VPN technology over a public communication network such as the Internet, Oracle's FastConnect technology that uses a private network instead of a public network, and others. The device or equipment in customer on-premise network 116 that forms one end point for communication channel 124 is referred to as the customer premise equipment (CPE), such as CPE 126 depicted in FIG. 1. On the CSPI 101 side, the endpoint may be a host machine executing DRG 122.

[0095] In certain embodiments, a Remote Peering Connection (RPC) can be added to a DRG, which allows a customer to peer one VCN with another VCN in a different region. Using such an RPC, customer VCN 104 can use DRG 122 to connect with a VCN 108 in another region. DRG 122 may also be used to communicate with other remote cloud networks 118, not hosted by CSPI 101 such as a Microsoft AZURE cloud, Amazon AWS cloud, and others.

[0096] As shown in FIG. 1, an Internet Gateway (IGW) 120 may be configured for customer VCN 104 and enables a compute instance on VCN 104 to communicate with public endpoints 114 accessible over a public network such as the Internet. IGW 120 is a gateway that connects a VCN to a public network such as the Internet. IGW 120 enables a public subnet (where the resources in the public subnet have public overlay IP addresses) within a VCN, such as VCN 104, direct access to public endpoints 112 on a public network 114 such as the Internet. Using IGW 120, connections can be initiated from a subnet within VCN 104 or from the Internet.

[0097] A Network Address Translation (NAT) gateway 128 can be configured for customer's VCN 104 and enables cloud resources in the customer's VCN, which do not have dedicated public overlay IP addresses, access to the Internet and it does so without exposing those resources to direct incoming Internet connections (e.g., L4-L7 connections). This enables a private subnet within a VCN, such as private Subnet-1 in VCN 104, with private access to public endpoints on the Internet. In NAT gateways, connections can be

initiated only from the private subnet to the public Internet and not from the Internet to the private subnet.

[0098] In certain embodiments, a Service Gateway (SGW) **126** can be configured for customer VCN **104** and provides a path for private network traffic between VCN **104** and supported services endpoints in a service network **110**. In certain embodiments, service network **110** may be provided by the CSP and may provide various services. An example of such a service network is Oracle's Services Network, which provides various services that can be used by customers. For example, a compute instance (e.g., a database system) in a private subnet of customer VCN **104** can back up data to a service endpoint (e.g., Object Storage) without needing public IP addresses or access to the Internet. In certain embodiments, a VCN can have only one SGW, and connections can only be initiated from a subnet within the VCN and not from service network **110**. If a VCN is peered with another, resources in the other VCN typically cannot access the SGW. Resources in on-premises networks that are connected to a VCN with FastConnect or VPN Connect can also use the service gateway configured for that VCN.

[0099] In certain implementations, SGW **126** uses the concept of a service Classless Inter-Domain Routing (CIDR) label, which is a string that represents all the regional public IP address ranges for the service or group of services of interest. The customer uses the service CIDR label when they configure the SGW and related route rules to control traffic to the service. The customer can optionally utilize it when configuring security rules without needing to adjust them if the service's public IP addresses change in the future.

[0100] A Local Peering Gateway (LPG) **132** is a gateway that can be added to customer VCN **104** and enables VCN **104** to peer with another VCN in the same region. Peering means that the VCNs communicate using private IP addresses, without the traffic traversing a public network such as the Internet or without routing the traffic through the customer's on-premises network **116**. In preferred embodiments, a VCN has a separate LPG for each peering it establishes. Local Peering or VCN Peering is a common practice used to establish network connectivity between different applications or infrastructure management functions.

[0101] Service providers, such as providers of services in service network **110**, may provide access to services using different access models. According to a public access model, services may be exposed as public endpoints that are publicly accessible by compute instance in a customer VCN via a public network such as the Internet and/or may be privately accessible via SGW **126**. According to a specific private access model, services are made accessible as private IP endpoints in a private subnet in the customer's VCN. This is referred to as a Private Endpoint (PE) access and enables a service provider to expose their service as an instance in the customer's private network. A Private Endpoint resource represents a service within the customer's VCN. Each PE manifests as a VNIC (referred to as a PE-VNIC, with one or more private IPs) in a subnet chosen by the customer in the customer's VCN. A PE thus provides a way to present a service within a private customer VCN subnet using a VNIC. Since the endpoint is exposed as a VNIC, all the features associated with a VNIC such as routing rules, security lists, etc., are now available for the PE VNIC.

[0102] A service provider can register their service to enable access through a PE. The provider can associate policies with the service that restricts the service's visibility to the customer tenancies. A provider can register multiple services under a single virtual IP address (VIP), especially for multi-tenant services. There may be multiple such private endpoints (in multiple VCNs) that represent the same service.

[0103] Compute instances in the private subnet can then use the PE VNIC's private IP address or the service DNS name to access the service. Compute instances in the customer VCN can access the service by sending traffic to the private IP address of the PE in the customer VCN. A Private Access Gateway (PAGW) **130** is a gateway resource that can be attached to a service provider VCN (e.g., a VCN in service network **110**) that acts as an ingress/egress point for all traffic from/to customer subnet private endpoints. PAGW **130** enables a provider to scale the number of PE connections without utilizing its internal IP address resources. A provider needs only configure one PAGW for any number of services registered in a single VCN. Providers can represent a service as a private endpoint in multiple VCNs of one or more customers. From the customer's perspective, the PE VNIC, which, instead of being attached to a customer's instance, appears attached to the service with which the customer wishes to interact. The traffic destined to the private endpoint is routed via PAGW **130** to the service. These are referred to as customer-to-service private connections (C2S connections).

[0104] The PE concept can also be used to extend the private access for the service to customer's on-premises networks and data centers, by allowing the traffic to flow through FastConnect/IPsec links and the private endpoint in the customer VCN. Private access for the service can also be extended to the customer's peered VCNs, by allowing the traffic to flow between LPG **132** and the PE in the customer's VCN.

[0105] A customer can control routing in a VCN at the subnet level, so the customer can specify which subnets in the customer's VCN, such as VCN **104**, use each gateway. A VCN's route tables are used to decide if traffic is allowed out of a VCN through a particular gateway. For example, in a particular instance, a route table for a public subnet within customer VCN **104** may send non-local traffic through IGW **120**. The route table for a private subnet within the same customer VCN **104** may send traffic destined for CSP services through SGW **126**. All remaining traffic may be sent via the NAT gateway **128**. Route tables only control traffic going out of a VCN.

[0106] Security lists associated with a VCN are used to control traffic that comes into a VCN via a gateway via inbound connections. All resources in a subnet use the same route table and security lists. Security lists may be used to control specific types of traffic allowed in and out of instances in a subnet of a VCN. Security list rules may comprise ingress (inbound) and egress (outbound) rules. For example, an ingress rule may specify an allowed source address range, while an egress rule may specify an allowed destination address range. Security rules may specify a particular protocol (e.g., TCP, ICMP), a particular port (e.g., 22 for SSH, 3389 for Windows RDP), etc. In certain implementations, an instance's operating system may enforce its own firewall rules that are aligned with the security list rules. Rules may be stateful (e.g., a connection

is tracked, and the response is automatically allowed without an explicit security list rule for the response traffic) or stateless.

[0107] Access from a customer VCN (i.e., by a resource or compute instance deployed on VCN **104**) can be categorized as public access, private access, or dedicated access. Public access refers to an access model where a public IP address or a NAT is used to access a public endpoint. Private access enables customer workloads in VCN **104** with private IP addresses (e.g., resources in a private subnet) to access services without traversing a public network such as the Internet. In certain embodiments, CSPI **101** enables customer VCN workloads with private IP addresses to access the (public service endpoints of) services using a service gateway. A service gateway thus offers a private access model by establishing a virtual link between the customer's VCN and the service's public endpoint residing outside the customer's private network.

[0108] Additionally, CSPI may offer dedicated public access using technologies such as FastConnect public peering where customer on-premises instances can access one or more services in a customer VCN using a FastConnect connection and without traversing a public network such as the Internet. CSPI also may offer dedicated private access using FastConnect private peering where customer on-premises instances with private IP addresses can access the customer's VCN workloads using a FastConnect connection. FastConnect is a network connectivity alternative to using the public Internet to connect a customer's on-premise network to CSPI and its services. FastConnect provides an easy, elastic, and economical way to create a dedicated and private connection with higher bandwidth options and a more reliable and consistent networking experience when compared to Internet-based connections.

[0109] FIG. 1 and the accompanying description above describes various virtualized components in an example virtual network. As described above, the virtual network is built on the underlying physical or substrate network. FIG. 2 depicts a simplified architectural diagram of the physical components in the physical network within CSPI **200** that provide the underlay for the virtual network according to certain embodiments. As shown, CSPI **200** provides a distributed environment comprising components and resources (e.g., compute, memory, and networking resources) provided by a cloud service provider (CSP). These components and resources are used to provide cloud services (e.g., IaaS services) to subscribing customers, i.e., customers that have subscribed to one or more services provided by the CSP. Based upon the services subscribed to by a customer, a subset of resources (e.g., compute, memory, and networking resources) of CSPI **200** are provisioned for the customer. Customers can then build their own cloud-based (i.e., CSPI-hosted) customizable and private virtual networks using physical compute, memory, and networking resources provided by CSPI **200**. As previously indicated, these customer networks are referred to as virtual cloud networks (VCNs). A customer can deploy one or more customer resources, such as compute instances, on these customer VCNs. Compute instances can be in the form of virtual machines, bare metal instances, and the like. CSPI **200** provides infrastructure and a set of complementary cloud services that enable customers to build and run a wide range of applications and services in a highly available hosted environment.

[0110] In the example embodiment depicted in FIG. 2, the physical components of CSPI **200** include one or more physical host machines or physical servers (e.g., **202**, **206**, **208**), network virtualization devices (NVDs) (e.g., **210**, **212**), top-of-rack (TOR) switches (e.g., **214**, **216**), and a physical network (e.g., **218**), and switches in physical network **218**. The physical host machines or servers may host and execute various compute instances that participate in one or more subnets of a VCN. The compute instances may include virtual machine instances, and bare metal instances. For example, the various compute instances depicted in FIG. 1 may be hosted by the physical host machines depicted in FIG. 2. The virtual machine compute instances in a VCN may be executed by one host machine or by multiple different host machines. The physical host machines may also host virtual host machines, container-based hosts or functions, and the like. The VNICS and VCN VR depicted in FIG. 1 may be executed by the NVDs depicted in FIG. 2. The gateways depicted in FIG. 1 may be executed by the host machines and/or by the NVDs depicted in FIG. 2.

[0111] The host machines or servers may execute a hypervisor (also referred to as a virtual machine monitor or VMM) that creates and enables a virtualized environment on the host machines. The virtualization or virtualized environment facilitates cloud-based computing. One or more compute instances may be created, executed, and managed on a host machine by a hypervisor on that host machine. The hypervisor on a host machine enables the physical set of computing resources of the host machine (e.g., compute, memory, and networking resources) to be shared between the various compute instances executed by the host machine.

[0112] For example, as depicted in FIG. 2, host machines **202** and **208** execute hypervisors **260** and **266**, respectively. These hypervisors may be implemented using software, firmware, or hardware, or combinations thereof. Typically, a hypervisor is a process or a software layer that sits on top of the host machine's operating system (OS), which in turn executes on the hardware processors of the host machine. The hypervisor provides a virtualized environment by enabling the physical set of computing resources (e.g., processing resources such as processors/cores, memory resources, networking resources) of the host machine to be shared among the various virtual machine compute instances executed by the host machine. For example, in FIG. 2, hypervisor **260** may sit on top of the OS of host machine **202** and enables the set of computing resources (e.g., processing, memory, and networking resources) of host machine **202** to be shared between compute instances (e.g., virtual machines) executed by host machine **202**. A virtual machine can have its own operating system (referred to as a guest operating system), which may be the same as or different from the OS of the host machine. The operating system of a virtual machine executed by a host machine may be the same as or different from the operating system of another virtual machine executed by the same host machine. A hypervisor thus enables multiple operating systems to be executed alongside each other while sharing the same set of computing resources of the host machine. The host machines depicted in FIG. 2 may have the same or different types of hypervisors.

[0113] A compute instance can be a virtual machine instance or a bare metal instance. In FIG. 2, compute instances **268** on host machine **202** and **274** on host machine

208 are examples of virtual machine instances. Host machine **206** is an example of a bare metal instance that is provided to a customer.

[0114] In certain instances, an entire host machine may be provisioned to a single customer, and all of the one or more compute instances (either virtual machines or bare metal instance) hosted by that host machine belong to that same customer. In other instances, a host machine may be shared between multiple customers (i.e., multiple tenants). In such a multi-tenancy scenario, a host machine may host virtual machine compute instances belonging to different customers. These compute instances may be members of different VCNs of different customers. In certain embodiments, a bare metal compute instance is hosted by a bare metal server without a hypervisor. When a bare metal compute instance is provisioned, a single customer or tenant maintains control of the physical CPU, memory, and network interfaces of the host machine hosting the bare metal instance and the host machine is not shared with other customers or tenants.

[0115] As previously described, each compute instance that is part of a VCN is associated with a VNIC that enables the compute instance to become a member of a subnet of the VCN. The VNIC associated with a compute instance facilitates the communication of packets or frames to and from the compute instance. A VNIC is associated with a compute instance when the compute instance is created. In certain embodiments, for a compute instance executed by a host machine, the VNIC associated with that compute instance is executed by an NVD connected to the host machine. For example, in FIG. 2, host machine **202** executes a virtual machine compute instance **268** that is associated with VNIC **276**, and VNIC **276** is executed by NVD **210** connected to host machine **202**. As another example, bare metal instance **272** hosted by host machine **206** is associated with VNIC **280** that is executed by NVD **212** connected to host machine **206**. As yet another example, VNIC **284** is associated with compute instance **274** executed by host machine **208**, and VNIC **284** is executed by NVD **212** connected to host machine **208**.

[0116] For compute instances hosted by a host machine, an NVD connected to that host machine also executes VCN VRs corresponding to VCNs of which the compute instances are members. For example, in the embodiment depicted in FIG. 2, NVD **210** executes VCN VR **277** corresponding to the VCN of which compute instance **268** is a member. NVD **212** may also execute one or more VCN VRs **283** corresponding to VCNs corresponding to the compute instances hosted by host machines **206** and **208**.

[0117] A host machine may include one or more network interface cards (NIC) that enable the host machine to be connected to other devices. A NIC on a host machine may provide one or more ports (or interfaces) that enable the host machine to be communicatively connected to another device. For example, a host machine may be connected to an NVD using one or more ports (or interfaces) provided on the host machine and on the NVD. A host machine may also be connected to other devices such as another host machine.

[0118] For example, in FIG. 2, host machine **202** is connected to NVD **210** using link **220** that extends between a port **234** provided by a NIC **232** of host machine **202** and between a port **236** of NVD **210**. Host machine **206** is connected to NVD **212** using link **224** that extends between a port **246** provided by a NIC **244** of host machine **206** and between a port **248** of NVD **212**. Host machine **208** is

connected to NVD **212** using link **226** that extends between a port **252** provided by a NIC **250** of host machine **208** and between a port **254** of NVD **212**.

[0119] The NVDs are in turn connected via communication links to top-of-the-rack (TOR) switches, which are connected to physical network **218** (also referred to as the switch fabric). In certain embodiments, the links between a host machine and an NVD, and between an NVD and a TOR switch are Ethernet links. For example, in FIG. 2, NVDs **210** and **212** are connected to TOR switches **214** and **216**, respectively, using links **228** and **230**. In certain embodiments, the links **220**, **224**, **226**, **228**, and **230** are Ethernet links. The collection of host machines and NVDs that are connected to a TOR is sometimes referred to as a rack.

[0120] Physical network **218** provides a communication fabric that enables TOR switches to communicate with each other. Physical network **218** can be a multi-tiered network. In certain implementations, physical network **218** is a multi-tiered Clos network of switches, with TOR switches **214** and **216** representing the leaf level nodes of the multi-tiered and multi-node physical switching network **218**. Different Clos network configurations are possible including but not limited to a 2-tier network, a 3-tier network, a 4-tier network, a 5-tier network, and in general a “n”-tiered network. An example of a Clos network is depicted in FIG. 5 and described below.

[0121] Various different connection configurations are possible between host machines and NVDs such as one-to-one configuration, many-to-one configuration, one-to-many configuration, and others. In a one-to-one configuration implementation, each host machine is connected to its own separate NVD. For example, in FIG. 2, host machine **202** is connected to NVD **210** via NIC **232** of host machine **202**. In a many-to-one configuration, multiple host machines are connected to one NVD. For example, in FIG. 2, host machines **206** and **208** are connected to the same NVD **212** via NICs **244** and **250**, respectively.

[0122] In a one-to-many configuration, one host machine is connected to multiple NVDs. FIG. 3 shows an example within CSPI **300** where a host machine is connected to multiple NVDs. As shown in FIG. 3, host machine **302** comprises a network interface card (NIC) **304** that includes multiple ports **306** and **308**. Host machine **300** is connected to a first NVD **310** via port **306** and link **320** and connected to a second NVD **312** via port **308** and link **322**. Ports **306** and **308** may be Ethernet ports and the links **320** and **322** between host machine **302** and NVDs **310** and **312** may be Ethernet links. NVD **310** is in turn connected to a first TOR switch **314** and NVD **312** is connected to a second TOR switch **316**. The links between NVDs **310** and **312**, and TOR switches **314** and **316** may be Ethernet links. TOR switches **314** and **316** represent the Tier-0 switching devices in multi-tiered physical network **318**.

[0123] The arrangement depicted in FIG. 3 provides two separate physical network paths to and from physical switch network **318** to host machine **302**: a first path traversing TOR switch **314** to NVD **310** to host machine **302**, and a second path traversing TOR switch **316** to NVD **312** to host machine **302**. The separate paths provide for enhanced availability (referred to as high availability) of host machine **302**. If there are problems in one of the paths (e.g., a link in one of the paths goes down) or devices (e.g., a particular NVD is not functioning), then the other path may be used for communications to/from host machine **302**.

[0124] In the configuration depicted in FIG. 3, the host machine is connected to two different NVDs using two different ports provided by a NIC of the host machine. In other embodiments, a host machine may include multiple NICs that enable connectivity of the host machine to multiple NVDs.

[0125] Referring back to FIG. 2, an NVD is a physical device or component that performs one or more network and/or storage virtualization functions. An NVD may be any device with one or more processing units (e.g., CPUs, Network Processing Units (NPUs), FPGAs, packet processing pipelines, etc.), memory including cache, and ports. The various virtualization functions may be performed by software/firmware executed by the one or more processing units of the NVD.

[0126] An NVD may be implemented in various different forms. For example, in certain embodiments, an NVD is implemented as an interface card referred to as a smartNIC or an intelligent NIC with an embedded processor onboard. A smartNIC is a separate device from the NICs on the host machines. In FIG. 2, the NVDs 210 and 212 may be implemented as smartNICs that are connected to host machines 202, and host machines 206 and 208, respectively.

[0127] A smartNIC is however just one example of an NVD implementation. Various other implementations are possible. For example, in some other implementations, an NVD or one or more functions performed by the NVD may be incorporated into or performed by one or more host machines, one or more TOR switches, and other components of CSPI 200. For example, an NVD may be embodied in a host machine where the functions performed by an NVD are performed by the host machine. As another example, an NVD may be part of a TOR switch, or a TOR switch may be configured to perform functions performed by an NVD that enables the TOR switch to perform various complex packet transformations that are used for a public cloud. A TOR that performs the functions of an NVD is sometimes referred to as a smart TOR. In yet other implementations, where virtual machines (VMs) instances, but not bare metal (BM) instances, are offered to customers, functions performed by an NVD may be implemented inside a hypervisor of the host machine. In some other implementations, some of the functions of the NVD may be offloaded to a centralized service running on a fleet of host machines.

[0128] In certain embodiments, such as when implemented as a smartNIC as shown in FIG. 2, an NVD may comprise multiple physical ports that enable it to be connected to one or more host machines and to one or more TOR switches. A port on an NVD can be classified as a host-facing port (also referred to as a “south port”) or a network-facing or TOR-facing port (also referred to as a “north port”). A host-facing port of an NVD is a port that is used to connect the NVD to a host machine. Examples of host-facing ports in FIG. 2 include port 236 on NVD 210, and ports 248 and 254 on NVD 212. A network-facing port of an NVD is a port that is used to connect the NVD to a TOR switch. Examples of network-facing ports in FIG. 2 include port 256 on NVD 210, and port 258 on NVD 212. As shown in FIG. 2, NVD 210 is connected to TOR switch 214 using link 228 that extends from port 256 of NVD 210 to the TOR switch 214. Likewise, NVD 212 is connected to TOR switch 216 using link 230 that extends from port 258 of NVD 212 to the TOR switch 216.

[0129] An NVD receives packets and frames from a host machine (e.g., packets and frames generated by a compute instance hosted by the host machine) via a host-facing port and, after performing the necessary packet processing, may forward the packets and frames to a TOR switch via a network-facing port of the NVD. An NVD may receive packets and frames from a TOR switch via a network-facing port of the NVD and, after performing the necessary packet processing, may forward the packets and frames to a host machine via a host-facing port of the NVD.

[0130] In certain embodiments, there may be multiple ports and associated links between an NVD and a TOR switch. These ports and links may be aggregated to form a link aggregator group of multiple ports or links (referred to as a LAG). Link aggregation allows multiple physical links between two endpoints (e.g., between an NVD and a TOR switch) to be treated as a single logical link. All the physical links in a given LAG may operate in full-duplex mode at the same speed. LAGs help increase the bandwidth and reliability of the connection between two endpoints. If one of the physical links in the LAG goes down, traffic is dynamically and transparently reassigned to one of the other physical links in the LAG. The aggregated physical links deliver higher bandwidth than each individual link. The multiple ports associated with a LAG are treated as a single logical port. Traffic can be load-balanced across the multiple physical links of a LAG. One or more LAGs may be configured between two endpoints. The two endpoints may be between an NVD and a TOR switch, between a host machine and an NVD, and the like.

[0131] An NVD implements or performs network virtualization functions. These functions are performed by software/firmware executed by the NVD. Examples of network virtualization functions include without limitation: packet encapsulation and de-capsulation functions; functions for creating a VCN network; functions for implementing network policies such as VCN security list (firewall) functionality; functions that facilitate the routing and forwarding of packets to and from compute instances in a VCN; and the like. In certain embodiments, upon receiving a packet, an NVD is configured to execute a packet processing pipeline for processing the packet and determining how the packet is to be forwarded or routed. As part of this packet processing pipeline, the NVD may execute one or more virtual functions associated with the overlay network such as executing VNICs associated with compute instances in the VCN, executing a Virtual Router (VR) associated with the VCN, the encapsulation and decapsulation of packets to facilitate forwarding or routing in the virtual network, execution of certain gateways (e.g., the Local Peering Gateway), the implementation of Security Lists, Network Security Groups, network address translation (NAT) functionality (e.g., the translation of Public IP to Private IP on a host by host basis), throttling functions, and other functions.

[0132] In certain embodiments, the packet processing data path in an NVD may comprise multiple packet pipelines, each composed of a series of packet transformation stages. In certain implementations, upon receiving a packet, the packet is parsed and classified to a single pipeline. The packet is then processed in a linear fashion, one stage after another, until the packet is either dropped or sent out over an interface of the NVD. These stages provide basic functional packet processing building blocks (e.g., validating headers, enforcing throttle, inserting new Layer-2 headers, enforcing

L4 firewall, VCN encapsulation/decapsulation, etc.) so that new pipelines can be constructed by composing existing stages, and new functionality can be added by creating new stages and inserting them into existing pipelines.

[0133] An NVD may perform both control plane and data plane functions corresponding to a control plane and a data plane of a VCN. Examples of a VCN Control Plane are also depicted in FIGS. 23, 20, 21, and 22 (see references 2316, 2016, 2116, and 2216) and described below. Examples of a VCN Data Plane are depicted in FIGS. 23, 20, 21, and 22 (see references 2418, 2018, 2118, and 2218) and described below. The control plane functions include functions used for configuring a network (e.g., setting up routes and route tables, configuring VNICs, etc.) that controls how data is to be forwarded. In certain embodiments, a VCN Control Plane is provided that computes all the overlay-to-substrate mappings centrally and publishes them to the NVDs and to the virtual network edge devices such as various gateways such as the DRG, the SGW, the IGW, etc. Firewall rules may also be published using the same mechanism. In certain embodiments, an NVD only gets the mappings that are relevant for that NVD. The data plane functions include functions for the actual routing/forwarding of a packet based upon configuration set up using control plane. A VCN data plane is implemented by encapsulating the customer's network packets before they traverse the substrate network. The encapsulation/decapsulation functionality is implemented on the NVDs. In certain embodiments, an NVD is configured to intercept all network packets in and out of host machines and perform network virtualization functions.

[0134] As indicated above, an NVD executes various virtualization functions including VNICs and VCN VRs. An NVD may execute VNICs associated with the compute instances hosted by one or more host machines connected to the VNIC. For example, as depicted in FIG. 2, NVD 210 executes the functionality for VNIC 276 that is associated with compute instance 268 hosted by host machine 202 connected to NVD 210. As another example, NVD 212 executes VNIC 280 that is associated with bare metal compute instance 272 hosted by host machine 206 and executes VNIC 284 that is associated with compute instance 274 hosted by host machine 208. A host machine may host compute instances belonging to different VCNs, which belong to different customers, and the NVD connected to the host machine may execute the VNICs (i.e., execute VNICs-related functionality) corresponding to the compute instances.

[0135] An NVD also executes VCN Virtual Routers corresponding to the VCNs of the compute instances. For example, in the embodiment depicted in FIG. 2, NVD 210 executes VCN VR 277 corresponding to the VCN to which compute instance 268 belongs. NVD 212 executes one or more VCN VRs 283 corresponding to one or more VCNs to which compute instances hosted by host machines 206 and 208 belong. In certain embodiments, the VCN VR corresponding to that VCN is executed by all the NVDs connected to host machines that host at least one compute instance belonging to that VCN. If a host machine hosts compute instances belonging to different VCNs, an NVD connected to that host machine may execute VCN VRs corresponding to those different VCNs.

[0136] In addition to VNICs and VCN VRs, an NVD may execute various software (e.g., daemons) and include one or more hardware components that facilitate the various network virtualization functions performed by the NVD. For

purposes of simplicity, these various components are grouped together as "packet processing components" shown in FIG. 2. For example, NVD 210 comprises packet processing components 286 and NVD 212 comprises packet processing components 288. For example, the packet processing components for an NVD may include a packet processor that is configured to interact with the NVD's ports and hardware interfaces to monitor all packets received by and communicated using the NVD and store network information. The network information may, for example, include network flow information identifying different network flows handled by the NVD and per flow information (e.g., per flow statistics). In certain embodiments, network flows information may be stored on a per VNIC basis. The packet processor may perform packet-by-packet manipulations as well as implement stateful NAT and L4 firewall (FW). As another example, the packet processing components may include a replication agent that is configured to replicate information stored by the NVD to one or more different replication target stores. As yet another example, the packet processing components may include a logging agent that is configured to perform logging functions for the NVD. The packet processing components may also include software for monitoring the performance and health of the NVD and, also possibly of monitoring the state and health of other components connected to the NVD.

[0137] FIG. 1 shows the components of an example virtual or overlay network including a VCN, subnets within the VCN, compute instances deployed on subnets, VNICs associated with the compute instances, a VR for a VCN, and a set of gateways configured for the VCN. The overlay components depicted in FIG. 1 may be executed or hosted by one or more of the physical components depicted in FIG. 2. For example, the compute instances in a VCN may be executed or hosted by one or more host machines depicted in FIG. 2. For a compute instance hosted by a host machine, the VNIC associated with that compute instance is typically executed by an NVD connected to that host machine (i.e., the VNIC functionality is provided by the NVD connected to that host machine). The VCN VR function for a VCN is executed by all the NVDs that are connected to host machines hosting or executing the compute instances that are part of that VCN. The gateways associated with a VCN may be executed by one or more different types of NVDs. For example, certain gateways may be executed by smartNICs, while others may be executed by one or more host machines or other implementations of NVDs.

[0138] As described above, a compute instance in a customer VCN may communicate with various different endpoints, where the endpoints can be within the same subnet as the source compute instance, in a different subnet but within the same VCN as the source compute instance, or with an endpoint that is outside the VCN of the source compute instance. These communications are facilitated using VNICs associated with the compute instances, the VCN VRs, and the gateways associated with the VCNs.

[0139] For communications between two compute instances on the same subnet in a VCN, the communication is facilitated using VNICs associated with the source and destination compute instances. The source and destination compute instances may be hosted by the same host machine or by different host machines. A packet originating from a source compute instance may be forwarded from a host machine hosting the source compute instance to an NVD

connected to that host machine. On the NVD, the packet is processed using a packet processing pipeline, which can include execution of the VNIC associated with the source compute instance. Since the destination endpoint for the packet is within the same subnet, execution of the VNIC associated with the source compute instance results in the packet being forwarded to an NVD executing the VNIC associated with the destination compute instance, which then processes and forwards the packet to the destination compute instance. The VNICs associated with the source and destination compute instances may be executed on the same NVD (e.g., when both the source and destination compute instances are hosted by the same host machine) or on different NVDs (e.g., when the source and destination compute instances are hosted by different host machines connected to different NVDs). The VNICs may use routing/forwarding tables stored by the NVD to determine the next hop for the packet.

[0140] For a packet to be communicated from a compute instance in a subnet to an endpoint in a different subnet in the same VCN, the packet originating from the source compute instance is communicated from the host machine hosting the source compute instance to the NVD connected to that host machine. On the NVD, the packet is processed using a packet processing pipeline, which can include execution of one or more VNICs, and the VR associated with the VCN. For example, as part of the packet processing pipeline, the NVD executes or invokes functionality corresponding to the VNIC (also referred to as executes the VNIC) associated with source compute instance. The functionality performed by the VNIC may include looking at the VLAN tag on the packet. Since the packet's destination is outside the subnet, the VCN VR functionality is next invoked and executed by the NVD. The VCN VR then routes the packet to the NVD executing the VNIC associated with the destination compute instance. The VNIC associated with the destination compute instance then processes the packet and forwards the packet to the destination compute instance. The VNICs associated with the source and destination compute instances may be executed on the same NVD (e.g., when both the source and destination compute instances are hosted by the same host machine) or on different NVDs (e.g., when the source and destination compute instances are hosted by different host machines connected to different NVDs).

[0141] If the destination for the packet is outside the VCN of the source compute instance, then the packet originating from the source compute instance is communicated from the host machine hosting the source compute instance to the NVD connected to that host machine. The NVD executes the VNIC associated with the source compute instance. Since the destination end point of the packet is outside the VCN, the packet is then processed by the VCN VR for that VCN. The NVD invokes the VCN VR functionality, which may result in the packet being forwarded to an NVD executing the appropriate gateway associated with the VCN. For example, if the destination is an endpoint within the customer's on-premise network, then the packet may be forwarded by the VCN VR to the NVD executing the DRG gateway configured for the VCN. The VCN VR may be executed on the same NVD as the NVD executing the VNIC associated with the source compute instance or by a different NVD. The gateway may be executed by an NVD, which may be a smartNIC, a host machine, or other NVD implementation. The packet is then processed by the gateway and

forwarded to a next hop that facilitates communication of the packet to its intended destination endpoint. For example, in the embodiment depicted in FIG. 2, a packet originating from compute instance 268 may be communicated from host machine 202 to NVD 210 over link 220 (using NIC 232). On NVD 210, VNIC 276 is invoked since it is the VNIC associated with source compute instance 268. VNIC 276 is configured to examine the encapsulated information in the packet and determine a next hop for forwarding the packet with the goal of facilitating communication of the packet to its intended destination endpoint, and then forward the packet to the determined next hop.

[0142] A compute instance deployed on a VCN can communicate with various different endpoints. These endpoints may include endpoints that are hosted by CSPI 200 and endpoints outside CSPI 200. Endpoints hosted by CSPI 200 may include instances in the same VCN or other VCNs, which may be the customer's VCNs, or VCNs not belonging to the customer. Communications between endpoints hosted by CSPI 200 may be performed over physical network 218. A compute instance may also communicate with endpoints that are not hosted by CSPI 200 or are outside CSPI 200. Examples of these endpoints include endpoints within a customer's on-premise network or data center, or public endpoints accessible over a public network such as the Internet. Communications with endpoints outside CSPI 200 may be performed over public networks (e.g., the Internet) (not shown in FIG. 2) or private networks (not shown in FIG. 2) using various communication protocols.

[0143] The architecture of CSPI 200 depicted in FIG. 2 is merely an example and is not intended to be limiting. Variations, alternatives, and modifications are possible in alternative embodiments. For example, in some implementations, CSPI 200 may have more or fewer systems or components than those shown in FIG. 2, may combine two or more systems, or may have a different configuration or arrangement of systems. The systems, subsystems, and other components depicted in FIG. 2 may be implemented in software (e.g., code, instructions, program) executed by one or more processing units (e.g., processors, cores) of the respective systems, using hardware, or combinations thereof. The software may be stored on a non-transitory storage medium (e.g., on a memory device).

[0144] FIG. 4 depicts connectivity between a host machine and an NVD for providing I/O virtualization for supporting multitenancy according to certain embodiments. As depicted in FIG. 4, host machine 402 executes a hypervisor 404 that provides a virtualized environment. Host machine 402 executes two virtual machine instances, VM1 406 belonging to customer/tenant #1 and VM2 408 belonging to customer/tenant #2. Host machine 402 comprises a physical NIC 410 that is connected to an NVD 412 via link 414. Each of the compute instances is attached to a VNIC that is executed by NVD 412. In the embodiment in FIG. 4, VM1 406 is attached to VNIC-VM1 420 and VM2 408 is attached to VNIC-VM2 422.

[0145] As shown in FIG. 4, NIC 410 comprises two logical NICs, logical NIC A 416 and logical NIC B 418. Each virtual machine is attached to and configured to work with its own logical NIC. For example, VM1 406 is attached to logical NIC A 416 and VM2 408 is attached to logical NIC B 418. Even though host machine 402 comprises only one physical NIC 410 that is shared by the multiple tenants, due

to the logical NICs, each tenant's virtual machine believes they have their own host machine and NIC.

[0146] In certain embodiments, each logical NIC is assigned its own VLAN ID. Thus, a specific VLAN ID is assigned to logical NIC A **416** for Tenant #1 and a separate VLAN ID is assigned to logical NIC B **418** for Tenant #2. When a packet is communicated from VM1 **406**, a tag assigned to Tenant #1 is attached to the packet by the hypervisor and the packet is then communicated from host machine **402** to NVD **412** over link **414**. In a similar manner, when a packet is communicated from VM2 **408**, a tag assigned to Tenant #2 is attached to the packet by the hypervisor and the packet is then communicated from host machine **402** to NVD **412** over link **414**. Accordingly, a packet **424** communicated from host machine **402** to NVD **412** has an associated tag **426** that identifies a specific tenant and associated VM. On the NVD, for a packet **424** received from host machine **402**, the tag **426** associated with the packet is used to determine whether the packet is to be processed by VNIC-VM1 **420** or by VNIC-VM2 **422**. The packet is then processed by the corresponding VNIC. The configuration depicted in FIG. 4 enables each tenant's compute instance to believe that they own their own host machine and NIC. The setup depicted in FIG. 4 provides for I/O virtualization for supporting multi-tenancy.

[0147] FIG. 5 depicts a simplified block diagram of a physical network **500** according to certain embodiments. The embodiment depicted in FIG. 5 is structured as a Clos network. A Clos network is a particular type of network topology designed to provide connection redundancy while maintaining high bisection bandwidth and maximum resource utilization. A Clos network is a type of non-blocking, multistage or multi-tiered switching network, where the number of stages or tiers can be two, three, four, five, etc. The embodiment depicted in FIG. 5 is a 3-tiered network comprising tiers 1, 2, and 3. The TOR switches **504** represent Tier-0 switches in the Clos network. One or more NVDs are connected to the TOR switches. Tier-0 switches are also referred to as edge devices of the physical network. The Tier-0 switches are connected to Tier-1 switches, which are also referred to as leaf switches. In the embodiment depicted in FIG. 5, a set of "n" Tier-0 TOR switches are connected to a set of "n" Tier-1 switches and together form a pod. Each Tier-0 switch in a pod is interconnected to all the Tier-1 switches in the pod, but there is no connectivity of switches between pods. In certain implementations, two pods are referred to as a block. Each block is served by or connected to a set of "n" Tier-2 switches (sometimes referred to as spine switches). There can be several blocks in the physical network topology. The Tier-2 switches are in turn connected to "n" Tier-3 switches (sometimes referred to as super-spine switches). Communication of packets over physical network **500** is typically performed using one or more Layer-3 communication protocols. Typically, all the layers of the physical network, except for the TORs layer are n-ways redundant thus allowing for high availability. Policies may be specified for pods and blocks to control the visibility of switches to each other in the physical network so as to enable scaling of the physical network.

[0148] A feature of a Clos network is that the maximum hop count to reach from one Tier-0 switch to another Tier-0 switch (or from an NVD connected to a Tier-0-switch to another NVD connected to a Tier-0 switch) is fixed. For example, in a 3-Tiered Clos network at most seven hops are

needed for a packet to reach from one NVD to another NVD, where the source and target NVDs are connected to the leaf tier of the Clos network. Likewise, in a 4-tiered Clos network, at most nine hops are needed for a packet to reach from one NVD to another NVD, where the source and target NVDs are connected to the leaf tier of the Clos network. Thus, a Clos network architecture maintains consistent latency throughout the network, which is important for communication within and between data centers. A Clos topology scales horizontally and is cost effective. The bandwidth/throughput capacity of the network can be easily increased by adding more switches at the various tiers (e.g., more leaf and spine switches) and by increasing the number of links between the switches at adjacent tiers.

[0149] In certain embodiments, each resource within CSPI is assigned a unique identifier called a Cloud Identifier (CID). This identifier is included as part of the resource's information and can be used to manage the resource, for example, via a Console or through APIs. An example syntax for a CID is:

[0150] `ocid1.<RESOURCE TYPE>.<REALM>.[REGION] [.FUTURE USE].<UNIQUE ID>`where,

[0151] `ocid1`: The literal string indicating the version of the CID;

[0152] `resource type`: The type of resource (for example, instance, volume, VCN, subnet, user, group, and so on);

[0153] `realm`: The realm the resource is in. Example values are "c1" for the commercial realm, "c2" for the Government Cloud realm, or "c3" for the Federal Government Cloud realm, etc. Each realm may have its own domain name;

[0154] `region`: The region the resource is in. If the region is not applicable to the resource, this part might be blank;

[0155] `future use`: Reserved for future use.

[0156] `unique ID`: The unique portion of the ID. The format may vary depending on the type of resource or service.

Multi-Cloud

[0157] FIG. 6 depicts a simplified high-level diagram **600** of a distributed environment comprising multiple cloud environments provided by different cloud service providers (CSPs). As depicted in FIG. 6, various cloud environments (also referred to as "clouds") may be provided by different CSPs, each cloud environment or cloud offering one or more cloud services that can be subscribed to by one or more customers of the corresponding CSP. The set of cloud services offered by a cloud environment provided by a CSP may include one or more different types of cloud services including but not restricted to Software-as-a-Service (SaaS) services, Infrastructure-as-a-Service (IaaS) services, Platform-as-a-Service (PaaS) services, Database-as-a-Service (DBaaS) services, and others. Examples of cloud environments provided by various CSPs include Oracle® Cloud Infrastructure (OCI) provided by Oracle Corporation, Microsoft® AZURE provided by Microsoft Corporation, Google Cloud™ provided by Google LLC, Amazon Web Services (AWS®) provided by Amazon Corporation, and others. The cloud services offered by a particular cloud environment may be different from the set of cloud services offered by another cloud environment.

[0158] In a typical cloud environment, a CSP provides cloud service provider infrastructure (CSPI) that is used to provide the one or more cloud services that are offered by that cloud environment to its customers. The CSPI provided by a CSP may include various types of hardware and software resources including compute resources, memory resources, networking resources, consoles for accessing the cloud services, and others. A customer of a cloud environment provided by a CSP may subscribe to one or more of the cloud services offered by that cloud environment. Various subscription models may be offered by the CSP to its customers. After a customer subscribes to a cloud service provided by a cloud environment, one or more users may be associated with the subscribing customer and such users can use the cloud service subscribed to by the customer. In certain implementations, when a customer subscribes to a cloud service provided by a particular cloud environment, a customer account or customer tenancy is created for that customer. One or more users can then be associated with the customer tenancy and such users can then use the services subscribed to by the customer under the customer tenancy. Information regarding the services subscribed to by a customer, the users associated with the customer tenancy, etc., is usually stored within the cloud environment and associated with the customer tenancy.

[0159] For example, two different cloud environments provided by two different CSPs are depicted in FIG. 6 (although a different number of cloud environments is possible). These include a Cloud Environment A (cloud A) 610 provided by a CSP A and a Cloud Environment B (cloud B) 640 provided by a CSP B.

[0160] Cloud A 610 includes infrastructure CSPI_A 612 provided by CSP A. This infrastructure CSPI_A 612 may be used to provide a set of services offered by cloud A 610. One or more customers (e.g., Cust_A1 616-1, Cust_A2 616-2) may subscribe to one or more of such services. One or more users 618-1 may be associated with Customer A1 616-1 and can use the services subscribed to by customer A1 616-1 in cloud A 610. In a similar manner, one or more users 618-2 may be associated with customer A2 616-2 and can use the services subscribed to by customer A2 616-2 in cloud A 610. In various use cases, the services subscribed to by customer A1 616-1 may be different from the services subscribed to by customer A2 616-2.

[0161] Similarly, cloud B 640 includes infrastructure CSPI_B 642 provided by CSP B. This infrastructure CSPI_B 642 may be used to provide a set of services offered by cloud B 640 (which may, but need not, be different from the services offered by Cloud A 610). One or more customers (e.g., Cust_B1 646-1) may subscribe to one or more services from Services B 644. One or more users 648-1 may be associated with Customer B1 646-1 and can use the services subscribed to by customer B1 646-1 in cloud B 640.

[0162] As depicted in FIG. 6, customer A1 616-1 is also a customer of CSP B and has subscribed to services available from CSPI_B 642. As such, customer A1 616-1 has tenancies in both cloud B 640 and cloud A 610.

[0163] In certain embodiments, CSP A and CSP B may agree to offer cross-cloud services of each. In the illustration of FIG. 6, CSP A offers one or more of its services to customers of CSP B via CSPI_B 642 (referred to herein as a cross-cloud services). These cross-cloud services include, for example, database services, storage services, compute services, and the like. As such, customer A1 616-1 (customer

of both clouds A and B) can request, subscribe, use, and/or manage one or more cross-cloud services of CSP A via its tenancy at CSPI_B 642. In comparison, customer B1 646-1 has no tenancy at CSPI_A 612. As such, cross-cloud services of CSP A may not be available to customer B1 646-1, unless customer B1 646-1 requests a tenancy to be provisioned with CSP A. This request can be submitted and managed via the portal of cloud B 640, as further described in the next figures.

[0164] To enable CSP A's cross-cloud services offering and availability via cloud B 640, cloud A 610 can implement an inter-cloud service 614. The inter-cloud service 614 can be configured to, among other things, enable the use of CSP A's cross-cloud services via cloud B. For example, the inter-cloud service 614 can communicate with services 644 of cloud B 640 and translate such communications into information suitable for use by intra-cloud services 615 of cloud A 610. More specifically, the services 644 can enable a portal of CSP B and the deployment and management of resources for a customer of CSP B within cloud B 640. Through this portal, a cross-cloud service of CSP A can be offered. Accordingly, a customer of CSP B (e.g., customer A1 616-1) can subscribe to and request the cross-cloud service of CSP A via the portal. Such subscription and cloud operation requests can be received by the inter-cloud service 614 that then translates them into information specific to cloud A 610. Such information can be passed to one or more of the intra-cloud services 615 that then provision the service. Information back from the intra-cloud services 615 can be translated and sent by the inter-cloud service 614 to cloud services 644 for use thereby.

[0165] In certain embodiments, the requested cross-cloud service can be provisioned by the one or more of the intra-cloud services 615 across resources in both cloud A 610 and cloud B 640. Doing so can support latency sensitive operations (or, at least, reduce processing latency).

[0166] As illustrated, infrastructure CSPI_A 612 of cloud A 610 includes an infrastructure 624 for a private cloud of customer A1 616-1 in cloud A 610 (e.g., for a VCN that is part of a tenancy of this customer at cloud A 610), an infrastructure 626 for a private cloud of customer A2 616-2 in cloud A 610, and other infrastructures 620. Each of these infrastructures includes hardware and/or software provided by CSP A and installed at locations (organized as regions) under the control of CSP A.

[0167] In comparison, infrastructure CSPI_B 642 of cloud B 640 includes an infrastructure 644 for the private cloud of customer A1 616-1 in cloud A (e.g., for the VCN that is part of the tenancy of this customer at cloud A 610), an infrastructure 646 for a private cloud of customer A1 616-1 in cloud B 640 (e.g., for a VNET that is part of this customer at cloud B 640), an infrastructure 648 for a private cloud of customer B1 646-1 in cloud B 640, and other infrastructures 650. The infrastructure 644 includes hardware and/or software provided by CSP A and installed at locations (organized as regions) under the control of CSP B (e.g., co-located with components of CSPI_B 642). In comparison, the infrastructures 646, 648, and 650 include hardware and/or software provided by CSP B and installed at locations (organized as regions) under the control of CSP B.

[0168] The infrastructure 644 and the infrastructure 646 can be networked together such that customer A1 616-1 can access, via its private cloud with CSP B its private cloud with CSP A, where the private cloud with CSP A is distrib-

uted between CSPI_A 612 and CSPI_B 642. This communicative coupling of the private clouds can be initiated by the inter-cloud service 614 and performed by one or more of the intra-cloud services 615. In this way, customer A 616-1 has two tenancies: a first one with CSP A that includes a first private cloud (e.g., a VCN) distributed between CSPI_A 612 and CSPI_B 642, and a second one with CSP B that includes a second private cloud (e.g., VNET) local to CSPI_B 642.

[0169] The requested cross-cloud service can be actually hosted on the first private cloud (e.g., at least in part on the infrastructure 644 within CSPI_B 642 and, possibly, in the infrastructure 624 within CSPI_A 612) and accessible via the second private cloud (e.g., to workflows hosted by the infrastructure 646). This distribution of components to enable a first private cloud of a first CSP to be hosted, at least in part, by a second cloud of a second CSP and to be linked to a second private cloud hosted by the second cloud is further described in the next figures.

[0170] FIG. 7 depicts an exemplary physical architecture for providing a cross-cloud service based on infrastructure distributed between multiple CSPs, according to some embodiments. In FIG. 7, a customer of a second CSP would manage the lifecycle of a cross-cloud service developed by a first CSP. For illustrative purposes, an Exadata service (which may also be referred to herein as Oracle DB service) and Oracle and Microsoft clouds are described. In this illustration, Oracle corresponds to the first CSP and Microsoft to the second CSP, whereas Exadata service corresponds to a cross-cloud service offered by the first CSP via a cloud of the second CSP. However, the embodiments are not limited as such and, instead, similarly apply to other CSPIs, CSPs and/or cross-cloud services.

[0171] A customer of the second CSP (e.g., an AZURE customer) can create and manage, via a portal of the second CSP (e.g., an AZURE portal), a virtual resource (e.g., an infrastructure and/or a VM cluster) with the cloud of the first CSP. This virtual resource can be provisioned to provide the cross-cloud service. The support of such offering can be implemented, at least in part, using the inter-cloud service 614 of FIG. 6. Further, the infrastructure supporting the virtual resource can be distributed between CSPIs of the two CSPs (e.g., between an AZURE datacenter and an OCI region). In particular, a first portion of the infrastructure provided by the first CSP is installed at the CSPI of the second CSP, while a remaining, second portion of the infrastructure provided by the first CSP is installed at the CSPI of the first CSP. The first portion can be referred to as being included or forming a child site within the cloud of the second CSP. The second portion can be referred to as being included or forming a child parent region for the child site, where this parent region is within the cloud of the first CSP.

[0172] On the left side of FIG. 7, a second CSPI_B 750 of the second CSP is shown. This CSPI_B 700 can represent, for example, a datacenter of the second CSP (e.g., an AZURE datacenter). Within the CSPI_B 750, a substrate network 730 is available and is provided by the second CSP. This substrate network 730 includes a set of computing resources, such as routers 732A, 732B, and so on (in the case of an AZURE datacenter, these routers can include MeetMe ToRs that support the MeetMe protocol for connection peering). The CSPI_B 750 also includes the child site 720 (which is set of computing resources, such as server blades, racks, or other physical hardware of the first CSP executing software of the CSP). The child site 720 includes, among

other things, routers 722A, 722B, and so on (in the case of OCI, these routers can include FastConnect routers that support the FastConnect protocol for connection peering), a connectivity fabric 724 (e.g., a physical fabric, such as JFAB, that provides connectivity to other physical fabrics and components) and physical resources (e.g., racks, such as OCI server blades optimized for the cross-cloud service such that Exadata service).

[0173] On the right side of FIG. 7, a first CSPI_A 700 of the first CSP is shown. This CSPI_A 700 can represent, for example, a datacenter of the first CSP (e.g., an OCI datacenter). Within the CSPI_A 700, a parent region 710 is illustrated. The parent region 710 can include a substrate network having multiple components. Among these components is a connectivity fabric 712 (e.g., a physical fabric, such as JFAB). The connectivity fabric 712 is connected with the connectivity fabric 724 such that the child site 720 is communicatively coupled with the parent region 710. Optionally, the parent region 710 is within a region within physical proximity to the child site 720 (or, equivalently, the CSPI_B 750), such that network latency for communications between the child site 720 and the parent region 710 is reduced.

[0174] The routers 732 within the substrate network 730 can be interconnected (e.g., via Ethernet cables) to the routers 722 within the child site 720. The connectivity fabric 724 of the child site 720 provides inter-connectivity between the routers 722 and the physical resources 726. Further, this connectivity fabric 724 is connected (e.g., using fiber optics, such as Dark Fiber) with the connectivity fabric 714 of the parent region 710 such that a data connectivity can exist between the child site 720 and the parent region 710.

[0175] The substrate network 730 can host a set of resources of the second CSP for a customer (e.g., to provide a VNET that includes compute instances having access to the Exadata service, as further illustrated in the next figures). These second CSP resources can be part of a proximity group within a certain latency (e.g., 100 μ s) to the cross-cloud service.

[0176] The child site 720 can host latency critical resources of the first CSP, where these first CSP resources support the cross-cloud service (e.g., the child site 720 can host OCI database resources and data plane resources in support of an Exadata service). The parent region 720 can host other resources (e.g., the inter-cloud service 614 and the intra-cloud services 715 of FIG. 6) that support the cross-cloud service (e.g., for the Exadata service, the parent region 720 can host Oracle resource provider (ORP), OCI tooling, OCI metrics and logging, OCI control plane, regional OCI services, and a customer facing console). Some of these resources (e.g., the as OCI database resources and data plane resources) can be deployed as part of a first private cloud of the customer with the first CSP (e.g., as a VCN in case of OCI) and can be perceived by the customer as being available to them via a second private cloud of the customer with the second CSP (e.g., a VNET in case of AZURE). This perception is possible by using the same IP address range in the two private clouds for the cross-cloud service.

[0177] In an example, the customer can have multiple private clouds with each CSP (e.g., multiple VNETs and/or multiple VCNs). In the case of multiple private clouds with the first CSP, the underlying physical resources may not be co-located in the same CSPI of the second CSP and, instead, can be included in or form different child sites. In this case,

these resources may not be directly interconnected (e.g., no direct physical connection may exist between the different child sites). Instead, an indirect connection can exist via parent region, where each of the different child sites is physically connected (e.g., via fiber optics) to the parent region, and where data flows between the two child sites through the parent region.

[0178] Further, the child site 720 can support a multi-tenancy architecture. In particular, multiple customers can each have one or more private clouds with each CSP (e.g., one or more VNETs and one or more VCNs). Each of such customers can have a separate access to a corresponding cross-cloud service via the child site 720.

[0179] FIG. 8 depicts an exemplary virtual architecture for providing a cross-cloud service based on infrastructure distributed between multiple CSPs, according to some embodiments. In FIG. 8, a first private cloud 810 of a customer with a first CSP (e.g., a VCN hosted by OCI) and a second cloud 820 of the customer with a second CSP (e.g., VNET hosted by AZURE) can be provisioned to provide a cross-cloud service of the first CSP via a cloud of the second CSP (e.g., an Exadata service offered by Oracle and available via AZURE to an AZURE customer).

[0180] In an example, the second private cloud 820 (e.g., VNET) is hosted by physical computing resources of CSPI_B 850 of the second CSP (e.g., an AZURE datacenter). The second private cloud 820 can include compute instances (not shown) that have access to an the cross-cloud service. The second private cloud 820 includes a subnet 822. The subnet 822 uses an IP address range. This range can be specified based on input of the customer via a portal of the second CSP (e.g., an AZURE portal). An IP address can be allocated from the IP address range to each compute instance launched to provide the cross-cloud service for the customer (e.g., a VM that belongs to a VM cluster and that provides a database instance). These compute instances can be hosted by the first private cloud 810 (shown, more generally in FIG. 8, as virtual resources 814A, 814B, 814C, and 814D). In FIG. 8, four computed instances are illustrated and indicated with rectangles labeled NICs 824A, 824B, 824C, and 824D (or collectively NICs 824). Although a NIC stands for a network interface card, the NICs 824 of FIG. 8 are not as such per se. Instead, the NIC 824 represent mapping information to map each IP address to the corresponding compute instance of the cross-cloud service. This mapping information can be stored as part of the configuration information of the second private cloud 820. If a compute instance of the second private cloud 820 sends traffic or requests traffic using an IP address, and this IP address corresponds to a NIC, the mapping information indicates that the traffic is to be sent to or is to be received from the first private cloud 810 (e.g., the VCN) and is cross-cloud service traffic.

[0181] The first private cloud 810 (e.g., the VCN) can be hosted by physical computing resources of the first CSP including, for example, physical computing resources of a child site and a parent region (similar to the ones illustrated in FIG. 7). The first private cloud 810 can include a subnet 812 that uses the same IP address range. Each compute instance launched to provide the cross-cloud service (e.g., each of the virtual resources 814A, 814B, 814C, and 814D, collectively referred to as virtual resources 814) is hosted by a set of physical computing resources of the first CSP. The set for a compute instance can include a physical computing resource in the child site and a physical computing resource

in the parent region. The child site's computing resource can include, for example, hardware (e.g., rack) executing software for the cross-cloud service as shown in FIG. 7 that provides latency critical virtual resources (e.g., database resources, data plane resources, etc.). The parent region's physical computing resource can include, for example, servers, server fleet, network virtualization devices, etc. that provide non-latency critical virtual resources (e.g., control plane resources, tooling resources, etc.).

[0182] Each compute instance can have an IP address from the IP address range. A one-to-one mapping exists between compute instances of the first private cloud 810 (e.g., the virtual resources 814) and the NICS 824 of the second private cloud 820. As such cross-cloud service traffic can be sent from the second private cloud 820 to the first private cloud 810 and vice versa using this one-to-one mapping.

[0183] The second private cloud 820 to the first private cloud 810 can be connected (e.g., peered) via one or more virtual routers (e.g., coupling a substrate network of the second CSP with the child site), gateways (e.g., to couple the child site with the parent region, a gateway 826 is implemented at the child site, and a gateway 816 is implemented at the parent region, such as by being parts of connectivity fabrics) and connection protocols (e.g., MeetMe and Fast-Connect protocols). Collectively, the gateways 816 and 826 can represent a dynamic routing gateway.

[0184] From a customer perspective, the customer need not be aware of the underlying physical architecture and inter-connections. Instead, it may suffice that the customer can have visibility at the virtual level, whereby the customer can perceive and manage their private clouds 810 and 820 (e.g., their network configurations). Managing the second private cloud 820 can be via a second portal of the second CSP. Managing the first private cloud 810 can be via the second portal of the second CSP or a first portal of the first CSP. In an embodiment, changes and/or operations related to the cross-cloud service are only enabled via the second CSP (e.g., via the second CSP, the customer can scale up or down, remove, terminate, add, etc. virtual resources at the first private cloud 810 launched for the cross-cloud service). In comparison, change and/or operations unrelated to the cross-cloud service are enabled (possibly only) via the first CSP (e.g., via the first CSP, the customer can scale up or down, remove, terminate, add, etc. virtual resources at the first private cloud 810 launched for a non-cross-cloud service).

[0185] FIG. 9 depicts exemplary virtual resources provisioned for a customer, according to some embodiments. The right hand side of FIG. 9 shows first resources of a first CSP (e.g., OCI resources) deployed for a customer. The left hand side of FIG. 9 shows second resources of a second CSP (e.g., to an AZURE datacenter) deployed for the customer. The first CSP (shown as CSP A) provide a cross-cloud service (e.g., an Exadata service) available via a cloud of the second CSP (shown as CSP B).

[0186] The first resources (which can be a combination of physical and virtual resources) can include a private cloud 920, a service tenancy 910, a VM cluster 926, and a gateway 928. The private cloud 920 (e.g., a VCN) can correspond to a tenancy of the customer with the first CSP and can include a primary subnet 922 having an IP range (e.g., Classless Inter-Domain Routing (CIDR) range) and, optionally, a backup subnet 924 using the same IP range. The service tenancy 910 can be available for multiple customers of the first CSP and can include a gateway 912. The VM cluster

926 includes a number of VMs (or compute instances) launched for the client to provide thereto the cross-cloud service. The VM cluster **926** can be hosted on physical resources within a child site. The gateway **928** (e.g., a service gateway) connects virtual resources of the primary subnet **922** (and, likewise of the backup subnet **924**) that are hosted on physical resources within a parent region with the VM cluster **926**. In comparison, the gateway **912** connects such virtual resources to a private cloud **960** of the customer with the second CSP. The gateway **912** can be implemented as a dynamic routing gateway (DRG) attached to the private cloud **920** in the customer tenancy. The connection can be via a gateway/router **914**. The gateway/router **914** can be provided, at least in part, in a child site and connect the child site to a parent region (thus, providing gateway functionalities) via a first connection protocol (e.g., FastConnect) and connect the child site to a substrate network of the second CSP (thus, providing routing functionalities) via a second connection protocol (e.g., MeetMe). For example, the gateway/router **914** can represent a FastConnect Virtual Circuit resource in the service tenancy **910** to connect the DRG to an AZURE MeetMe router (an example of a router **968**).

[0187] The second resources (which can be a combination of physical and virtual resources) can include the private cloud **960** and the router **968**. The private cloud **960** can include a primary subnet **922** having an IP range and, optionally, a backup subnet **964** using the same IP range. The IP range can be the same as the one of the primary subnet **922**. The router **968** can be connected to the gateway/router **914** such that to provide a data connection from the primary subnet **962** (e.g., and the backup subnet **964**) to the private cloud **920**. The primary subnet **962** can include NICs **966A** through **966K** that are connected to the router **968**.

[0188] The IP addresses used for the different virtual resources to provide the cross-cloud service can be selected by the first CSP (e.g., by an inter or intra-cloud service thereof) from the IP range. The first CSP (e.g., The inter or intra-cloud service thereof) can create domain name system (DNS) corresponding to the IP addresses and can be provided to the private cloud **960**. Such DNS record can be used to set up a private DNS zone for the customer, where this private DNS zone can be used with the private cloud **960**.

[0189] To illustrate, the subnets **922**, **924**, **962**, **964** use IP addresses from 10.0.10.0/24. Each NIC **966** has an IP address from this range (e.g., NICA **966A** uses 10.0.10.10, NIC K **966K** uses 10.0.10.11). The VMs in the VM clusters use the same IP addresses (e.g., a first VM uses 10.0.10.10, whereas a second VM uses 10.0.10.11).

[0190] FIG. 10 depicts an exemplary architecture for provisioning and managing a cross-cloud service based on an infrastructure distributed between multiple CSPs, according to some embodiments. In FIG. 10, a customer of a second CSP (shown as a CSP B) would manage the lifecycle of a cross-cloud service developed by a first CSP (shown as CSP A). In the interest of clarity and for illustrative purposes, OCI and AZURE are described as examples of CSPs, and Exadata service is described as an example of the cross-cloud service. However, the embodiments are not limited as such and, instead, similarly apply to other CSPs and/or cross-cloud services.

[0191] A customer (e.g., an AZURE customer) can operate a customer device **1000** to create and manage a set of virtual resources for the cross-cloud service (e.g., an Oracle DB Infrastructure resources) via a portal of the second CSP

(shown as a CSP_B portal **1052**). In an example, this portal **1052** interacts with one or more services of the second CSP (shown as CSP_B services **1054**, examples thereof can include an AZURE resource manager (ARM) which exposes APIs to manage the lifecycle of Oracle DB Infrastructure resources (DB Infrastructure resources include Exadata Infrastructure and Cloud VM Cluster); other examples include AZURE RPaaS to simplify development and operation of an AZURE resource provider, such as asynchronous operations and their progress updates). To manage resources built on top of set of virtual resources (DB Home, Database, Pluggable Database), the CSP_B services **1054** interact with an inter-cloud service **1012** of the first CSP such that the customer is redirected to this service **1012** (e.g., Oracle Resource Provider (ORP) in the case of OCI).

[0192] The inter-cloud service **1012** is configured to handle translation of identifiers assigned by the second CSP (e.g., AZURE identifiers) to identifiers assigned by the first CSP (e.g., OCI identifiers) including identities, resource IDs, and subscription IDs, translation of second CSP states (e.g., AZURE states) to first CSP states (e.g., OCI states), and vice versa, and to delegate the request to intra-cloud services **1014** of the first CSP to execute (e.g., to a resource control plane **1016**, such as an OCI DBaaS control plane). The inter-cloud service **1012** is also configured to coordinate any second CSP specific integrations with other second CSP services (e.g., with AZURE network Resource Provider). The inter-cloud service **1012** can also be configured to perform or cause other intra-cloud services **1014** to perform operations including linking cloud accounts, publishing observability information and vending tokens to access other cloud customer environments.

[0193] The intra-cloud services **1014** can provide a portal of the first CSP (shown as CSP_A portal **1012**) accessible to the customer device **1000** of the customer to manage other services of the customer with the first CSP. The two portal **1052** and **1012** can enable similar functionalities (e.g., by presenting inputs and outputs fields) and yet be different. For example, the CSP_B portal **1052** (e.g., an AZURE portal) can have a presentation format controlled by the second CSP. Additionally, the CSP_B portal **1052** can enable functionalities specific to the second CSP and unrelated to the first CSP, in addition to the functionalities related to the cross-cloud service. In comparison, the CSP_A portal **1012** (e.g., an OCI portal) can have a presentation format controlled by the first CSP. Additionally, the CSP_A portal **1012** can enable functionalities specific to the first CSP and unrelated to the second CSP, in addition to the functionalities related to the cross-cloud service. As far as the cross-cloud service, the CSP_B portal **1052** can expose information available from the second CSP, where this information can be provided by the inter-cloud service **1012**.

[0194] In an example of an OCI and AZURE use case, the inter-cloud service **1012** exposes Oracle DB Products. The inter-cloud service **1012** can be registered with ARM through RPaaS and is configured implement the AZURE Resource Provider Contract (RPC), which is a set of operations that all AZURE Resource Providers support. The AZURE Resource Provider as a Service (RPaaS) internal service is used to support the bulk of RPC operations.

[0195] Generally, the inter-cloud service **1012** can be configured to have a second CSP identity (e.g., an AZURE RP identity equivalent to an OCI service principal), as well as the ability to operate on a second CSP customer envi-

ronment using the second CSP flows (e.g., AZURE OBO flows). The inter-cloud service **1012** can also be configured to persist second CSP-specific metadata for a first CSP resource, such as the AZURE identifier to OCID mapping for DB resources. The inter-cloud service **1012** can also be configured have a first CSP identity to obtain a scope to operate on the first CSP customer environment. The inter-cloud service **1012** can also be configured to act as a thin adaptor layer, accepting second CSP formatted requests that have already been authenticated by the services **1054**, translating them to a first CSP request and delegating the request to the intra-cloud services.

[0196] As such, the inter-cloud service **1012** performs multiple operations. These operations include translating identifiers from one cloud to another cloud, and vice versa. The operations also include obtaining a first CSP identity for incoming requests from the services **1054** to use in calling the intra-cloud services **1014**. These operations also include translating second CSP format requests into first CSP format requests and call the intra-cloud services **1014**. The operations also include limiting/quota/capacity validation pass-through or conciliation, implicitly creating first CSP prerequisites for network connectivity for network connected resources (e.g., creating OCI DRG, VCN and subnet that a VMCluster is attached to), causing the linking of private clouds, and configuring DNS entries with the second CSP for resources that have a DNS record associated with them.

[0197] In an example, the inter-cloud service **1012** can be hosted in one or more child sites. The inter-cloud service **1012** can cause a virtual resource **1016** to be hosted in a child site. In an Exadata service use case, the virtual resource **1016** can include a database as a service (DBaaS) data plane via a DBaaS control plane. This control plane is hosted in a parent region, as part of the intra-cloud services **1014**. In this use case, a compute instance (e.g., a VM) can be instantiated for the customer on a private cloud of the customer with the second CSP (e.g., a VNET in AZURE). The compute instance can perform database operations by placing calls to the DBaaS data plane. Such operations include queries, storage, etc. or any operation that the Exadata service supports. The calls and responses thereto can be internal to the private cloud with the second CSP. Upon usage, the DBaaS data plane can report usage information (e.g., for metrics analysis, billing, etc.) to one or more of the intra-cloud services **1014** (e.g., to an observability service). This usage information can then be provided to a monitoring service of the second CSP.

[0198] FIG. 11 depicts an exemplary user experience flow **1100** to provision resources, according to some embodiments. In an example, the user experience flow **1100** involves a customer device **1100**, a CSPI_B **1150** of a second CSP, and an inter-cloud service of a first CSP. The first CSP can offer a cross-cloud service via the second CSP. A virtual resource can be provisioned to provide at least a part of the cross-cloud service, where the provisioning follows the user experience flow **1100**.

[0199] User input is received by the CSPI_B **1150** (e.g., by a portal, similar to the CSP_B portal **1052** of FIG. 10) from the customer device **1100** to create a subnet and to mark it as being delegated such that it can be used for the cross-cloud service. The subnet and the customer's second private cloud with the second CSP are created, and an indication of this creation is provided to the customer device **1100**.

[0200] Next, user input is received by inter-cloud service **1110** from the computing device **1100** (where this input can be at the portal) and indicates a request to create a cross-cloud service infrastructure (e.g., an Exadata service). The user input can indicate various parameters for the provisioning, such as a region and an availability zone. The inter-cloud service **1110** can indicate a start of the provisioning to the computing device **1100** (e.g., via the portal) and can execute a provisioning workflow by which the cross-cloud service infrastructure is created in the relevant child site. While this infrastructure is being created, the inter-cloud service **1110** sets its state to "provisioning." Once created, the inter-cloud service **1110** updates the state to "succeeded" and provides. Indications of these states can be provided to the computing device **1100** (e.g., via the portal). Similarly, computing device **1100** can check the current state via the portal, where a state querying request can be made from the CSPI_B **1150** to the inter-cloud service **1110** and the inter-cloud service **1110** would respond back with the state information.

[0201] Thereafter, user input is received by the inter-cloud service **1110** from the CSPI_B **1150** (where this input is provided at the portal) and corresponds to a request from the computing device **1100** to create a VM cluster resource. The request can indicate a subnet. The inter-cloud service **1110** then executes a set of checks and starts provisioning the VM cluster. This provisioning is further described in the next figure. The inter-cloud service **1110** sets the state to "provisioning." Once created, the inter-cloud service **1110** updates the state to "succeeded." Here also, indications of these states can be provided to the computing device **1100** (e.g., via the portal). Similarly, computing device **1100** can check the current state via the portal, where a state querying request can be made from the CSPI_B **1150** to the inter-cloud service **1110** and the inter-cloud service **1110** would respond back with the state information.

[0202] FIG. 12 depicts an exemplary control plane provisioning flow **1200**, according to some embodiments. In an example, the control plane provisioning flow **1200** involves an inter-cloud service **1200**, a control plane **1202** (e.g., an example of an intra-cloud service), a private cloud **1204** of a customer with a first CSP, a gateway **1206** (e.g., a DRG), a connection module **1208** (e.g., one provided by a connectivity fabric and supporting a connection protocol, such as FastConnect), a cross-cloud service infrastructure **1210** with the first CSP, and a CSPI_B **1250** of a second CSP. The first CSP can offer a cross-cloud service via the second CSP. Here, the control plane **1202** can execute part of the control plane provisioning flow **1200** upon being triggered by the inter-cloud service **1200** to provision VM cluster resources. The inter-cloud service **1200** can make such a call to the control plane **1202** upon receiving input from the CSPI_B **1250** (from ARM in AZURE) in response to a user input a portal of the second CSP.

[0203] As described herein above, the inter-cloud service **1220** receives a request to create a VM cluster for the cross-cloud service (e.g., an Exadata VM cluster) and passes this request with other information (e.g., first identifiers assigned by the first CSP and mapped to second identifiers assigned by the second CSP) to the control plane **1202**. In turn, the control plane **1202** retrieves details from CSPI_B **1250** (e.g., from ARM in AZURE) about a subnet, including the CIDR, for example and verifies that pre-requisites (e.g.,

the subnet has been provisioned, the region of a second private cloud with the second CSP (e.g., the VNET in AZURE), etc.) are met. The control plane **1202** then creates a first private cloud with the first CSP (e.g., a VCN in OCI) in the customer tenancy (e.g., based on the first identifiers) and creates a subnet in the first private cloud. This subnet has the same CIDR as the subnet in the second private cloud.

[0204] Next, the control plane **1202** creates/configures the gateway **1206** (e.g., the DRG) in the service tenancy and request to attach the gateway **1206** to the first private cloud in the customer's tenancy. The control plane **1202** also configures the routing information in the gateway **1206** (e.g., by creating a DRG routing table for traffic between the two private clouds or MeetMeRouter). Similarly, the control plane **1202** also configures the routing information in the first private cloud (e.g., by creating a VCN routing table for traffic to the DRG and the VM Cluster). Once the routing information has been created, the control plane **1202** can inform inter-cloud service **1200** about the success. In response, the inter-cloud service **1200** can configure a network security group for the first private cloud and request a VM cluster to be created. The VM cluster is then provisioned in the relevant child site, whereby IP addresses from the CIDR are assigned to it. DNS records corresponding to the IP addresses are also generated. The inter-cloud service **1200** can receive such DNS records and request attachment of the first private network to the second private network (e.g., the VCN to the VNET).

[0205] The control plane **1202** then calls an API of the second CSP, where this API allows IP address injection into the customer subnet in a form of a NIC. Each IP address can represent a VM Cluster node IP. The control plane **1202** can register all IP addresses that the VM cluster will have. The API call can include a device identifier (e.g., GUID of the MeetMe router). This identifier can be based on a mapping of GUIDs mapped to physical locations (child sites). The call can also include a subnet resource identifier (e.g., the private network's subnet in the customer tenancy), an IP address (e.g., the IP address of the NIC that is being created, where this address is in the subnet range), and a resource name (the name of the NIC that will be visible to the user). The API can return a virtual local area network identifier (VLAN ID), which is usable during a virtual circuit creation. The control plane **1202** then creates a virtual circuit that uses a connection (e.g., FastConnect connection) to a router of the CSPI_B **1250** (e.g., to a MeetMe router) and provides the DNS records such that a private DNS zone can be added to the customer's second private network. At this point, the VM cluster is created successfully. Compute instances for the cross-cloud service (e.g., Exadata database instances) can then be hosted in the VM cluster.

Examples of Cloud Infrastructure

[0206] As noted above, infrastructure as a service (IaaS) is one particular type of cloud computing. IaaS can be configured to provide virtualized set of computing resources over a public network (e.g., the Internet). In an IaaS model, a cloud computing provider can host the infrastructure components (e.g., servers, storage devices, network nodes (e.g., hardware), deployment software, platform virtualization (e.g., a hypervisor layer), or the like). In some cases, an IaaS provider may also supply a variety of services to accompany those infrastructure components (e.g., billing, monitoring, logging, security, load balancing and clustering,

etc.). Thus, as these services may be policy-driven, IaaS users may be able to implement policies to drive load balancing to maintain application availability and performance.

[0207] In some instances, IaaS customers may access resources and services through a wide area network (WAN), such as the Internet, and can use the cloud provider's services to install the remaining elements of an application stack. For example, the user can log in to the IaaS platform to create virtual machines (VMs), install operating systems (OSs) on each VM, deploy middleware such as databases, create storage buckets for workloads and backups, and even install enterprise software into that VM. Customers can then use the provider's services to perform various functions, including balancing network traffic, troubleshooting application issues, monitoring performance, managing disaster recovery, etc.

[0208] In most cases, a cloud computing model will require the participation of a cloud provider. The cloud provider may, but need not be, a third-party service that specializes in providing (e.g., offering, renting, selling) IaaS. An entity might also opt to deploy a private cloud, becoming its own provider of infrastructure services.

[0209] In some examples, IaaS deployment is the process of putting a new application, or a new version of an application, onto a prepared application server or the like. It may also include the process of preparing the server (e.g., installing libraries, daemons, etc.). This is often managed by the cloud provider, below the hypervisor layer (e.g., the servers, storage, network hardware, and virtualization). Thus, the customer may be responsible for handling (OS), middleware, and/or application deployment (e.g., on self-service virtual machines (e.g., that can be spun up on demand) or the like).

[0210] In some examples, IaaS provisioning may refer to acquiring computers or virtual hosts for use, and even installing needed libraries or services on them. In most cases, deployment does not include provisioning, and the provisioning may need to be performed first.

[0211] In some cases, there are two different problems for IaaS provisioning. First, there is the initial challenge of provisioning the initial set of infrastructure before anything is running. Second, there is the challenge of evolving the existing infrastructure (e.g., adding new services, changing services, removing services, etc.) once everything has been provisioned. In some cases, these two challenges may be addressed by enabling the configuration of the infrastructure to be defined declaratively. In other words, the infrastructure (e.g., what components are needed and how they interact) can be defined by one or more configuration files. Thus, the overall topology of the infrastructure (e.g., what resources depend on which, and how they each work together) can be described declaratively. In some instances, once the topology is defined, a workflow can be generated that creates and/or manages the different components described in the configuration files.

[0212] In some examples, an infrastructure may have many interconnected elements. For example, there may be one or more virtual private clouds (VPCs) (e.g., a potentially on-demand pool of configurable and/or shared set of computing resources), also known as a core network. In some examples, there may also be one or more security group rules provisioned to define how the security of the network will be set up and one or more virtual machines (VMs).

Other infrastructure elements may also be provisioned, such as a load balancer, a database, or the like. As more and more infrastructure elements are desired and/or added, the infrastructure may incrementally evolve.

[0213] In some instances, continuous deployment techniques may be employed to enable deployment of infrastructure code across various virtual computing environments. Additionally, the described techniques can enable infrastructure management within these environments. In some examples, service teams can write code that is desired to be deployed to one or more, but often many, different production environments (e.g., across various different geographic locations, sometimes spanning the entire world). However, in some examples, the infrastructure on which the code will be deployed must first be set up. In some instances, the provisioning can be done manually, a provisioning tool may be utilized to provision the resources, and/or deployment tools may be utilized to deploy the code once the infrastructure is provisioned.

[0214] Techniques for reclaiming resources that have been provisioned but are no longer in use (e.g., terminated resources) is critical for optimizing performance and efficiency of a system. By freeing up memory, processing power, and network bandwidth, resource reclaiming increases availability of memory, processing power, and network bandwidth. Resource reclaiming can also improve energy usage because less resources may sit idle. Additionally, resource reclaiming can improve security of systems by controlling which resources are available to certain services, subscriptions, and/or user accounts. By reducing resources being available to fewer services, subscriptions, and accounts, unauthorized use of the resources can be minimized. Further, resource reclaiming can increase the likelihood that resources are available and accessible for high-priority tasks, thereby improving system responsiveness and throughput.

[0215] Additionally, reclaiming unused resources reduces the risk of system slowdowns or crashes caused by resource exhaustion, enhancing overall reliability and stability. It also allows for better scalability, enabling systems to handle increased workloads without requiring significant upgrades or over-provisioning. From a networking perspective, freeing up bandwidth ensures smoother data flow and minimizes latency, particularly in environments where real-time communication or large-scale data transfers are critical. FIGS. 13-19 include further description of techniques for managing resources throughout their lifecycle to lead to benefits like those just described.

[0216] FIG. 13 depicts an example of an architecture 1300 that includes a resource management system 1352 for cross-cloud services between multiple cloud environments, according to some embodiments. Architecture 1300 can include a first cloud environment 1302 of a first CSP (e.g., Oracle's OCI) and a second cloud environment 1318 of a second CSP (e.g., Microsoft's Azure). The first cloud environment 1302 and the second cloud environment 1318 can be implemented according to the distributed environment described with respect to FIGS. 6-9. The first cloud environment 1302 and the second cloud environment 1318 can include one or more private clouds (e.g., a VCN in the case of Oracle's OCI and a VNET in the case of Microsoft's Azure). Additionally, a cross-cloud service between the first cloud environment 1302 and the second cloud environment

1318 can be provisioned according to the experience and provisioning flows described with respect to FIGS. 10-12.

[0217] The first cloud environment 1302 may include a console 1314, a multi-cloud platform 1312, analytic services 1316, a resource provider 1304, an intra-cloud service 1310, a network control plane 1308, and/or a service control plane 1306. The second cloud environment 1318 may include a portal 1320 that includes a blade 1322, a resource provider as a service (RPaaS) 1328 that may include a configuration 1330, a resource manager 1324 that may include a configuration 1326, a network provider 1334, a service data plane 1332, and/or a resource usage processing system 1356.

[0218] The first cloud environment 1302 can be configured to receive requests for cross-cloud services, evaluate permission statuses for such requests, generate instructions for provisioning such services in one or more other cloud environments such as the second cloud environment 1318, deploy such services in the one or more other cloud environments, and manage the deployed services. In some implementations, the intra-cloud services 1310 of the first cloud environment 1302 can be configured to provide one or more of the intra-cloud services 1310 (e.g., an Exadata intra-cloud service) as one or more cross-cloud services (e.g., an Exadata cross-cloud service) to customers having a tenancy in the second cloud environment 1318.

[0219] Provisioning an intra-cloud service offered by the first cloud environment 1302 as a cross-cloud service between the first cloud environment 1302 and the second cloud environment 1318 can at least be facilitated by the resource provider 1304 of the first cloud environment 1302. To provision a cross-cloud service, the resource provider 1304 can be configured to send a provisioning request for the requested cross-cloud service to the service control plane 1306 and the network control plane 1308 of the first cloud environment 1302 and, in response, the service control plane 1306 and the network control plane 1308 can deploy an intra-cloud service of the intra-cloud services 1310 as a cross-cloud service to the second cloud environment 1318 (e.g., to the service data plane 1332 and network provider 1334 of the second cloud environment 1318). The service (e.g., the cross-cloud service) provisioned may be associated with a service identifier. The service identifier may identify the service. Each instance of the same service (e.g., a database management service, VM service) may be associated with the same service identifier. Other services may be associated with other service identifiers. Once the service is provisioned, the service may be associated with a service instance identifier. The service instance identifier may uniquely identify the instance of the service from other instances of the same service provisioned to the second cloud environment and/or a user account (e.g., a first user account associated with the first cloud environment 1302 or a second user account associated with the second cloud environment 1318).

[0220] In some implementations, the first cloud environment 1302 can include multiple parent regions and the second cloud environment 1318 can include multiple child sites corresponding to the multiple parent regions. In some implementations, the service data plane 1332 of the second cloud environment 1318 can serve as and/or form one or more child sites of a respective parent region. In some implementations, each respective parent region of the first cloud environment 1302 can include a resource provider such as the resource provider 1304 that together with one or

more child sites associated with the respective parent region facilitates the provisioning and lifecycle management of one or more cross-cloud services. The one or more cross-cloud services can include one or more of the intra-cloud services **1310** of the first cloud environment **1302**. For example, the resource provider **1304** can be a resource provider for a parent region in the first cloud environment **1302** and can provision and manage the lifecycle of an intra-cloud service of the intra-cloud services **1310** as a cross-cloud service between the service control plane **1306** of the first cloud environment **1302** and the service data plane **1332** of the second cloud environment **1318**. Child sites can at least be implemented according to the physical architecture described above with respect to FIG. 7.

[0221] In some implementations, provisioning of cross-cloud services between the first cloud environment **1302** and the second cloud environment **1318** can be facilitated by the resource provider **1304** of the first cloud environment **1302** and the resource manager **1324** of the second cloud environment **1318**. For example, the resource manager **1324** of the second cloud environment **1318** may send a cross-cloud service provisioning request to the resource provider **1304** of the first cloud environment **1302** and, in response, the resource provider **1304** can facilitate the provisioning of the cross-cloud service together with the second cloud environment **1318**. The resource provider **1304** may manage the lifecycle of provisioned cross-cloud services and the resources used by the services. For example, the resource provider **1304** may maintain a mapping of relationships between a tenancy, a subscription, a service, and/or a resource. The mapping may be used to determine which resources are associated with specific services, tenancies, and/or subscriptions. Other relationships between resources, services, tenancies, and/or subscriptions may be represented by the mapping (e.g., described below with respect to FIGS. **15** and **15**). The resource provider **1304** may be used to provision, activate, suspend, and/or terminate a service and/or a resource. Provisioning a resource can include establishing a mapping between (e.g., associating) a resource with a service, a subscription, and/or a tenancy. Activating a resource can include enabling a resource and/or service to be accessible to a user account. The resource may be associated with a configuration **1326** that was previously configured (e.g., before suspending the resource). A resource may be activated after it is provisioned. Suspending a resource may maintain the configuration **1326** of the resource and may cause the resource to be inaccessible and/or unavailable to a tenancy or account until the resource is activated. Terminating a resource may cause the resource to be unavailable and inaccessible to the tenancy, service, and/or user account. The terminated resource may have configurations **1326** of the resource removed from memory (e.g., also referred to as “reclaiming” the resource). The resource provider **1304** may dissociate the resource with the service, tenancy, and/or account after the resource is terminated.

[0222] Managing the lifecycle of provisioned cross-cloud services between the first cloud environment **1302** and the second cloud environment **1318** can be facilitated by the resource provider **1304** of the first cloud environment **1302** and the resource manager **1324** of the second cloud environment **1318**. For example, the resource manager **1324** of the second cloud environment **1318** may send a request for managing the lifecycle of a provisioned cross-cloud service

(e.g., a request to terminate, suspend, or activate the cross-cloud service) to the resource provider **1304** and, in response, the resource provider **1304** can facilitate the lifecycle management function for the cross-cloud service together with the second cloud environment **1318**. In some implementations, to facilitate the provisioning and lifecycle management, the resource manager **1324** can be configured to communicate with the resource provider **1304** (e.g., using APIs of the first cloud environment **1302** that are exposed by the resource provider **1304** to the resource manager **1324**).

[0223] In some implementations, a customer of the second cloud environment **1318** and/or the second CSP desiring to provision a cross-cloud service from within the second cloud environment **1318** and/or manage a cross-cloud service that has been provisioned in conjunction with the second cloud environment **1318** can initiate a request **1336** to do so. The customer can initiate the request **1336** via the portal **1320** of the second cloud environment **1318**. In some implementations, the portal **1320** can include one or more graphical user interfaces that can be accessed via a client device such as a computer (e.g., through an application, operating system, and/or software program executing on the client device). The one or more graphical user interfaces or portions thereof can be generated by, populated by, and/or otherwise supplied by the resource manager **1324** of the second cloud environment **1318**. Customers of the second cloud environment **1318** can access the portal **1320** to manipulate and/or interact with the one or more graphical user interfaces to initiate the request **1336** and perform other functions such as manage their tenancy within the second cloud environment **1318**.

[0224] In some implementations, the one or more graphical user interfaces or portions thereof can be generated by, populated by, and/or otherwise supplied by the resource provider **1304** of the first cloud environment **1302**. The one or more graphical user interfaces or portions thereof can be provided to and/or declared to the portal **1320** using a blade **1322** of the portal **1320**. The blade **1322** can serve as and/or function as an extension, plugin, add-on, and the like of the portal **1320**. Customers of both the first cloud environment **1302** and the second cloud environment **1318** can access the portal **1320** to manipulate and/or interact with the one or more graphical user interfaces to initiate the request **1336** and perform other functions such as manage their tenancies within the first cloud environment **1302** and the second cloud environment **1318**.

[0225] In some implementations, requests received via the portal **1320** to provision a cross-cloud service can be routed to the resource manager **1324** of the second cloud environment **1318** which in turn can route the request **1336** to the resource provider **1304** of the first cloud environment **1302** (e.g., via a first set of APIs of the first cloud environment **1302** that are exposed within the resource manager **1324**). On the other hand, requests received via the portal **1320** to manage the lifecycle of a provisioned cross-cloud service (e.g., viewing analytics, viewing consumption, viewing costs, viewing logs, provisioning a service, activating a service, suspending a service, terminating a service, etc.) can be routed to a console **1314** of the first cloud environment **1302** which in turn can route the request **1336** within the first cloud environment **1302** (e.g., via a second set of APIs that are exposed within first cloud environment **1302**). For example, the console **1314** can be configured to route the request **1336** to analytic services **1316** of the first cloud

environment **1302** for viewing analytics, consumption, costs, logs, and the like pertaining to the provisioned cross-cloud service. In an example, the console **1314** can be configured to route the request **1336** to resource provider **1304** of the first cloud environment **1302** for managing the lifecycle (e.g., provisioning, activating, suspending, terminating) of one or more resources. In some implementations, respective customers of the second cloud environment **1318** may be assigned respective identifiers (e.g., a first cloud environment account identifier, a second cloud environment account identifier) such that each request **1336** initiated by a respective customer of the second cloud environment **1318** can be associated with the identifier for that respective customer. In this way, access to the portal and request initiation can be controlled and managed by the resource manager **1324** based on roles and/or permissions associated with each customer identifier (e.g., a user account, a customer account).

[0226] To facilitate the provisioning of a cross-cloud service and lifecycle management of a provisioned cross-cloud service, the resource provider **1304** can be linked to the resource manager **1324** and a resource provider as a service (RPaaS) **1328** of the second cloud environment **1318**. Linking the resource provider **1304** to the second cloud environment **1318** enables the resource provider **1304**, resource manager **1324**, and RPaaS **1328** to coordinate resources and operations (e.g., provisioning, activating, suspending, terminating, etc.). The resource provider **1304** can have an identity that is associated with the second cloud environment **1318**. The identity can be configured to replicate an identity that the resource provider **1304** has with the first cloud environment **1302**. To facilitate the linking, the resource provider **1304** can provide a configuration **1326** to the resource manager **1324** and a configuration **1330** to the RPaaS **1328**. The configuration **1326** and the configuration **1330** can include an identifier for the resource provider **1304** and can define API specifications, connection endpoints, and/or locations of intra-cloud services associated with the resource provider **1304**. Configuration **1326** may include resource configurations such resource allocations, resource characteristics, and/or configurations made by an account to the resource after the resource was provisioned.

[0227] The resource provider **1304** can be configured to provision a cross-cloud service and/or manage of the lifecycle of a provisioned cross-cloud service based on operations performed by the multi-cloud platform **1312** of the first cloud environment **1302**. The multi-cloud platform **1312** can be configured to perform operations that are common to linking and integrating the first cloud environment **1302** to the second cloud environment **1318** and other cloud environments of other CSPs. For example, the multi-cloud platform **1312** can be configured to link the customer's second account for the second CSP to the customer's first account for the first CSP, publish observation information collected from the first cloud environment **1302** and/or the second cloud environment **1318**, generate vending tokens for accessing the customer's other cloud environments, and the like. Additionally, the multi-cloud platform **1312** can be configured to create, define, supply, and/or otherwise implement a contract between the resource provider **1304**, the resource manager **1324**, and the RPaaS **1328**. The contract can identify resources supported by and/or operations to be performed by the resource provider **1304**, the resource manager **1324**, and the RPaaS **1328**. In some implementa-

tions, the contract can allow the resource provider **1304**, the resource manager **1324**, and the RPaaS **1328** to operate on the same tenancy within the first cloud environment **1302**. For example, the contract can define provisioning and/or lifecycle management events that are to be sent from the resource manager **1324** and/or RPaaS **1328** upon occurrence of such events where the resource provider **1304** can be configured to perform provisioning and/or lifecycle management operations (e.g., suspending, terminating, activating) asynchronously based on the reception and/or a change in status of such events.

[0228] In certain embodiments, the events may include a state change to the state of a subscription. For example, a subscription associated with a service, tenancy, and/or resource may have a subscription state that indicates whether the subscription is an active subscription, a terminated subscription, or a suspended subscription. An active subscription state may indicate that the subscription is active and that services and resources associated with the subscription should be accessible and available to user accounts associated with the subscription. In certain embodiments, the subscription may have an active state when the subscription is paid for and has not been terminated by a user account. A terminated subscription state may indicate that the subscription is terminated and that services and resources associated with the subscription should not be accessible (e.g., resources should be reclaimed). In certain embodiments, a resource associated with a terminated subscription or service may be accessible to another subscription or service if the same resource is also associated with the active subscription or service, respectively. A suspended subscription state may indicate that the subscription is suspended and that services and resources associated with the subscription should be inaccessible to the user account. In certain embodiments, a subscription may be suspended for a period of time before the subscription is terminated (e.g., a suspended subscription state changes to a terminated subscription state after a period of time). A suspended subscription state may indicate that the subscription has not been paid for by the user account.

[0229] The multi-cloud platform **1312** can be further configured to perform generalized cloud management operations including, but not limited to: (i) mapping subscriptions and tenancies of the second cloud environment **1318** to the subscriptions and tenancies of the first cloud environment **1302**; (ii) generating and managing policy statements that govern tenancies in respective cloud environments (e.g., a statement that facilitates the multi-cloud platform **1312** and the resource provider **1304** to operate in the same tenancy); (iii) generating and managing access tokens that facilitate cross-cloud access and/or communication between respective cloud environments; and/or (iv) mapping observability information of the first cloud environment **1302** to the second cloud environment **1318** (e.g., writing an event to the second cloud environment **1318** when a backup is completed on the first cloud environment **1302**, writing resource logs of the first cloud environment **1302** to the second cloud environment **1318**, and the like). In this way, the first cloud environment **1302** and the second cloud environment **1318** can avoid overlapping operations and resources, which in turn can increase efficiency.

[0230] In some implementations, upon receiving the request **1336** to provision a cross-cloud service, the resource provider **1304** can be configured to map the request **1336** to

an identifier of the first cloud environment **1302** and pass the request along to the service control plane **1306**. The service control plane **1306** can be configured to provision and connect resources: provisioning the relevant resources of the first cloud environment **1302**; and connecting these resources to the customer's tenancy in the second cloud environment **1318** (e.g., the customer's VNET in the second cloud environment **1318**). Under the provisioning process, the service control plane **1306** creates a VCN for the customer in the first cloud environment **1302** and creates one or more subnets within the VCN. In some implementations, the VCN can function as a shadow tenancy for the customer's tenancy in the second cloud environment **1318**. One or more of the subnets within the VCN and one or more subnets in the VNET can use the same CIDR. The service control plane **1306** also creates a DRG in the first cloud environment **1302** and attaches the DRG to the customer's VCN and configures routing information for the DRG and the VCN (e.g., to interconnect these two resources). The service control plane **1306** also provisions a VM cluster in the service data plane **1332**. IP address(es) of this VM cluster are from the CIDR and are mapped to corresponding DNS records. Under the connecting process, the service control plane **1306** registers these IP address(es) with the second cloud environment **1318** and creates a virtual circuit between the DRG and the network provider **1334** and sends the DNS records such that a private DNS zone can be set up in second cloud environment **1318**.

[0231] Additionally, or alternatively, upon receiving the request **1336** to provision a cross-cloud service, the resource provider **1304** is configured to: (i) map an identifier assigned to the customer by the second CSP to an identifier assigned to the customer by the first CSP; (ii) obtain the identity of the resource provider **1304** that is associated with the request **1336** (i.e., the second cloud environment **1318** from which the request **1336** originates); (iii) translate the format of the request **1336** from that of the second cloud environment **1318** to that of the first cloud environment **1302** and route the formatted request along with the mapped identifier to the service control plane **1306** and network control plane **1308**; (iv) establish network connectivity prerequisites for network connectivity between the first cloud environment **1302** and the second cloud environment **1318** (e.g., establish DRG, VCN, and subnets for connecting to the network provider **1334** and VMs of the service data plane **1332** of the second cloud environment **1318**); (v) establish network connectivity between the network control plane **1308** and the network provider **1334** based on the network connectivity prerequisites (e.g., linking the delegated subnets of the second cloud environment **1318** to the VCN of the first cloud environment **1302** and configuring DNS entries in the second cloud environment **1318**).

[0232] Additionally, once the cross-cloud service is provisioned, the resource provider **1304** can be configured to persist metadata between the first cloud environment **1302** and the second cloud environment **1318** (e.g., the mapping between the identifier for the second cloud environment **1318** and the first cloud environment **1302**) and act as a thin adaptor layer that accepts requests that are formatted for the second cloud environment **1318** and that have already been authenticated by resource manager **1324** (e.g., requests to manage the lifecycle of provisioned cross-cloud services), translates them into requests that are formatted for the first

cloud environment **1302**, and delegates the translated requests to the service control plane **1306**.

[0233] In some implementations, upon receiving the request **1336** to terminate a cross-cloud service, the resource provider **1304** can be configured to map the request **1336** to an identifier of the first cloud environment **1302** and pass the request along to the service control plane **1306**. The service control plane **1306** can be configured to terminate resources: disconnecting these resources from the customer's tenancy in the second cloud environment **1318** (e.g., the customer's VNET in the second cloud environment **1318**), disassociating the resources from the customer's tenancy in the second cloud environment **1318**, and/or disassociating the resources from the second cloud environment **1318**. The service control plane may terminate (e.g., deprovision, reclaim) a VCN of customer in the first cloud environment **1302**. The service control plane **1306** may unregister IP addresses with the second cloud environment **1318**.

[0234] In some implementations, upon receiving the request **1336** to activate a cross-cloud service, the resource provider **1304** can be configured to map the request **1336** to an identifier of the first cloud environment **1302** and pass the request along to the service control plane **1306**. The service control plane **1306** may allow access to one or more services or resources. The service control plane **1306** may instruct the service data plane **1332** to allow access to one or more services or resources.

[0235] In some implementations, upon receiving the request **1336** to suspend a cross-cloud service, the resource provider **1304** can be configured to map the request **1336** to an identifier of the first cloud environment **1302** and pass the request along to the service control plane **1306**. The service control plane **1306** may prevent access to one or more services or resources, causing the resources to be inaccessible. The service control plane **1306** may instruct the service data plane **1332** to cause one or more services or resources to be inaccessible and/or unavailable to a tenancy, customer of the second cloud environment **1318**, and/or the first cloud environment **1302**.

[0236] FIG. 14 depicts an example of a resource management system **1450**, according to some embodiments. The resource management system **1450** may be part of the resource provider **1304** described above and may include a resource mapping system **1408** and an instruction system **1412**. The resource management system **1450** can cause a resource to be available, unavailable, accessible, or inaccessible to one or more tenancies, cloud environments, user accounts, and/or services.

[0237] The resource mapping system **1408** may maintain a data structure (e.g., a searchable table, a hash table, a mapping, an array, etc.). The data structure may store resource IDs, tenancy IDs, subscription IDs, and/or subscription states. The resource mapping system **1408** may store entries in the data structure. An entry may represent which tenancy, subscription, and/or subscription state are associated with/mapped to a resource ID. The mapping between the resource IDs, tenancy IDs, subscription IDs, and/or subscription states can be used to determine one or more resource IDs using at least a tenancy ID and/or a subscription ID. The mapping between the resource IDs, tenancy IDs, subscription IDs, and/or subscription states can be used to determine one or more subscription IDs and/or tenancy IDs using at least the resource ID.

[0238] The resource mapping system 1408 may add entries to the association mapping maintained by the data structure. In certain embodiments, when a resource is provisioned, the resource mapping system 1408 may receive the resource ID 1416, the tenancy ID 1402, the subscription ID 1404, and/or the subscription state 1406 and insert an entry into the data structure representing the association between the resource ID 1416, the tenancy ID 1402, the subscription ID 1404, and/or the subscription state 1406. The subscription state 1406 may be received by the resource management system 1450 after a change to the subscription state 1406 has been detected. A change to the subscription state 1406 may be detected responsive to the subscription state 1406 changing or may be periodically checked and therefore detected during a period check. The tenancy ID 1402, subscription ID 1404, and/or the subscription state 1406 may be received from a billing system, a console (e.g., console 1314 described above), a service control plane (e.g., service control plane 1306 described above), and/or a multi-cloud platform (e.g., multi-cloud platform 1312 described above).

[0239] In certain embodiments, when a resource is provisioned, the resource mapping system 1408 may receive the tenancy ID 1402, the subscription ID 1404, and/or the subscription state 1406, generate a resource ID 1416 for the resource being provisioned to the tenancy and insert an entry into the data structure representing the association between the resource ID 1416, the tenancy ID 1402, the subscription ID 1404, and/or the subscription state 1406. In certain embodiments, the resource mapping system 1408 receives a set of one or more resource IDs to be associated with the combination of the tenancy ID 1402, the subscription ID 1404, and/or the subscription state 1406. After adding an entry, the resource ID 1416, and any other resource IDs added, may be output from the resource mapping system 1408 as output resource ID 1410. The resource mapping system 1408 may also output an indication that a provision occurred (e.g., the subscription state 1406). The output resource ID 1410 and/or indication that the provision occurred may be transmitted to the instruction system 1412 from the resource mapping system 1408.

[0240] In certain embodiments, the resource ID 1416 may be received by the resource mapping system without the resource mapping system receiving the subscription ID 1404. In such embodiments, a default subscription ID may be associated with the resource ID 1416.

[0241] The resource mapping system 1408 may modify entries of the mapping, thereby modifying the associations represented by the entries. For example, the resource mapping system 1408 may maintain an entry representing the association of the resource ID 1416, the tenancy ID 1402, the subscription ID 1404, and the subscription state 1406. The resource mapping system 1408 may modify the entry to change the subscription state based on receiving the subscription state 1406, the subscription ID 1404, the tenancy ID 1402, and the resource ID 1416. The subscription state may change when an event related to the subscription associated with the tenancy occurs. Modifying the entry can also include modifying a tenancy ID field with the tenancy ID 1402, a subscription ID field to include the subscription ID 1404, and/or a resource ID field to include the subscription ID 1404. Each of the tenancy ID field, the subscription ID field, and the resource ID field may include any number of respective IDs. After modifying the entry, the output resource ID 1410 associated with any modified fields, may

be output from the resource mapping system 1408. The resource mapping system 1408 may also output an indication that a modification occurred (e.g., the subscription state 1406). The output resource ID 1410 and/or indication that the modification occurred may be transmitted to the instruction system 1412 from the resource mapping system 1408.

[0242] The resource mapping system 1408 may remove entries from the mapping. The entries may be removed as part of a resource reclaiming process. Removing entries may cause one or more resource IDs to be disassociated from any number of tenancy IDs, subscription IDs, and/or subscription states. After removing the entry, the resource ID associated with any removed entry may be output as output resource ID 1410 from the resource mapping system 1408. The resource mapping system 1408 may also output an indication that a removal occurred (e.g., the subscription state 1406). The output resource ID 1410 and/or indication that the removal occurred may be transmitted to the instruction system 1412 from the resource mapping system 1408. In certain embodiments, the resource mapping system may output any combination of the output resource ID 1410, the tenancy ID 1402, the subscription ID 1404, and/or the subscription state 1406 based on the input received by the resource mapping system 1408 to determine relationships.

[0243] The instruction system 1412 may generate a resource management instruction 1414 based on the received output resource ID 1410 (or multiple output resource IDs 1410). The resource management instruction 1414 may be generated based at least in part on the indication received, the tenancy ID 1402 associated with the output resource ID 1410, and/or the subscription ID 1404 associated with the output resource ID (1410). The resource management instruction 1414 may be transmitted to a service control plane (e.g., service control plane 1306 described above with respect to FIG. 13) and be used to manage the lifecycle of the resource associated with the output resource ID 1410.

[0244] In certain embodiments, the resource management system maintains the mappings between the resource ID 1416, the tenancy ID 1402, the subscription ID 1404, and/or the subscription state 1406. Additionally, or alternatively, a control plane (e.g., the service control plane 1306), certain services, or another system component may maintain the mappings.

[0245] FIG. 15 depicts an example of a relationship mapping 1500 between a tenancy 1502, subscriptions (e.g., first subscription 1506 and second subscription 1512), and resources (e.g., first resource 1518, second resource 1526, third resource 1530, and fourth resource 1522), according to some embodiments. The relationship mapping 1500 may be stored and maintained by a data structure, as described above.

[0246] Tenancy 1502 may be associated with and identifier using a tenancy ID 1504 (e.g., tenancy ID 1402 described above). The tenancy ID 1504 may be used to identify the tenancy from other tenancies. The tenancy ID 1504 may be associated with any number of subscriptions. The tenancy ID 1504 can be associated with a subscription ID (e.g., subscription ID 1404 described above) that is associated with and identifies the subscription. In the illustrated example, the tenancy ID 1504 is associated with the first subscription ID 1510 and the second subscription ID 1516.

[0247] A subscription state may be associated with a subscription and/or a subscription ID. In the illustrated example, the first subscription **1506** is associated with first subscription state **1508** and the first subscription ID **1510**. Additionally, in the illustrated example, the second subscription **1512** is associated with the second subscription state **1514** and the second subscription ID **1516**. The subscription state may be like the subscription state **1406** described above. The first subscription state **1508** may include a subscription state representing that the first subscription **1506** is activated, suspended, or terminated. The second subscription state **1514** may include a subscription state representing that the second subscription **1512** is activated, suspended, or terminated.

[0248] A subscription ID may be associated with a resource identifier. For example, the first resource ID **1520** may be associated with the first subscription ID **1510**. The association can indicate that the first resource **1518** should be accessible and available when the first subscription **1506** is in an active state. The relationship mapping **1500** further illustrates that a resource may be dependent on another resource. For example, the fourth resource **1522** may depend on the first resource **1518**.

[0249] In certain embodiments, multiple subscription IDs are associated with a common resource ID. For example, the first resource ID **1520** and the second resource ID **1528** may be associated with the first subscription ID **1510**. In such embodiments, the second resource **1526** may remain accessible and available as long as at least one of the first subscription **1506** and the second subscription **1512** remain active, as indicated by the first subscription state **1508** and the second subscription state **1514**, respectively. The second resource **1526** may remain accessible to the tenancy **1502** in connection with processes performed in association with the active subscription and not be accessible to the tenancy **1502** in connection with processes performed associated with a suspended or terminated subscription.

[0250] In certain embodiments, the subscription ID is associated with a service ID and the service ID is associated with one or more resource IDs. For simplicity of illustration, the illustrated example shows a set number of tenancies, subscriptions, and resources. It should be appreciated that the relationship mapping **1500** may include zero or more tenancies associated with zero or more subscriptions associated with zero or more services associated with zero or more resources.

[0251] Some examples are provided below that illustrate how subscription states and the mapped relationships can indicate that one or more resource should be accessible to a user account and/or unavailable.

[0252] In a first example, the first subscription state **1508** may indicate that the first subscription **1506** is active. As a result, the first resource **1518** and the fourth resource **1522** may be accessible by a user account of the tenancy **1502**. Continuing the first example, the first subscription state **1508** may be changed to suspended. As a result of the first subscription state **1508** changing to a suspended state, the first resource **1518** and the fourth resource **1522** may be inaccessible to the tenancy **1502**. The first resource **1518** and the fourth resource **1522** may remain configured (e.g., according to resource specific configurations). If the first subscription state **1508** is changed back to an active state before the first subscription state **1508** is changed to a terminated state, the configurations of the first resource **1518**

and the fourth resource **1522** may remain and the first resource **1518** and the fourth resource **1522** may become accessible to the tenancy **1502** again. In certain embodiments, if the fourth resource **1522** is dependent on first resource **1518**, the first resource may not be suspended or terminated without the fourth resource **1522** first being suspended or terminated (e.g., to ensure that resources that are available and accessible can be used. The first resource **1518** may be accessible in association with the services associated with the fourth resource **1522** and not by the services not associated with the first resource **1518**.

[0253] In second example, the first subscription state **1508** may indicate that the first subscription **1506** is active. As a result, the first resource **1518** and the fourth resource **1522** may be accessible by a user account of the tenancy **1502**. Continuing the second example, the first subscription state **1508** may be changed to suspended. As a result of the first subscription state **1508** changing to a suspended state, the first resource **1518** and the fourth resource **1522** may be inaccessible to the tenancy **1502**. The first resource **1518** and the fourth resource **1522** may remain configured (e.g., according to resource specific configurations). If the first subscription state **1508** is changed to a terminated state, the resource configurations may be removed and the first resource **1518** and the fourth resource **1522** may be deprovisioned from the first subscription **1506** and/or the tenancy **1502**. The first resource **1518** and the fourth resource **1522** may remain unavailable and therefore inaccessible until subsequent provisioning of the first resource **1518** and the fourth resource **1522** occurs.

[0254] In third example, the first subscription state **1508** may indicate that the first subscription **1506** is active and the second subscription state **1514** may indicate that the second subscription **1512** is terminated or suspended. As a result, the second resource **1526** that is associated with the second subscription **15102** may be made accessible to processes associated with the second subscription **1512** while remaining inaccessible to processes associated with the first subscription **1506**. In such an example, the second resource **1526** remains accessible to a subset of the subscriptions associated with the tenancy **1502**. The second resource **1526** is not suspended or deprovisioned because the first subscription **1506** should still have the second resource **1526** accessible. The third resource **1530** may be deprovisioned or suspended based on the second subscription state **1514** because no other subscriptions are associated with the third resource **1530**.

[0255] FIG. 16 depicts an example flow **1600** for suspending a resource, according to some embodiments. The resource may be included in a set of resources being suspended. The example flow **1600** may be carried out using an architecture like architecture **1300** described above that includes a resource provider **1304**, a resource management system **1450** that includes a resource mapping system **14058** and a instruction system, **1412**, and a service control plane **1306**.

[0256] At step **S1602**, the resource provider **1304** may determine that an event associated a subscription ID occurred. For example, an indication from a billing system may be received by the resource provider **1304** that indicates the subscription identified by the subscription ID was not paid for and that the subscription should be suspended. In certain embodiments, the resource provider **1304** may receive an indication from a console (e.g., console **1314**

described above) that indicates the subscription should be suspended. For example, a user account associated with the subscription ID may have used a user interface to input the indication the subscription associated with the subscription ID should be suspended.

[0257] At step S1604, the resource provider 1304 may transmit a subscription ID (e.g., subscription state 1406, described above) associated with the subscription, the subscription state (e.g., subscription state 1406, described above), and/or a tenancy ID (e.g., tenancy ID 1402 described above) to the resource mapping system 1408.

[0258] At step S1606, the resource mapping system 1408 may use the tenancy ID, the subscription ID, and/or the subscription state to determine a set of one or more resource IDs (e.g., output resource ID 1410) that are associated with the combination of information received by the resource mapping system 1418 at step S1604.

[0259] At step S1608, the resource ID may be transmitted to the instruction system 1412 from the resource mapping system 1408. An indication of the subscription state may also be transmitted to the instruction system 1412 from the resource mapping system 1408.

[0260] At step S1610, the instruction system 1412 may generate an instruction based on the resource ID and/or the indication of the subscription state. For example, in the case of the subscription state indicating that the subscription is suspended, the instruction(s) may be generated by the instruction system 1412 to cause one or more resources associated with the subscription ID to be inaccessible to user accounts of the tenancy.

[0261] At step S1612, the instruction(s) may be transmitted to the service control plane 1306. In certain embodiments, a first instruction is transmitted to the service control plane 1306 that causes a resource(s) to stop. A second instruction may be transmitted to the service control plane 1306 after the first instruction and cause the resource(s) to be locked.

[0262] FIG. 17 depicts an example flow 1700 for activating (e.g., unsuspending) a resource, according to some embodiments. The resource may be included in a set of resources being activated. The example flow 1700 may be carried out using an architecture like architecture 1300 described above that includes a resource provider 1304, a resource management system 1450 that includes a resource mapping system 1408 and an instruction system, 1412, and a service control plane 1306.

[0263] At step S1702, the resource provider 1304 may determine that an event associated a subscription ID occurred. For example, an indication from a billing system may be received by the resource provider 1304 that indicates the subscription identified by the subscription ID was paid for and that the subscription should be activated (e.g., because the subscription was suspended). In certain embodiments, the resource provider 1304 may receive an indication from a console (e.g., console 1314 described above) that indicates the subscription should be activated. For example, a user account associated with the subscription ID may have used a user interface to input the indication that the subscription associated with the subscription ID should be activated.

[0264] At step S1704, the resource provider 1304 may transmit a subscription ID (e.g., subscription state 1406, described above) associated with the subscription, the subscription state (e.g., subscription state 1406, described

above), and/or a tenancy ID (e.g., tenancy ID 1402 described above) to the resource mapping system 1408.

[0265] At step S1706, the resource mapping system 1408 may use the tenancy ID, the subscription ID, and/or the subscription state to determine a set of one or more resource IDs (e.g., output resource ID 1410) that are associated with the combination of information received by the resource mapping system at step S1704.

[0266] At step S1708, the resource ID may be transmitted to the instruction system 1412 from the resource mapping system 1408. An indication of the subscription state may also be transmitted to the instruction system 1412 from the resource mapping system 1408.

[0267] At step S1710, the instruction system 1412 may generate an instruction based on the resource ID and/or the indication of the subscription state. For example, in the case of the subscription state indicating that the subscription is activated, the instruction(s) may be generated by the instruction system 1412 to cause one or more resources associated with the subscription ID to be available and accessible to user accounts of the tenancy.

[0268] At step S1712, the instruction(s) may be transmitted to the service control plane 1306. In certain embodiments, a first instruction is transmitted to the service control plane 1306 that causes one or more resource to be activated. A second instruction may be transmitted to the service control plane 1306 after the first instruction and cause the resource(s) to be unlocked (e.g., accessible by a user account of the tenancy).

[0269] FIG. 18 depicts an example flow 1800 for terminating a resource, according to some embodiments. The resource may be included in a set of resources being terminated. The example flow 1700 may be carried out using an architecture like architecture 1300 described above that includes a resource provider 1304, a resource management system 1450 that includes a resource mapping system 1408 and an instruction system, 1412, and a service control plane 1306.

[0270] At step S1802, the resource provider 1304 may determine that an event associated a subscription ID occurred. For example, an indication from a billing system may be received by the resource provider 1304 that indicates the subscription identified by the subscription ID was not paid for and that the subscription should be terminated (e.g., because the subscription was activated or because the subscription was suspended for longer than a predefined period of time). In certain embodiments, the resource provider 1304 may receive an indication from a console (e.g., console 1314 described above) that indicates the subscription should be terminated. For example, a user account associated with the subscription ID may have used a user interface to input the indication that the subscription associated with the subscription ID should be terminated.

[0271] At step S1804, the resource provider 1304 may transmit a subscription ID (e.g., subscription state 1406, described above) associated with the subscription, the subscription state (e.g., subscription state 1406, described above), and/or a tenancy ID (e.g., tenancy ID 1402 described above) to the resource mapping system 1408.

[0272] At step S1806, the resource mapping system 1408 may use the tenancy ID, the subscription ID, and/or the subscription state to determine a set of one or more resource IDs (e.g., output resource ID 1410) that are associated with the combination of information received by the resource

mapping system at step S1804. The resource mapping system may remove the mapping entry from the data structure maintained by the resource mapping system 1408 as described above with respect to FIG. 13.

[0273] At step S1808, the resource ID may be transmitted to the instruction system 1412 from the resource mapping system 1408. An indication of the subscription state may also be transmitted to the instruction system 1412 from the resource mapping system 1408.

[0274] At step S1810, the instruction system 1412 may generate an instruction based on the resource ID and/or the indication of the subscription state. For example, in the case of the subscription state indicating that the subscription is terminated, the instruction(s) may be generated by the instruction system 1412 to cause one or more resources associated with the subscription ID to be unavailable to user accounts of the tenancy. The one or more resources may be unavailable to the user account because the one or more user accounts may be deleted from memory and/or no longer associated with the subscription identifier and/or tenancy. If a resource is unavailable, the resource is also inaccessible. In certain embodiments, when a subscription state is terminated, the resource may remain available but be inaccessible because the resource is still provisioned and accessible to another subscription.

[0275] At step S1812, the instruction(s) may be transmitted to the service control plane 1306. In certain embodiments, an instruction is transmitted to the service control plane 1306 that causes one or more resource to be terminated. The resources may be deleted.

[0276] FIG. 19 depicts an example process 1900 for reclaiming a resource of a cloud environment, according to some embodiments. The process 1900 may be carried out using an architecture like architecture 1300 described above.

[0277] At 1902, a subscription of a user account (e.g., a tenancy) may be associated with a CSP (e.g., a CSP of the first cloud environment 1302 described above) and can enable the user account to access services of the CSP and resources (e.g., virtual resources as described herein) provided by the services. The subscription may be associated with the resources. In certain embodiments, the association is stored by a data structure that maps the associations between the resources, subscriptions, services, and/or tenancies. The associations may be mapped using identifiers (IDs) of the resources, subscriptions, services, and/or tenancies.

[0278] At 1904, a device associated with the user account may be enabled to access the resource. A resource provider (e.g., resource provider 1304) and a service control plane (e.g., service control plane 1306) may enable the access to the resource. Whether the user account is enabled to access the resource may be based at least in part on whether the user account is associated with a subscription and/or a service that is associated with the resource. Access to a resource can enable the user account to access (e.g., use) an available resource. The resource may be available after the resource has been provisioned to the user account. The resource may be provisioned to the user account based on the services configured to be used by the user account and the subscriptions of the user account. The resource may be available and accessible as part of a private cloud associated with the user account. The private cloud (e.g., second cloud environment 1318) may be operated by a second CSP.

[0279] A resource may be accessible because the resource is associated with an active state. The resource may be associated with an active state when a subscription associated with the resource is associated with a subscription state that is active. A resource may not be accessible by the user account when the resource has not been provisioned or when the subscription associated with the resource is associated with a suspended state or a terminated state. A resource that is unavailable may also not be accessible (e.g., when the resource has not been provisioned).

[0280] At 1906, a change to the subscription may be determined. The change to the subscription may be a change to a subscription state associated with the subscription. The subscription state may be a state that indicates whether the subscription is active, suspended, or terminated. An active subscription may enable the user account to access available resources. A suspended subscription may prevent the user account from accessing available resources. A terminated subscription may not have resource available to the user account associated with the subscription (e.g., the resources have been reclaimed).

[0281] A change to the subscription may be determined by receiving a subscription identifier that identifies the subscription and a subscription state to associate with the subscription. The subscription state may cause the identified subscription state to have an updated subscription state. For example, the subscription state associated with the subscription may be changed from active to suspended, suspended to terminated, suspended to active, or active to terminated.

[0282] In certain embodiments, the subscription state is received from the private cloud. The subscription state may have been received by a user interface of the private cloud. The subscription state may be periodically pulled from or pushed from the private cloud to the CSP and be used to determine if the subscription state has changed since a last known subscription state.

[0283] In certain embodiments, the change to the subscription is determined by receiving an indication associated with the subscription from a subscription billing system. The subscription billing system may maintain entries that indicate whether a subscription is active, suspended, or terminated (e.g., a subscription state). The subscription state may be transmitted by the billing system (e.g., periodically, upon request, upon a change occurring) to the CSP and used to determine a change to the subscription. The billing system may indicate that the subscription is active if the subscription has been paid for, the subscription is suspended if the subscription has not been paid for (e.g., not paid for a specific amount of time less than the time before the subscription is terminated), the subscription is terminated (e.g., not paid for longer than a predefined period of time, terminated based on input from a user interface).

[0284] In certain embodiments, the subscription state is determined based at least in part on a length of time the subscription has been associated with a second subscription state. For example, after the subscription state has been in a suspended state for the length of time (e.g., 30 days), the subscription state may be updated/changed to a terminated subscription state.

[0285] At 1908, the resource may be disassociated from the subscription. Disassociating the resource from the subscription can be performed by removing associations between the resource and the subscription in the mapping. Disassociating the resource from the subscription can be

performed by removing an entry from the mapping that indicates the subscription is associated with the resource. After the subscription is dissociated from the resource, the virtual resource may be no longer accessible and/or available to the user account via the private cloud. In certain embodiments, the resource may remain accessible and/or available for use in association with a different subscription.

[0286] In certain embodiments, the resource is terminated upon the resource being disassociated with the subscription if the resource is not associated with another subscription (e.g., a subscription associated with the user account). In certain embodiments, the resource is terminated upon the resource being disassociated with the subscription if no other resources depend on the resource. If other subscriptions depend on the resource, the resource may remain accessible to the other resources but not the subscription disassociated from the resource. The resource may be subsequently terminated if the other subscriptions are terminated.

[0287] Disassociating the resource may include transmitting a resource management instruction (e.g., resource management instruction **1414** to a control plane (e.g., service control plane **1306**). The service control plane may then terminate the resource. Terminating the resource may include deleting the resource configurations. Terminating the resource may include reclaiming, by the CSP, at least one of computing power, storage, or bandwidth allocated to the user account so that at least one of other user accounts or other subscriptions can use the at least one of the computing power, the storage, or the bandwidth.

[0288] In certain embodiments, when a subscription is terminated, all resources associated with the subscription are terminated. In certain embodiments, when a subscription is terminated (e.g., disassociated from the subscription), all resource associated with the subscription are terminated besides the resources associated with another subscription. When the resource(s) are terminated, the resource(s) may not be available and may not be accessible via the private cloud by the user account. Terminated resources may become available and accessible to a different subscription.

[0289] In certain embodiments, the subscription may be changed based on a subscription identifier, the subscription state, and an identifier of the virtual resource being received. The virtual resource can be associated with the subscription identifier based at least in part on the subscription state (e.g., by a resource management system **1450**).

Examples of Cloud Infrastructure

[0290] As noted above, IaaS is one particular type of cloud computing. IaaS can be configured to provide virtualized computing resources over a public network (e.g., the Internet). In an IaaS model, a cloud computing provider can host the infrastructure components (e.g., servers, storage devices, network nodes (e.g., hardware), deployment software, platform virtualization (e.g., a hypervisor layer), or the like). In some cases, an IaaS provider may also supply a variety of services to accompany those infrastructure components (example services include billing software, monitoring software, logging software, load balancing software, clustering software, etc.). Thus, as these services may be policy-driven, IaaS users may be able to implement policies to drive load balancing to maintain application availability and performance.

[0291] In some instances, IaaS customers may access resources and services through a wide area network (WAN), such as the Internet, and can use the cloud provider's services to install the remaining elements of an application stack. For example, the user can log in to the IaaS platform to create virtual machines (VMs), install operating systems (OSs) on each VM, deploy middleware such as databases, create storage buckets for workloads and backups, and even install enterprise software into that VM. Customers can then use the provider's services to perform various functions, including balancing network traffic, troubleshooting application issues, monitoring performance, managing disaster recovery, etc.

[0292] In most cases, a cloud computing model will require the participation of a cloud provider. The cloud provider may, but need not be, a third-party service that specializes in providing (e.g., offering, renting, selling) IaaS. An entity might also opt to deploy a private cloud, becoming its own provider of infrastructure services.

[0293] In some examples, IaaS deployment is the process of putting a new application, or a new version of an application, onto a prepared application server or the like. It may also include the process of preparing the server (e.g., installing libraries, daemons, etc.). This is often managed by the cloud provider, below the hypervisor layer (e.g., the servers, storage, network hardware, and virtualization). Thus, the customer may be responsible for handling (OS), middleware, and/or application deployment (e.g., on self-service virtual machines (e.g., that can be spun up on demand) or the like).

[0294] In some examples, IaaS provisioning may refer to acquiring computers or virtual hosts for use, and even installing needed libraries or services on them. In most cases, deployment does not include provisioning, and the provisioning may need to be performed first.

[0295] In some cases, there are two different challenges for IaaS provisioning. First, there is the initial challenge of provisioning the initial set of infrastructure before anything is running. Second, there is the challenge of evolving the existing infrastructure (e.g., adding new services, changing services, removing services, etc.) once everything has been provisioned. In some cases, these two challenges may be addressed by enabling the configuration of the infrastructure to be defined declaratively. In other words, the infrastructure (e.g., what components are needed and how they interact) can be defined by one or more configuration files. Thus, the overall topology of the infrastructure (e.g., what resources depend on which, and how they each work together) can be described declaratively. In some instances, once the topology is defined, a workflow can be generated that creates and/or manages the different components described in the configuration files.

[0296] In some examples, an infrastructure may have many interconnected elements. For example, there may be one or more virtual private clouds (VPCs) (e.g., a potentially on-demand pool of configurable and/or shared computing resources), also known as a core network. In some examples, there may also be one or more inbound/outbound traffic group rules provisioned to define how the inbound and/or outbound traffic of the network will be set up and one or more virtual machines (VMs). Other infrastructure elements may also be provisioned, such as a load balancer, a database,

or the like. As more and more infrastructure elements are desired and/or added, the infrastructure may incrementally evolve.

[0297] In some instances, continuous deployment techniques may be employed to enable deployment of infrastructure code across various virtual computing environments. Additionally, the described techniques can enable infrastructure management within these environments. In some examples, service teams can write code that is desired to be deployed to one or more, but often many, different production environments (e.g., across various different geographic locations, sometimes spanning the entire world). However, in some examples, the infrastructure on which the code will be deployed must first be set up. In some instances, the provisioning can be done manually, a provisioning tool may be utilized to provision the resources, and/or deployment tools may be utilized to deploy the code once the infrastructure is provisioned.

[0298] FIG. 20 is a block diagram 2000 illustrating an example pattern of an IaaS architecture, according to at least one embodiment. Service operators 2002 can be communicatively coupled to a secure host tenancy 2004 that can include a virtual cloud network (VCN) 2006 and a secure host subnet 2008. In some examples, the service operators 2002 may be using one or more client computing devices, which may be portable handheld devices (e.g., an iPhone®, cellular telephone, an iPad®, computing tablet, a personal digital assistant (PDA)) or wearable devices (e.g., a Google Glass® head mounted display), running software such as Microsoft Windows Mobile®, and/or a variety of mobile operating systems such as iOS, Windows Phone, Android, BlackBerry 8, Palm OS, and the like, and being Internet, e-mail, short message service (SMS), BlackBerry®, or other communication protocol enabled. Alternatively, the client computing devices can be general purpose personal computers including, by way of example, personal computers and/or laptop computers running various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems. The client computing devices can be workstation computers running any of a variety of commercially-available UNIX® or UNIX-like operating systems, including without limitation the variety of GNU/Linux operating systems, such as for example, Google Chrome OS. Alternatively, or in addition, client computing devices may be any other electronic device, such as a thin-client computer, an Internet-enabled gaming system (e.g., a Microsoft Xbox gaming console with or without a Kinect® gesture input device), and/or a personal messaging device, capable of communicating over a network that can access the VCN 2006 and/or the Internet.

[0299] The VCN 2006 can include a local peering gateway (LPG) 2010 that can be communicatively coupled to a secure shell (SSH) VCN 2012 via an LPG 2010 contained in the SSH VCN 2012. The SSH VCN 2012 can include an SSH subnet 2014, and the SSH VCN 2012 can be communicatively coupled to a control plane VCN 2016 via the LPG 2010 contained in the control plane VCN 2016. Also, the SSH VCN 2012 can be communicatively coupled to a data plane VCN 2018 via an LPG 2010. The control plane VCN 2016 and the data plane VCN 2018 can be contained in a service tenancy 2019 that can be owned and/or operated by the IaaS provider.

[0300] The control plane VCN 2016 can include a control plane demilitarized zone (DMZ) tier 2020 that acts as a

perimeter network (e.g., portions of a corporate network between the corporate intranet and external networks). The DMZ-based servers may have restricted responsibilities and help keep breaches contained. Additionally, the DMZ tier 2020 can include one or more load balancer (LB) subnet(s) 2022, a control plane app tier 2024 that can include app subnet(s) 2026, a control plane data tier 2028 that can include database (DB) subnet(s) 2030 (e.g., frontend DB subnet(s) and/or backend DB subnet(s)). The LB subnet(s) 2022 contained in the control plane DMZ tier 2020 can be communicatively coupled to the app subnet(s) 2026 contained in the control plane app tier 2024 and an Internet gateway 2034 that can be contained in the control plane VCN 2016, and the app subnet(s) 2026 can be communicatively coupled to the DB subnet(s) 2030 contained in the control plane data tier 2028 and a service gateway 2036 and a network address translation (NAT) gateway 2038. The control plane VCN 2016 can include the service gateway 2036 and the NAT gateway 2038.

[0301] The control plane VCN 2016 can include a data plane mirror app tier 2040 that can include app subnet(s) 2026. The app subnet(s) 2026 contained in the data plane mirror app tier 2040 can include a virtual network interface controller (VNIC) 2042 that can execute a compute instance 2044. The compute instance 2044 can communicatively couple the app subnet(s) 2026 of the data plane mirror app tier 2040 to app subnet(s) 2026 that can be contained in a data plane app tier 2046.

[0302] The data plane VCN 2018 can include the data plane app tier 2046, a data plane DMZ tier 2048, and a data plane data tier 2050. The data plane DMZ tier 2048 can include LB subnet(s) 2022 that can be communicatively coupled to the app subnet(s) 2026 of the data plane app tier 2046 and the Internet gateway 2034 of the data plane VCN 2018. The app subnet(s) 2026 can be communicatively coupled to the service gateway 2036 of the data plane VCN 2018 and the NAT gateway 2038 of the data plane VCN 2018. The data plane data tier 2050 can also include the DB subnet(s) 2030 that can be communicatively coupled to the app subnet(s) 2026 of the data plane app tier 2046.

[0303] The Internet gateway 2034 of the control plane VCN 2016 and of the data plane VCN 2018 can be communicatively coupled to a metadata management service 2052 that can be communicatively coupled to public Internet 2054. Public Internet 2054 can be communicatively coupled to the NAT gateway 2038 of the control plane VCN 2016 and of the data plane VCN 2018. The service gateway 2036 of the control plane VCN 2016 and of the data plane VCN 2018 can be communicatively coupled to cloud services 2056.

[0304] In some examples, the service gateway 2036 of the control plane VCN 2016 or of the data plane VCN 2018 can make application programming interface (API) calls to cloud services 2056 without going through public Internet 2054. The API calls to cloud services 2056 from the service gateway 2036 can be one-way: the service gateway 2036 can make API calls to cloud services 2056, and cloud services 2056 can send requested data to the service gateway 2036. But, cloud services 2056 may not initiate API calls to the service gateway 2036.

[0305] In some examples, the secure host tenancy 2004 can be directly connected to the service tenancy 2019, which may be otherwise isolated. The secure host subnet 2008 can communicate with the SSH subnet 2014 through an LPG

2010 that may enable two-way communication over an otherwise isolated system. Connecting the secure host subnet **2008** to the SSH subnet **2014** may give the secure host subnet **2008** access to other entities within the service tenancy **2019**.

[0306] The control plane VCN **2016** may allow users of the service tenancy **2019** to set up or otherwise provision desired resources. Desired resources provisioned in the control plane VCN **2016** may be deployed or otherwise used in the data plane VCN **2018**. In some examples, the control plane VCN **2016** can be isolated from the data plane VCN **2018**, and the data plane mirror app tier **2040** of the control plane VCN **2016** can communicate with the data plane app tier **2046** of the data plane VCN **2018** via VNICs **2042** that can be contained in the data plane mirror app tier **2040** and the data plane app tier **2046**.

[0307] In some examples, users of the system, or customers, can make requests, for example create, read, update, or delete (CRUD) operations, through public Internet **2054** that can communicate the requests to the metadata management service **2052**. The metadata management service **2052** can communicate the request to the control plane VCN **2016** through the Internet gateway **2034**. The request can be received by the LB subnet(s) **2022** contained in the control plane DMZ tier **2020**. The LB subnet(s) **2022** may determine that the request is valid, and in response to this determination, the LB subnet(s) **2022** can transmit the request to app subnet(s) **2026** contained in the control plane app tier **2024**. If the request is validated and requires a call to public Internet **2054**, the call to public Internet **2054** may be transmitted to the NAT gateway **2038** that can make the call to public Internet **2054**. Metadata that may be desired to be stored by the request can be stored in the DB subnet(s) **2030**.

[0308] In some examples, the data plane mirror app tier **2040** can facilitate direct communication between the control plane VCN **2016** and the data plane VCN **2018**. For example, changes, updates, or other suitable modifications to configuration may be desired to be applied to the resources contained in the data plane VCN **2018**. Via a VNIC **2042**, the control plane VCN **2016** can directly communicate with, and can thereby execute the changes, updates, or other suitable modifications to configuration to, resources contained in the data plane VCN **2018**.

[0309] In some embodiments, the control plane VCN **2016** and the data plane VCN **2018** can be contained in the service tenancy **2019**. In this case, the user, or the customer, of the system may not own or operate either the control plane VCN **2016** or the data plane VCN **2018**. Instead, the IaaS provider may own or operate the control plane VCN **2016** and the data plane VCN **2018**, both of which may be contained in the service tenancy **2019**. This embodiment can enable isolation of networks that may prevent users or customers from interacting with other users,' or other customers,' resources. Also, this embodiment may allow users or customers of the system to store databases privately without needing to rely on public Internet **2054**, which may not have a desired level of threat prevention, for storage.

[0310] In other embodiments, the LB subnet(s) **2022** contained in the control plane VCN **2016** can be configured to receive a signal from the service gateway **2036**. In this embodiment, the control plane VCN **2016** and the data plane VCN **2018** may be configured to be called by a customer of the IaaS provider without calling public Internet **2054**. Customers of the IaaS provider may desire this embodiment

since database(s) that the customers use may be controlled by the IaaS provider and may be stored on the service tenancy **2019**, which may be isolated from public Internet **2054**.

[0311] FIG. **21** is a block diagram **2100** illustrating another example pattern of an IaaS architecture, according to at least one embodiment. Service operators **2102** (e.g., service operators **2002** of FIG. **20**) can be communicatively coupled to a secure host tenancy **2104** (e.g., the secure host tenancy **2004** of FIG. **20**) that can include a virtual cloud network (VCN) **2106** (e.g., the VCN **2006** of FIG. **20**) and a secure host subnet **2108** (e.g., the secure host subnet **2008** of FIG. **20**). The VCN **2106** can include a local peering gateway (LPG) **2110** (e.g., the LPG **2010** of FIG. **20**) that can be communicatively coupled to a secure shell (SSH) VCN **2112** (e.g., the SSH VCN **2012** of FIG. **20**) via an LPG **2010** contained in the SSH VCN **2112**. The SSH VCN **2112** can include an SSH subnet **2114** (e.g., the SSH subnet **2014** of FIG. **20**), and the SSH VCN **2112** can be communicatively coupled to a control plane VCN **2116** (e.g., the control plane VCN **2016** of FIG. **20**) via an LPG **2110** contained in the control plane VCN **2116**. The control plane VCN **2116** can be contained in a service tenancy **2119** (e.g., the service tenancy **2019** of FIG. **20**), and the data plane VCN **2118** (e.g., the data plane VCN **2018** of FIG. **20**) can be contained in a customer tenancy **2121** that may be owned or operated by users, or customers, of the system.

[0312] The control plane VCN **2116** can include a control plane DMZ tier **2120** (e.g., the control plane DMZ tier **2020** of FIG. **20**) that can include LB subnet(s) **2122** (e.g., LB subnet(s) **2022** of FIG. **20**), a control plane app tier **2124** (e.g., the control plane app tier **2024** of FIG. **20**) that can include app subnet(s) **2126** (e.g., app subnet(s) **2026** of FIG. **20**), a control plane data tier **2128** (e.g., the control plane data tier **2028** of FIG. **20**) that can include database (DB) subnet(s) **2130** (e.g., similar to DB subnet(s) **2030** of FIG. **20**). The LB subnet(s) **2122** contained in the control plane DMZ tier **2120** can be communicatively coupled to the app subnet(s) **2126** contained in the control plane app tier **2124** and an Internet gateway **2134** (e.g., the Internet gateway **2034** of FIG. **20**) that can be contained in the control plane VCN **2116**, and the app subnet(s) **2126** can be communicatively coupled to the DB subnet(s) **2130** contained in the control plane data tier **2128** and a service gateway **2136** (e.g., the service gateway **2036** of FIG. **20**) and a network address translation (NAT) gateway **2138** (e.g., the NAT gateway **2038** of FIG. **20**). The control plane VCN **2116** can include the service gateway **2136** and the NAT gateway **2138**.

[0313] The control plane VCN **2116** can include a data plane mirror app tier **2140** (e.g., the data plane mirror app tier **2040** of FIG. **20**) that can include app subnet(s) **2126**. The app subnet(s) **2126** contained in the data plane mirror app tier **2140** can include a virtual network interface controller (VNIC) **2142** (e.g., the VNIC of **2042**) that can execute a compute instance **2144** (e.g., similar to the compute instance **2044** of FIG. **20**). The compute instance **2144** can facilitate communication between the app subnet(s) **2126** of the data plane mirror app tier **2140** and the app subnet(s) **2126** that can be contained in a data plane app tier **2146** (e.g., the data plane app tier **2046** of FIG. **20**) via the VNIC **2142** contained in the data plane mirror app tier **2140** and the VNIC **2142** contained in the data plane app tier **2146**.

[0314] The Internet gateway 2134 contained in the control plane VCN 2116 can be communicatively coupled to a metadata management service 2152 (e.g., the metadata management service 2052 of FIG. 20) that can be communicatively coupled to public Internet 2154 (e.g., public Internet 2054 of FIG. 20). Public Internet 2154 can be communicatively coupled to the NAT gateway 2138 contained in the control plane VCN 2116. The service gateway 2136 contained in the control plane VCN 2116 can be communicatively coupled to cloud services 2156 (e.g., cloud services 2056 of FIG. 20).

[0315] In some examples, the data plane VCN 2118 can be contained in the customer tenancy 2121. In this case, the IaaS provider may provide the control plane VCN 2116 for each customer, and the IaaS provider may, for each customer, set up a unique compute instance 2144 that is contained in the service tenancy 2119. Each compute instance 2144 may allow communication between the control plane VCN 2116, contained in the service tenancy 2119, and the data plane VCN 2118 that is contained in the customer tenancy 2121. The compute instance 2144 may allow resources, that are provisioned in the control plane VCN 2116 that is contained in the service tenancy 2119, to be deployed or otherwise used in the data plane VCN 2118 that is contained in the customer tenancy 2121.

[0316] In other examples, the customer of the IaaS provider may have databases that live in the customer tenancy 2121. In this example, the control plane VCN 2116 can include the data plane mirror app tier 2140 that can include app subnet(s) 2126. The data plane mirror app tier 2140 can reside in the data plane VCN 2118, but the data plane mirror app tier 2140 may not live in the data plane VCN 2118. That is, the data plane mirror app tier 2140 may have access to the customer tenancy 2121, but the data plane mirror app tier 2140 may not exist in the data plane VCN 2118 or be owned or operated by the customer of the IaaS provider. The data plane mirror app tier 2140 may be configured to make calls to the data plane VCN 2118 but may not be configured to make calls to any entity contained in the control plane VCN 2116. The customer may desire to deploy or otherwise use resources in the data plane VCN 2118 that are provisioned in the control plane VCN 2116, and the data plane mirror app tier 2140 can facilitate the desired deployment, or other usage of resources, of the customer.

[0317] In some embodiments, the customer of the IaaS provider can apply filters to the data plane VCN 2118. In this embodiment, the customer can determine what the data plane VCN 2118 can access, and the customer may restrict access to public Internet 2154 from the data plane VCN 2118. The IaaS provider may not be able to apply filters or otherwise control access of the data plane VCN 2118 to any outside networks or databases. Applying filters and controls by the customer onto the data plane VCN 2118, contained in the customer tenancy 2121, can help isolate the data plane VCN 2118 from other customers and from public Internet 2154.

[0318] In some embodiments, cloud services 2156 can be called by the service gateway 2136 to access services that may not exist on public Internet 2154, on the control plane VCN 2116, or on the data plane VCN 2118. The connection between cloud services 2156 and the control plane VCN 2116 or the data plane VCN 2118 may not be live or continuous. Cloud services 2156 may exist on a different network owned or operated by the IaaS provider. Cloud

services 2156 may be configured to receive calls from the service gateway 2136 and may be configured to not receive calls from public Internet 2154. Some cloud services 2156 may be isolated from other cloud services 2156, and the control plane VCN 2116 may be isolated from cloud services 2156 that may not be in the same region as the control plane VCN 2116. For example, the control plane VCN 2116 may be located in “Region 1,” and cloud service “Deployment 19,” may be located in Region 1 and in “Region 2.” If a call to Deployment 19 is made by the service gateway 2136 contained in the control plane VCN 2116 located in Region 1, the call may be transmitted to Deployment 19 in Region 1. In this example, the control plane VCN 2116, or Deployment 19 in Region 1, may not be communicatively coupled to, or otherwise in communication with, Deployment 19 in Region 2.

[0319] FIG. 22 is a block diagram 2200 illustrating another example pattern of an IaaS architecture, according to at least one embodiment. Service operators 2202 (e.g., service operators 2002 of FIG. 20) can be communicatively coupled to a secure host tenancy 2204 (e.g., the secure host tenancy 2004 of FIG. 20) that can include a virtual cloud network (VCN) 2206 (e.g., the VCN 2006 of FIG. 20) and a secure host subnet 2208 (e.g., the secure host subnet 2008 of FIG. 20). The VCN 2206 can include an LPG 2210 (e.g., the LPG 2010 of FIG. 20) that can be communicatively coupled to an SSH VCN 2212 (e.g., the SSH VCN 2012 of FIG. 20) via an LPG 2210 contained in the SSH VCN 2212. The SSH VCN 2212 can include an SSH subnet 2214 (e.g., the SSH subnet 2014 of FIG. 20), and the SSH VCN 2212 can be communicatively coupled to a control plane VCN 2216 (e.g., the control plane VCN 2016 of FIG. 20) via an LPG 2210 contained in the control plane VCN 2216 and to a data plane VCN 2218 (e.g., the data plane 2018 of FIG. 20) via an LPG 2210 contained in the data plane VCN 2218. The control plane VCN 2216 and the data plane VCN 2218 can be contained in a service tenancy 2219 (e.g., the service tenancy 2019 of FIG. 20).

[0320] The control plane VCN 2216 can include a control plane DMZ tier 2220 (e.g., the control plane DMZ tier 2020 of FIG. 20) that can include load balancer (LB) subnet(s) 2222 (e.g., LB subnet(s) 2022 of FIG. 20), a control plane app tier 2224 (e.g., the control plane app tier 2024 of FIG. 20) that can include app subnet(s) 2226 (e.g., similar to app subnet(s) 2026 of FIG. 20), a control plane data tier 2228 (e.g., the control plane data tier 2028 of FIG. 20) that can include DB subnet(s) 2230. The LB subnet(s) 2222 contained in the control plane DMZ tier 2220 can be communicatively coupled to the app subnet(s) 2226 contained in the control plane app tier 2224 and to an Internet gateway 2234 (e.g., the Internet gateway 2034 of FIG. 20) that can be contained in the control plane VCN 2216, and the app subnet(s) 2226 can be communicatively coupled to the DB subnet(s) 2230 contained in the control plane data tier 2228 and to a service gateway 2236 (e.g., the service gateway of FIG. 20) and a network address translation (NAT) gateway 2238 (e.g., the NAT gateway 2038 of FIG. 20). The control plane VCN 2216 can include the service gateway 2236 and the NAT gateway 2238.

[0321] The data plane VCN 2218 can include a data plane app tier 2246 (e.g., the data plane app tier 2046 of FIG. 20), a data plane DMZ tier 2248 (e.g., the data plane DMZ tier 2048 of FIG. 20), and a data plane data tier 2250 (e.g., the data plane data tier 2050 of FIG. 20). The data plane DMZ

tier **2248** can include LB subnet(s) **2222** that can be communicatively coupled to trusted app subnet(s) **2260** and untrusted app subnet(s) **2262** of the data plane app tier **2246** and the Internet gateway **2234** contained in the data plane VCN **2218**. The trusted app subnet(s) **2260** can be communicatively coupled to the service gateway **2236** contained in the data plane VCN **2218**, the NAT gateway **2238** contained in the data plane VCN **2218**, and DB subnet(s) **2230** contained in the data plane data tier **2250**. The untrusted app subnet(s) **2262** can be communicatively coupled to the service gateway **2236** contained in the data plane VCN **2218** and DB subnet(s) **2230** contained in the data plane data tier **2250**. The data plane data tier **2250** can include DB subnet(s) **2230** that can be communicatively coupled to the service gateway **2236** contained in the data plane VCN **2218**.

[0322] The untrusted app subnet(s) **2262** can include one or more primary VNICS **2264(1)-(N)** that can be communicatively coupled to tenant virtual machines (VMs) **2266(1)-(N)**. Each tenant VM **2266(1)-(N)** can be communicatively coupled to a respective app subnet **2267(1)-(N)** that can be contained in respective container egress VCNs **2268(1)-(N)** that can be contained in respective customer tenancies **2270(1)-(N)**. Respective secondary VNICS **2272(1)-(N)** can facilitate communication between the untrusted app subnet(s) **2262** contained in the data plane VCN **2218** and the app subnet contained in the container egress VCNs **2268(1)-(N)**. Each container egress VCNs **2268(1)-(N)** can include a NAT gateway **2238** that can be communicatively coupled to public Internet **2254** (e.g., public Internet **2054** of FIG. 20).

[0323] The Internet gateway **2234** contained in the control plane VCN **2216** and contained in the data plane VCN **2218** can be communicatively coupled to a metadata management service **2252** (e.g., the metadata management system **2052** of FIG. 20) that can be communicatively coupled to public Internet **2254**. Public Internet **2254** can be communicatively coupled to the NAT gateway **2238** contained in the control plane VCN **2216** and contained in the data plane VCN **2218**. The service gateway **2236** contained in the control plane VCN **2216** and contained in the data plane VCN **2218** can be communicatively coupled to cloud services **2256**.

[0324] In some embodiments, the data plane VCN **2218** can be integrated with customer tenancies **2270**. This integration can be useful or desirable for customers of the IaaS provider in some cases such as a case that may desire support when executing code. The customer may provide code to run that may be destructive, may communicate with other customer resources, or may otherwise cause undesirable effects. In response to this, the IaaS provider may determine whether to run code given to the IaaS provider by the customer.

[0325] In some examples, the customer of the IaaS provider may grant temporary network access to the IaaS provider and request a function to be attached to the data plane app tier **2246**. Code to run the function may be executed in the VMs **2266(1)-(N)**, and the code may not be configured to run anywhere else on the data plane VCN **2218**. Each VM **2266(1)-(N)** may be connected to one customer tenancy **2270**. Respective containers **2271(1)-(N)** contained in the VMs **2266(1)-(N)** may be configured to run the code. In this case, there can be a dual isolation (e.g., the containers **2271(1)-(N)** running code, where the containers **2271(1)-(N)** may be contained in at least the VM **2266(1)-(N)** that are contained in the untrusted app subnet(s) **2262**), which may help prevent incorrect or otherwise undesirable

code from damaging the network of the IaaS provider or from damaging a network of a different customer. The containers **2271(1)-(N)** may be communicatively coupled to the customer tenancy **2270** and may be configured to transmit or receive data from the customer tenancy **2270**. The containers **2271(1)-(N)** may not be configured to transmit or receive data from any other entity in the data plane VCN **2218**. Upon completion of running the code, the IaaS provider may kill or otherwise dispose of the containers **2271(1)-(N)**.

[0326] In some embodiments, the trusted app subnet(s) **2260** may run code that may be owned or operated by the IaaS provider. In this embodiment, the trusted app subnet(s) **2260** may be communicatively coupled to the DB subnet(s) **2230** and be configured to execute CRUD operations in the DB subnet(s) **2230**. The untrusted app subnet(s) **2262** may be communicatively coupled to the DB subnet(s) **2230**, but in this embodiment, the untrusted app subnet(s) may be configured to execute read operations in the DB subnet(s) **2230**. The containers **2271(1)-(N)** that can be contained in the VM **2266(1)-(N)** of each customer and that may run code from the customer may not be communicatively coupled with the DB subnet(s) **2230**.

[0327] In other embodiments, the control plane VCN **2216** and the data plane VCN **2218** may not be directly communicatively coupled. In this embodiment, there may be no direct communication between the control plane VCN **2216** and the data plane VCN **2218**. However, communication can occur indirectly through at least one method. An LPG **2210** may be established by the IaaS provider that can facilitate communication between the control plane VCN **2216** and the data plane VCN **2218**. In another example, the control plane VCN **2216** or the data plane VCN **2218** can make a call to cloud services **2256** via the service gateway **2236**. For example, a call to cloud services **2256** from the control plane VCN **2216** can include a request for a service that can communicate with the data plane VCN **2218**.

[0328] FIG. 23 is a block diagram **2300** illustrating another example pattern of an IaaS architecture, according to at least one embodiment. Service operators **2302** (e.g., service operators **2002** of FIG. 20) can be communicatively coupled to a secure host tenancy **2304** (e.g., the secure host tenancy **2004** of FIG. 20) that can include a virtual cloud network (VCN) **2306** (e.g., the VCN **2006** of FIG. 20) and a secure host subnet **2308** (e.g., the secure host subnet **2008** of FIG. 20). The VCN **2306** can include an LPG **2310** (e.g., the LPG **2010** of FIG. 20) that can be communicatively coupled to an SSH VCN **2312** (e.g., the SSH VCN **2012** of FIG. 20) via an LPG **2310** contained in the SSH VCN **2312**. The SSH VCN **2312** can include an SSH subnet **2314** (e.g., the SSH subnet **2014** of FIG. 20), and the SSH VCN **2312** can be communicatively coupled to a control plane VCN **2316** (e.g., the control plane VCN **2016** of FIG. 20) via an LPG **2310** contained in the control plane VCN **2316** and to a data plane VCN **2318** (e.g., the data plane **2018** of FIG. 20) via an LPG **2310** contained in the data plane VCN **2318**. The control plane VCN **2316** and the data plane VCN **2318** can be contained in a service tenancy **2319** (e.g., the service tenancy **2019** of FIG. 20).

[0329] The control plane VCN **2316** can include a control plane DMZ tier **2320** (e.g., the control plane DMZ tier **2020** of FIG. 20) that can include LB subnet(s) **2322** (e.g., LB subnet(s) **2022** of FIG. 20), a control plane app tier **2324** (e.g., the control plane app tier **2024** of FIG. 20) that can

include app subnet(s) **2326** (e.g., app subnet(s) **2026** of FIG. **20**), a control plane data tier **2328** (e.g., the control plane data tier **2028** of FIG. **20**) that can include DB subnet(s) **2330** (e.g., DB subnet(s) **2230** of FIG. **22**). The LB subnet(s) **2322** contained in the control plane DMZ tier **2320** can be communicatively coupled to the app subnet(s) **2326** contained in the control plane app tier **2324** and to an Internet gateway **2334** (e.g., the Internet gateway **2034** of FIG. **20**) that can be contained in the control plane VCN **2316**, and the app subnet(s) **2326** can be communicatively coupled to the DB subnet(s) **2330** contained in the control plane data tier **2328** and to a service gateway **2336** (e.g., the service gateway of FIG. **20**) and a network address translation (NAT) gateway **2338** (e.g., the NAT gateway **2038** of FIG. **20**). The control plane VCN **2316** can include the service gateway **2336** and the NAT gateway **2338**.

[0330] The data plane VCN **2318** can include a data plane app tier **2346** (e.g., the data plane app tier **2046** of FIG. **20**), a data plane DMZ tier **2348** (e.g., the data plane DMZ tier **2048** of FIG. **20**), and a data plane data tier **2350** (e.g., the data plane data tier **2050** of FIG. **20**). The data plane DMZ tier **2348** can include LB subnet(s) **2322** that can be communicatively coupled to trusted app subnet(s) **2360** (e.g., trusted app subnet(s) **2260** of FIG. **22**) and untrusted app subnet(s) **2362** (e.g., untrusted app subnet(s) **2262** of FIG. **22**) of the data plane app tier **2346** and the Internet gateway **2334** contained in the data plane VCN **2318**. The trusted app subnet(s) **2360** can be communicatively coupled to the service gateway **2336** contained in the data plane VCN **2318**, the NAT gateway **2338** contained in the data plane VCN **2318**, and DB subnet(s) **2330** contained in the data plane data tier **2350**. The untrusted app subnet(s) **2362** can be communicatively coupled to the service gateway **2336** contained in the data plane VCN **2318** and DB subnet(s) **2330** contained in the data plane data tier **2350**. The data plane data tier **2350** can include DB subnet(s) **2330** that can be communicatively coupled to the service gateway **2336** contained in the data plane VCN **2318**.

[0331] The untrusted app subnet(s) **2362** can include primary VNICS **2364(1)-(N)** that can be communicatively coupled to tenant virtual machines (VMs) **2366(1)-(N)** residing within the untrusted app subnet(s) **2362**. Each tenant VM **2366(1)-(N)** can run code in a respective container **2367(1)-(N)**, and be communicatively coupled to an app subnet **2326** that can be contained in a data plane app tier **2346** that can be contained in a container egress VCN **2368**. Respective secondary VNICS **2372(1)-(N)** can facilitate communication between the untrusted app subnet(s) **2362** contained in the data plane VCN **2318** and the app subnet contained in the container egress VCN **2368**. The container egress VCN can include a NAT gateway **2338** that can be communicatively coupled to public Internet **2354** (e.g., public Internet **2054** of FIG. **20**).

[0332] The Internet gateway **2334** contained in the control plane VCN **2316** and contained in the data plane VCN **2318** can be communicatively coupled to a metadata management service **2352** (e.g., the metadata management system **2052** of FIG. **20**) that can be communicatively coupled to public Internet **2354**. Public Internet **2354** can be communicatively coupled to the NAT gateway **2338** contained in the control plane VCN **2316** and contained in the data plane VCN **2318**. The service gateway **2336** contained in the control plane VCN **2316** and contained in the data plane VCN **2318** can be communicatively coupled to cloud services **2356**.

[0333] In some examples, the pattern illustrated by the architecture of block diagram **2300** of FIG. **23** may be considered an exception to the pattern illustrated by the architecture of block diagram **2200** of FIG. **22** and may be desirable for a customer of the IaaS provider if the IaaS provider cannot directly communicate with the customer (e.g., a disconnected region). The respective containers **2367(1)-(N)** that are contained in the VMs **2366(1)-(N)** for each customer can be accessed in real-time by the customer. The containers **2367(1)-(N)** may be configured to make calls to respective secondary VNICS **2372(1)-(N)** contained in app subnet(s) **2326** of the data plane app tier **2346** that can be contained in the container egress VCN **2368**. The secondary VNICS **2372(1)-(N)** can transmit the calls to the NAT gateway **2338** that may transmit the calls to public Internet **2354**. In this example, the containers **2367(1)-(N)** that can be accessed in real-time by the customer can be isolated from the control plane VCN **2316** and can be isolated from other entities contained in the data plane VCN **2318**. The containers **2367(1)-(N)** may also be isolated from resources from other customers.

[0334] In other examples, the customer can use the containers **2367(1)-(N)** to call cloud services **2356**. In this example, the customer may run code in the containers **2367(1)-(N)** that requests a service from cloud services **2356**. The containers **2367(1)-(N)** can transmit this request to the secondary VNICS **2372(1)-(N)** that can transmit the request to the NAT gateway that can transmit the request to public Internet **2354**. Public Internet **2354** can transmit the request to LB subnet(s) **2322** contained in the control plane VCN **2316** via the Internet gateway **2334**. In response to determining the request is valid, the LB subnet(s) can transmit the request to app subnet(s) **2326** that can transmit the request to cloud services **2356** via the service gateway **2336**.

[0335] It should be appreciated that IaaS architectures **2000**, **2000**, **2200**, **2300** depicted in the figures may have other components than those depicted. Further, the embodiments shown in the figures are only some examples of a cloud infrastructure system that may incorporate an embodiment of the disclosure. In some other embodiments, the IaaS systems may have more or fewer components than shown in the figures, may combine two or more components, or may have a different configuration or arrangement of components.

[0336] In certain embodiments, the IaaS systems described herein may include a suite of applications, middleware, and database service offerings that are delivered to a customer in a self-service, subscription-based, elastically scalable, reliable, highly available, and secure manner. An example of such an IaaS system is the Oracle Cloud Infrastructure (OCI) provided by the present assignee.

[0337] FIG. **24** illustrates an example computer system **2400**, in which various embodiments may be implemented. The system **2400** may be used to implement any of the computer systems described above. As shown in the figure, computer system **2400** includes a processing unit **2404** that communicates with a number of peripheral subsystems via a bus subsystem **2402**. These peripheral subsystems may include a processing acceleration unit **2406**, an I/O subsystem **2408**, a storage subsystem **2418** and a communications subsystem **2424**. Storage subsystem **2418** includes tangible computer-readable storage media **2422** and a system memory **2410**.

[0338] Bus subsystem **2402** provides a mechanism for letting the various components and subsystems of computer system **2400** communicate with each other as intended. Although bus subsystem **2402** is shown schematically as a single bus, alternative embodiments of the bus subsystem may utilize multiple buses. Bus subsystem **2402** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. For example, such architectures may include an Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus, which can be implemented as a Mezzanine bus manufactured to the IEEE P1386.1 standard.

[0339] Processing unit **2404**, which can be implemented as one or more integrated circuits (e.g., a conventional microprocessor or microcontroller), controls the operation of computer system **2400**. One or more processors may be included in processing unit **2404**. These processors may include single core or multicore processors. In certain embodiments, processing unit **2404** may be implemented as one or more independent processing units **2432** and/or **2434** with single or multicore processors included in each processing unit. In other embodiments, processing unit **2404** may also be implemented as a quad-core processing unit formed by integrating two dual-core processors into a single chip.

[0340] In various embodiments, processing unit **2404** can execute a variety of programs in response to program code and can maintain multiple concurrently executing programs or processes. At any given time, some or all of the program code to be executed can be resident in processor(s) **2404** and/or in storage subsystem **2418**. Through suitable programming, processor(s) **2404** can provide various functionalities described above. Computer system **2400** may additionally include a processing acceleration unit **2406**, which can include a digital signal processor (DSP), a special-purpose processor, and/or the like.

[0341] I/O subsystem **2408** may include user interface input devices and user interface output devices. User interface input devices may include a keyboard, pointing devices such as a mouse or trackball, a touchpad or touch screen incorporated into a display, a scroll wheel, a click wheel, a dial, a button, a switch, a keypad, audio input devices with voice command recognition systems, microphones, and other types of input devices. User interface input devices may include, for example, motion sensing and/or gesture recognition devices such as the Microsoft Kinect® motion sensor that enables users to control and interact with an input device, such as the Microsoft Xbox® 360 game controller, through a natural user interface using gestures and spoken commands. User interface input devices may also include eye gesture recognition devices such as the Google Glass® blink detector that detects eye activity (e.g., ‘blinking’ while taking pictures and/or making a menu selection) from users and transforms the eye gestures as input into an input device (e.g., Google Glass®). Additionally, user interface input devices may include voice recognition sensing devices that enable users to interact with voice recognition systems (e.g., Siri® navigator), through voice commands.

[0342] User interface input devices may also include, without limitation, three dimensional (3D) mice, joysticks or pointing sticks, gamepads and graphic tablets, and audio/

visual devices such as speakers, digital cameras, digital camcorders, portable media players, webcams, image scanners, fingerprint scanners, barcode reader 3D scanners, 3D printers, laser rangefinders, and eye gaze tracking devices. Additionally, user interface input devices may include, for example, medical imaging input devices such as computed tomography, magnetic resonance imaging, position emission tomography, medical ultrasonography devices. User interface input devices may also include, for example, audio input devices such as MIDI keyboards, digital musical instruments, and the like.

[0343] User interface output devices may include a display subsystem, indicator lights, or non-visual displays such as audio output devices, etc. The display subsystem may be a cathode ray tube (CRT), a flat-panel device, such as that using a liquid crystal display (LCD) or plasma display, a projection device, a touch screen, and the like. In general, use of the term “output device” is intended to include all possible types of devices and mechanisms for outputting information from computer system **2400** to a user or other computer. For example, user interface output devices may include, without limitation, a variety of display devices that visually convey text, graphics, and audio/video information such as monitors, printers, speakers, headphones, automotive navigation systems, plotters, voice output devices, and modems.

[0344] Computer system **2400** may comprise a storage subsystem **2418** that provides a tangible non-transitory computer-readable storage medium for storing software and data constructs that provide the functionality of the embodiments described in this disclosure. The software can include programs, code modules, instructions, scripts, etc., that when executed by one or more cores or processors of processing unit **2404** provide the functionality described above. Storage subsystem **2418** may also provide a repository for storing data used in accordance with the present disclosure.

[0345] As depicted in the example in FIG. 24, storage subsystem **2418** can include various components including a system memory **2410**, computer-readable storage media **2422**, and a computer readable storage media reader **2420**. System memory **2410** may store program instructions that are loadable and executable by processing unit **2404**. System memory **2410** may also store data that is used during the execution of the instructions and/or data that is generated during the execution of the program instructions. Various different kinds of programs may be loaded into system memory **2410** including but not limited to client applications, Web browsers, mid-tier applications, relational database management systems (RDBMS), virtual machines, containers, etc.

[0346] System memory **2410** may also store an operating system **2416**. Examples of operating system **2416** may include various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems, a variety of commercially-available UNIX® or UNIX-like operating systems (including without limitation the variety of GNU/Linux operating systems, the Google Chrome® OS, and the like) and/or mobile operating systems such as iOS, Windows® Phone, Android® OS, BlackBerry® OS, and Palm® OS operating systems. In certain implementations where computer system **2400** executes one or more virtual machines, the virtual machines along with their guest oper-

ating systems (GOSs) may be loaded into system memory **2410** and executed by one or more processors or cores of processing unit **2404**.

[0347] System memory **2410** can come in different configurations depending upon the type of computer system **2400**. For example, system memory **2410** may be volatile memory (such as random access memory (RAM)) and/or non-volatile memory (such as read-only memory (ROM), flash memory, etc.) Different types of RAM configurations may be provided including a static random access memory (SRAM), a dynamic random access memory (DRAM), and others. In some implementations, system memory **2410** may include a basic input/output system (BIOS) containing basic routines that help to transfer information between elements within computer system **2400**, such as during start-up.

[0348] Computer-readable storage media **2422** may represent remote, local, fixed, and/or removable storage devices plus storage media for temporarily and/or more permanently containing, storing, computer-readable information for use by computer system **2400** including instructions executable by processing unit **2404** of computer system **2400**.

[0349] Computer-readable storage media **2422** can include any appropriate media known or used in the art, including storage media and communication media, such as but not limited to, volatile and non-volatile, removable, and non-removable media implemented in any method or technology for storage and/or transmission of information. This can include tangible computer-readable storage media such as RAM, ROM, electronically erasable programmable ROM (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disk (DVD), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or other tangible computer readable media.

[0350] By way of example, computer-readable storage media **2422** may include a hard disk drive that reads from or writes to non-removable, nonvolatile magnetic media, a magnetic disk drive that reads from or writes to a removable, nonvolatile magnetic disk, and an optical disk drive that reads from or writes to a removable, nonvolatile optical disk such as a CD ROM, DVD, and Blu-Ray® disk, or other optical media. Computer-readable storage media **2422** may include, but is not limited to, Zip® drives, flash memory cards, universal serial bus (USB) flash drives, secure digital (SD) cards, DVD disks, digital video tape, and the like. Computer-readable storage media **2422** may also include, solid-state drives (SSD) based on non-volatile memory such as flash-memory based SSDs, enterprise flash drives, solid state ROM, and the like, SSDs based on volatile memory such as solid state RAM, dynamic RAM, static RAM, DRAM-based SSDs, magnetoresistive RAM (MRAM) SSDs, and hybrid SSDs that use a combination of DRAM and flash memory based SSDs. The disk drives and their associated computer-readable media may provide non-volatile storage of computer-readable instructions, data structures, program modules, and other data for computer system **2400**.

[0351] Machine-readable instructions executable by one or more processors or cores of processing unit **2404** may be stored on a non-transitory computer-readable storage medium. A non-transitory computer-readable storage medium can include physically tangible memory or storage devices that include volatile memory storage devices and/or non-volatile storage devices. Examples of non-transitory

computer-readable storage medium include magnetic storage media (e.g., disk or tapes), optical storage media (e.g., DVDs, CDs), various types of RAM, ROM, or flash memory, hard drives, floppy drives, detachable memory drives (e.g., USB drives), or other type of storage device.

[0352] Communications subsystem **2424** provides an interface to other computer systems and networks. Communications subsystem **2424** serves as an interface for receiving data from and transmitting data to other systems from computer system **2400**. For example, communications subsystem **2424** may enable computer system **2400** to connect to one or more devices via the Internet. In some embodiments communications subsystem **2424** can include radio frequency (RF) transceiver components for accessing wireless voice and/or data networks (e.g., using cellular telephone technology, advanced data network technology, such as 3G, 4G or EDGE (enhanced data rates for global evolution), WiFi (IEEE 802.11 family standards, or other mobile communication technologies, or any combination thereof), global positioning system (GPS) receiver components, and/or other components. In some embodiments communications subsystem **2424** can provide wired network connectivity (e.g., Ethernet) in addition to or instead of a wireless interface.

[0353] In some embodiments, communications subsystem **2424** may also receive input communication in the form of structured and/or unstructured data feeds **2426**, event streams **2428**, event updates **2430**, and the like on behalf of one or more users who may use computer system **2400**.

[0354] By way of example, communications subsystem **2424** may be configured to receive data feeds **2426** in real-time from users of social networks and/or other communication services such as Twitter® feeds, Facebook® updates, web feeds such as Rich Site Summary (RSS) feeds, and/or real-time updates from one or more third party information sources.

[0355] Additionally, communications subsystem **2424** may also be configured to receive data in the form of continuous data streams, which may include event streams **2428** of real-time events and/or event updates **2430**, that may be continuous or unbounded in nature with no explicit end. Examples of applications that generate continuous data may include, for example, sensor data applications, financial tickers, network performance measuring tools (e.g., network monitoring and traffic management applications), click-stream analysis tools, automobile traffic monitoring, and the like.

[0356] Communications subsystem **2424** may also be configured to output the structured and/or unstructured data feeds **2426**, event streams **2428**, event updates **2430**, and the like to one or more databases that may be in communication with one or more streaming data source computers coupled to computer system **2400**.

[0357] Computer system **2400** can be one of various types, including a handheld portable device (e.g., an iPhone® cellular phone, an iPad® computing tablet, a PDA), a wearable device (e.g., a Google Glass® head mounted display), a PC, a workstation, a mainframe, a kiosk, a server rack, or any other data processing system.

[0358] Due to the ever-changing nature of computers and networks, the description of computer system **2400** depicted in the figure is intended only as a specific example. Many other configurations having more or fewer components than the system depicted in the figure are possible. For example,

customized hardware might also be used and/or particular elements might be implemented in hardware, firmware, software (including applets), or a combination. Further, connection to other computing devices, such as network input/output devices, may be employed. Based on the disclosure and teachings provided herein, a person of ordinary skill in the art will appreciate other ways and/or methods to implement the various embodiments.

[0359] Although specific embodiments have been described, various modifications, alterations, alternative constructions, and equivalents are also encompassed within the scope of the disclosure. Embodiments are not restricted to operation within certain specific data processing environments, but are free to operate within a plurality of data processing environments. Additionally, although embodiments have been described using a particular series of transactions and steps, it should be apparent to those skilled in the art that the scope of the present disclosure is not limited to the described series of transactions and steps. Various features and aspects of the above-described embodiments may be used individually or jointly.

[0360] Further, while embodiments have been described using a particular combination of hardware and software, it should be recognized that other combinations of hardware and software are also within the scope of the present disclosure. Embodiments may be implemented only in hardware, or only in software, or using combinations thereof. The various processes described herein can be implemented on the same processor or different processors in any combination. Accordingly, where components or services are described as being configured to perform certain operations, such configuration can be accomplished, e.g., by designing electronic circuits to perform the operation, by programming programmable electronic circuits (such as microprocessors) to perform the operation, or any combination thereof. Processes can communicate using a variety of techniques including but not limited to conventional techniques for inter process communication, and different pairs of processes may use different techniques, or the same pair of processes may use different techniques at different times.

[0361] The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. It will, however, be evident that additions, subtractions, deletions, and other modifications and changes may be made thereunto without departing from the broader spirit and scope as set forth in the claims. Thus, although specific disclosure embodiments have been described, these are not intended to be limiting. Various modifications and equivalents are within the scope of the following claims.

[0362] The use of the terms “a” and “an” and “the” and similar referents in the context of describing the disclosed embodiments (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. The term “connected” is to be construed as partly or wholly contained within, attached to, or joined together, even if there is something intervening. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein and each separate value is incorporated into

the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate embodiments and does not pose a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

[0363] Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is intended to be understood within the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

[0364] Preferred embodiments of this disclosure are described herein, including the best mode known for carrying out the disclosure. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. Those of ordinary skill should be able to employ such variations as appropriate and the disclosure may be practiced otherwise than as specifically described herein. Accordingly, this disclosure includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the disclosure unless otherwise indicated herein.

[0365] All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0366] In the foregoing specification, aspects of the disclosure are described with reference to specific embodiments thereof, but those skilled in the art will recognize that the disclosure is not limited thereto. Various features and aspects of the above-described disclosure may be used individually or jointly. Further, embodiments can be utilized in any number of environments and applications beyond those described herein without departing from the broader spirit and scope of the specification. The specification and drawings are, accordingly, to be regarded as illustrative rather than restrictive.

What is claimed is:

1. A method comprising:

associating a subscription of a user account with a cloud service provider (CSP) with a virtual resource of the CSP;

enabling a device associated with the user account to access the virtual resource, the virtual resource available and accessible as part of a private cloud associated with the user account;

determining a change to the subscription; and

disassociating, based on the change, the virtual resource from the subscription such that the virtual resource is unavailable via the private cloud in association with the

subscription and such that the virtual resource is available and accessible for use in association with a different subscription.

2. The method of claim 1, wherein the virtual resource is associated with the subscription and not with any other subscription associated with the user account.

3. The method of claim 2, wherein the virtual resource is not depended on by other virtual resources that are accessible.

4. The method of claim 1, wherein disassociating the virtual resource comprises transmitting a resource management instruction, indicating the virtual resource should be deleted, to a control plane that manages virtual resources associated with the user account.

5. The method of claim 4, wherein the private cloud is a second CSP.

6. The method of claim 1, wherein determining the change to the subscription comprises:

receiving a subscription identifier that identifies the subscription and a subscription state to associate with the subscription.

7. The method of claim 6, wherein determining the change to the subscription further comprises receiving the subscription state from the private cloud.

8. The method of claim 6, wherein determining the change to the subscription further comprises:

receiving an indication associated with the subscription from a subscription billing system.

9. The method of claim 6, wherein the subscription state is determined based at least in part on a length of time the subscription has been associated with a second subscription state.

10. The method of claim 6, further comprising:

receiving the subscription identifier, the subscription state, and an identifier of the virtual resource; and associating the virtual resource with the subscription identifier based at least in part on the subscription state.

11. The method of claim 1, wherein the unavailable virtual resource comprises a terminated virtual resource.

12. The method of claim 11, wherein the terminated virtual resource has a configuration associated with the virtual resource removed; and wherein the method further comprises:

reclaiming, by the CSP, at least one of computing power, storage, or bandwidth allocated to the user account so that at least one of other user accounts or other subscriptions can use the at least one of the computing power, the storage, or the bandwidth.

13. The method of claim 1, wherein determining the change to the subscription comprises determining the subscription is terminated; and

wherein the method further comprises:

disassociating, based on the change, all virtual resources from the subscription such that all the virtual resource are unavailable via the private cloud in association with the subscription and such that the virtual resources are available and accessible for use in association with the different subscription.

14. A system comprising:

one or more storage media storing instructions; and one or more processors configured to execute the instructions to cause the system to:

associate a subscription of a user account with a cloud service provider (CSP) with a virtual resource of the CSP;

enable a device associated with the user account to access the virtual resource, the virtual resource available and accessible as part of a private cloud associated with the user account;

determine a change to the subscription; and

disassociate, based on the change, the virtual resource from the subscription such that the virtual resource is unavailable via the private cloud in association with the subscription and such that the virtual resource is available and accessible for use in association with a different subscription.

15. The system of claim 14, wherein the virtual resource is associated with the subscription and not with any other subscription associated with the user account.

16. The system of claim 14, wherein the virtual resource is not depended on by other virtual resources that are accessible.

17. The system of claim 14, wherein the executing the instructions further cause the system to:

reclaim at least one of computing power, storage, or bandwidth allocated to the user account so that at least one of other user accounts or other subscriptions can use the at least one of the computing power, the storage, or the bandwidth.

18. One or more non-transitory computer-readable media storing instructions which, when executed by one or more processors, cause a system to perform operations comprising:

associating a subscription of a user account with a cloud service provider (CSP) with a virtual resource of the CSP;

enabling a device associated with the user account to access the virtual resource, the virtual resource available and accessible as part of a private cloud associated with the user account;

determining a change to the subscription; and

disassociating, based on the change, the virtual resource from the subscription such that the virtual resource is unavailable via the private cloud in association with the subscription and such that the virtual resource is available and accessible for use in association with a different subscription.

19. The one or more non-transitory computer-readable media of claim 18, wherein executing the instructions further cause the system to perform operations comprising:

receiving a subscription identifier that identifies the subscription and a subscription state to associate with the subscription.

20. The one or more non-transitory computer-readable media of claim 18, wherein the virtual resource is associated with the subscription and not with any other subscription associated with the user account.

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